

Consumer Electronics

JAN-MAY2026

References:

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- A.Dhake (2000) “Color Television”
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Introduction to Consumer Electronics

- Consumer Electronics deals with the **design, operation, and application of electronic devices** meant for everyday personal and household use.
- These devices are developed to make human life more comfortable, efficient, and entertaining.
- Common examples **include televisions, smartphones, audio systems, home appliances, computers, digital cameras, and smart home devices.**

1961 Radio and TV shop



- **1920s, radio broadcasting became the basis of the mass production of radio receivers.**
- **The vacum tubes that had made radios practical with record players .**
- **This was to amplify the sound so that it could be played through a loudspeaker.**
- **Television was invented soon after, but remained insignificant in the consumer market until the 1950s.**

The first working transistor, was invented by John Bardeen and Walter Houser Brattain at Bell Labs in 1947.

The invention and development of the earliest transistors at Bell led to transistor radios, in turn promoting the emergence of the home entertainment.

integrated circuits (ICs) followed when manufacturers built circuits (usually for military purposes) on a single substrate using electrical connections between circuits within the chip itself.

Consumer products



Top 10 Consumer Electronic Companies

- **1. Apple**
- **2. Microsoft**
- **3. Samsung**
- **4. DELL**
- **5. Sony**
- **6. Hitachi**
- **7. LG**
- **8. Panasonic**
- **9. HP**
- **10. Fujitsu**

Top 10 Consumer Electronics Companies Globally By Market Cap (2025)



Consumer product Design

- Consumer product design is the structured process of creating functional, aesthetically pleasing products that meet consumer needs and market demands.
- Consumer product design involves the creation of everyday products by integrating various disciplines such as industrial design, engineering, and marketing.
- The primary goal is to enhance user experience while ensuring that products are manufacturable at scale. Effective product design not only meets consumer needs but also differentiates a brand in a competitive market.

Ideas to design consumer products

1. Observe Real-Life Problems

How to do it?

Observe daily activities at home, office, travel, hospitals, temples, markets

- Notice:
 - Repeated complaints
 - Wasted time, energy, or money

Create a Idea: “Why is this still inconvenient?”

Example:

- People forgetting to switch off appliances → Smart auto-off plugs
- Elderly struggling with buttons → Voice-based controls

2. Talk to Users (User-Centric Thinking)

- Users reveal problems may be designers' mistakes.
- Short interviews (5–10 minutes)
- Ask open questions:
 - “What frustrates you daily?”
 - “What product do you wish existed?”
- Focus on emotions, not just functions

Remember: Users describe problems better than solutions.

- Smarter?
- Smarter?
- More energy-efficient?
- More energy-efficient?

3. Study Existing Products

- Innovation is often improvement, not invention.
- Analyze:
- What works well?
- What users complain about in reviews
- High price, poor durability, complexity, maintenance issues

Methods:

- Reverse engineering
- Feature comparison tables
- Tear-down analysis (hardware)

Can it be cheaper, Smaller? Smarter? More energy-efficient?

4. **Combine Technologies** (Cross-Domain Innovation)

- New ideas emerge by combining unrelated technologies.

Examples:

- IoT + Home appliances → Smart homes
- AI + Cameras → Face recognition doorbells
- Sensors + Health → Wearable health devices

Many successful products are technology combinations.

5. Use Structured Creativity Tools

When ideas don't come naturally, use frameworks.

(a) SCAMPER Technique

- Substitute
- Combine
- Adapt
- Modify
- Put to another use
- Eliminate
- Rearrange

Example:

Eliminate buttons → Touch / voice interface

Design Thinking (5 Steps)

- **Empathize**
- **Define problem**
- **Methodology**
- **Prototype**
- **Test**

This method is widely used by Apple, IDEO, Google.

Watch trends like:

- Energy efficiency
- Sustainability
- AI assistants
- Health monitoring
- Aging population needs

7. Think “Who Is Ignored?”

- Many products fail to serve:
- Elderly
- Children
- Disabled
- Rural users

Designing for **ignored users** creates powerful innovations.

Example:

- Simple phones for seniors
- Low-power devices for villages

8. Prototype

- Ideas improve when tested.

Tools:

- Breadboards
- Arduino / ESP32
- 3D printing
- Simulation software
- A rough prototype reveals:
 - Design flaws
 - User feedback
 - New ideas

Learn from Failures & Feedback

- Every failed idea teaches something.
- Keep an idea notebook
- Record rejected concepts
- Revise them later

Many good ideas succeed only after 2–3 redesigns

10. Ask the Right Questions Daily

Train your mind with questions like:

- “What can be simplified?”
- “What wastes energy?”
- “What causes confusion?”
- “What can be automated?”

- Thank you

Consumer Electronics

Consumer Product planning

Product Planning is the ongoing process of identifying and articulating market requirements that define a **product's** feature set.

Product planning serves as the basis for decisions about price, distribution and promotion.

The first step in product planning and development is generation of ideas for the development of new/innovative products.

Ideas may come from internal sources like company's own Research and Development (R&D) , managers, sales-force personnel etc.;

Even from external sources like, customers, dealers, competitors, consultants, scientists etc.

Objectives

Making significant contribution to **marketing objectives**.

To fulfil the customer needs:

To spotlight firms strengths and weaknesses:

To fortify better resource utilization:

To guarantee firm's survival:

To increase firm's sale:

Every firm depends on sales as the only source of income to meet the outgoings & retain a decent margin to justify its hard work. Every firm wants to target maximum sales by targeting right customer class at the right time.

Creativity

Product Innovation: adopting **new ideas**

Product Diversification :colours, sizes, designs, quality ,styles etc.

Product Standardisation: varieties or types of products in a given class, cost & human resources.

Components of Product Planning:

Product Innovation: An innovation is the idea of producing new product or developing new process, which is considered to be very useful. Innovation is radical to both growth & change.

Most marketing firms and industrial units will be unable to survive at the time of competitive environment, if they are unable to serve innovative products.

Innovations generate demand and thus motivate firms to produce innovative and technically advanced products with an aim to serve the present & potential market at a profit.

It is means to translate ideas into products and products into profit.

continue

Product Diversification: Product Diversification implies to product expansion in depth and in width.

Depth of product line implies to assortments of colours, sizes, designs, quality & styles etc.

Diversification encourages growth & development
stability to an organization for a longer duration.

It results in creating higher profits.

Diversification acts as a catalyst in the situation of financial crunch & managerial inefficiency

continue

Product Standardisation: Standardisation refers to a limitation of the **numbers of varieties or types originating a given class.**

Uniform quality may be manufactured by the manufacturer reducing unnecessary varieties.

Standardisation helps in contributing towards cost & human resources, thus using the scarce resources very effectively

conserving the scarce inputs for the use of future generation

continue

Product elimination: As a part of business cycle, every company goes through a stage where some of its products are to be eliminated from its product line.

Product elimination is not an easy task which cannot be done overnight. If the things are taken lightly, it may affect the growth of the company and may demoralize the firm.

The biggest problem of product elimination is psychological in nature.

The marketer is reluctant to give up a product, because he has invested his money, time & efforts during different phases of business cycle.

Phases of product planning

Developing the product concept

Studying the market

Market research

Product introduction

Product life cycle

Product elimination.

Stages of New Product Development:

Idea Generation

Screening

Business Analysis

Concept Testing

Product Test

Market Test.

Review and Revision

Commercialization

Product Reliability(Manufacturer point of view)

Is the project defined and do we want to make the investment in this project/product?

Do we have a robust design concept (ready to move forward into detailed design)?

Is the detailed design sufficiently complete (ready for verification)?

Is the completed design fully verified (ready for qualification)?

Is the design validated and production process qualified (ready for production)?

Is production achieving the intended outcome envisioned at the beginning of the project (revenue, margins, customer satisfaction, etc.)?

Are we effectively ending-the-life of the product when it no longer makes financial sense to support it?

continue

Product Reliability is defined as the probability that a device will perform its required function, subjected to stated conditions, for a specific period of time.

Product Reliability is quantified as MTBF (Mean Time Between Failures) for repairable product and MTTF (Mean Time To Failure) for non-repairable product.

Product reliability is important not only to the manufacturer, but also to the consumer. When consumers purchase products, they have certain expectations as to how well those products will perform and for how long. When manufacturers offer product warranties, they are standing behind the reliability of their products.

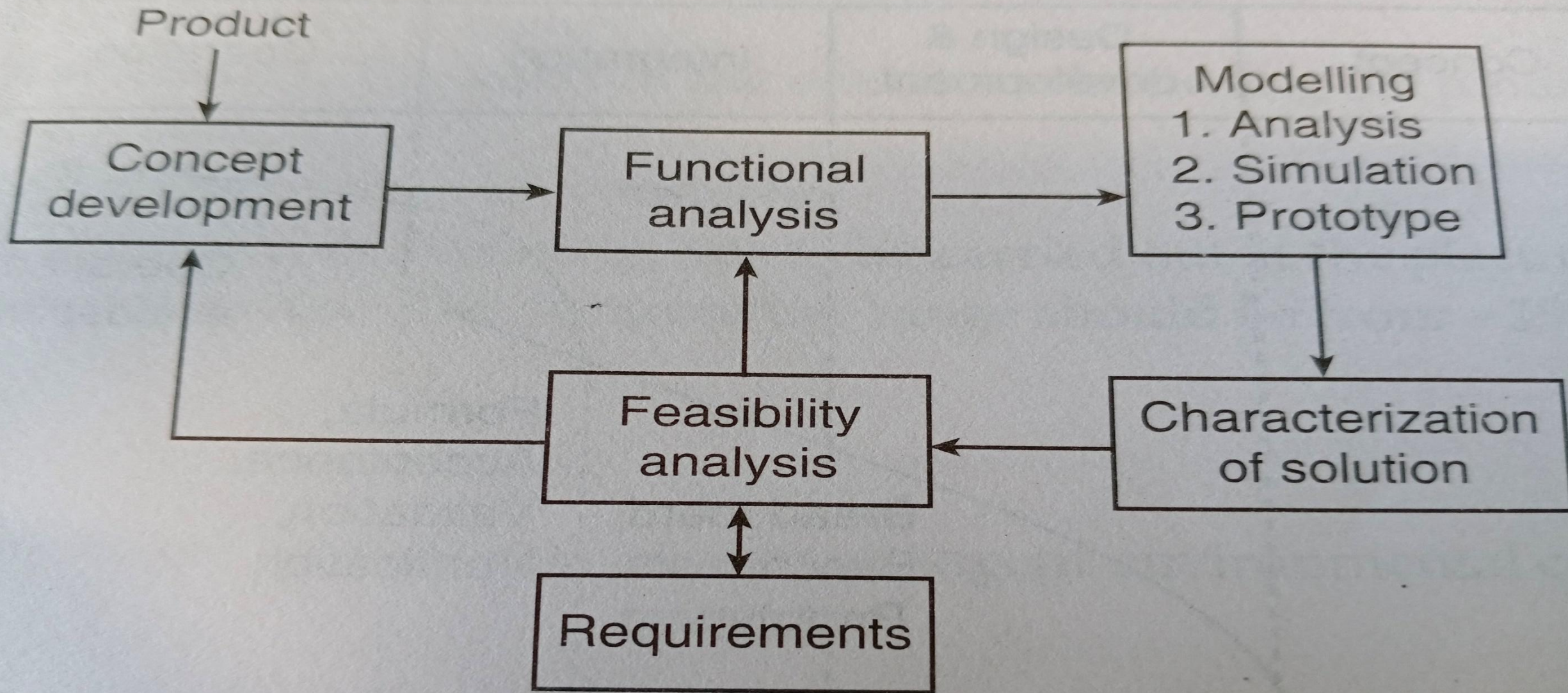
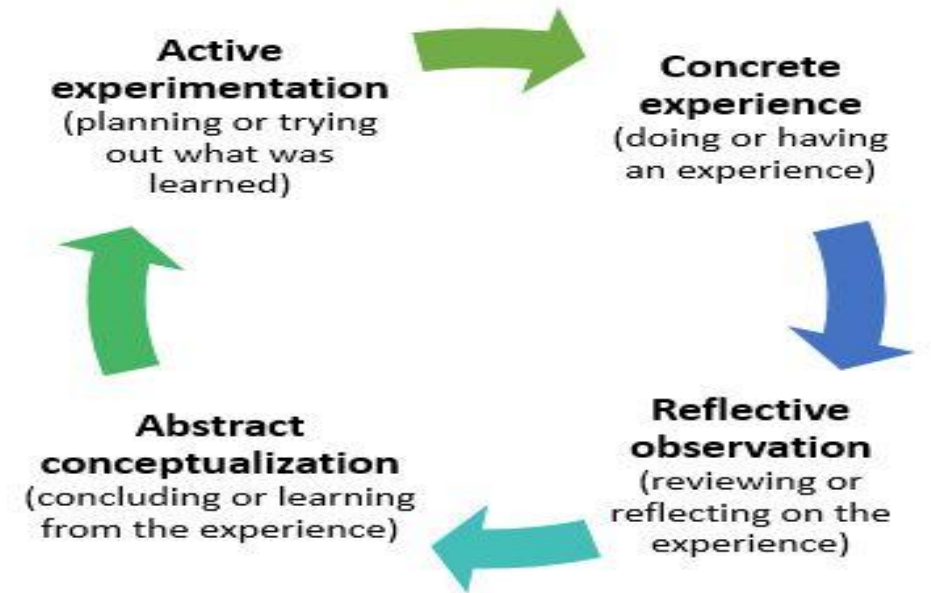
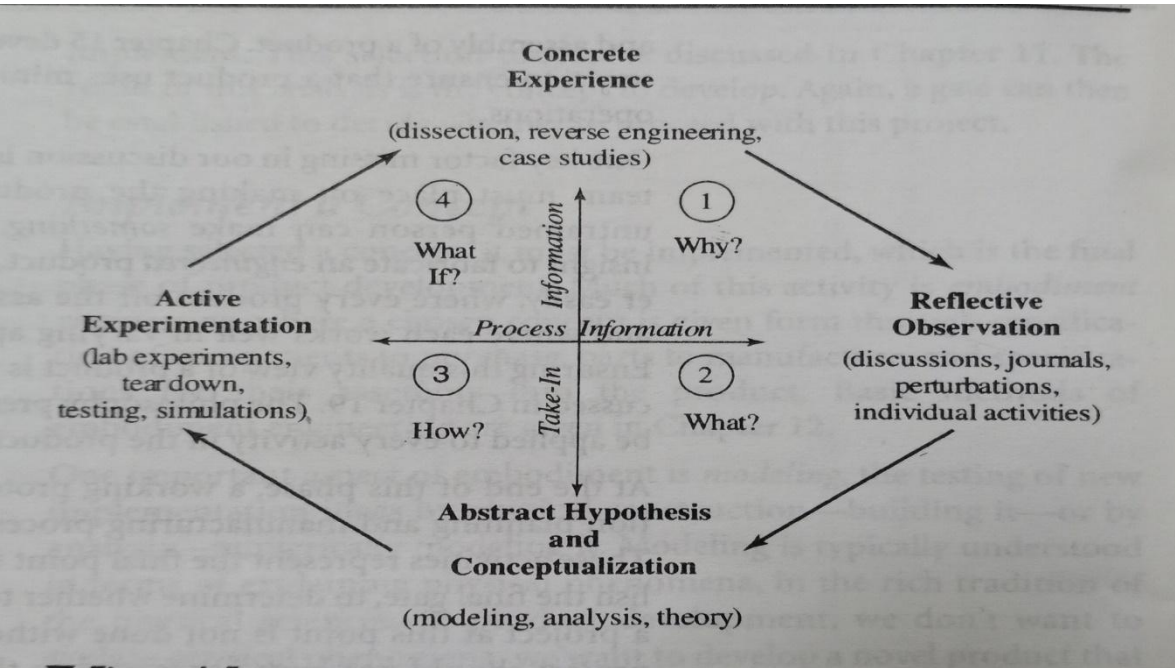


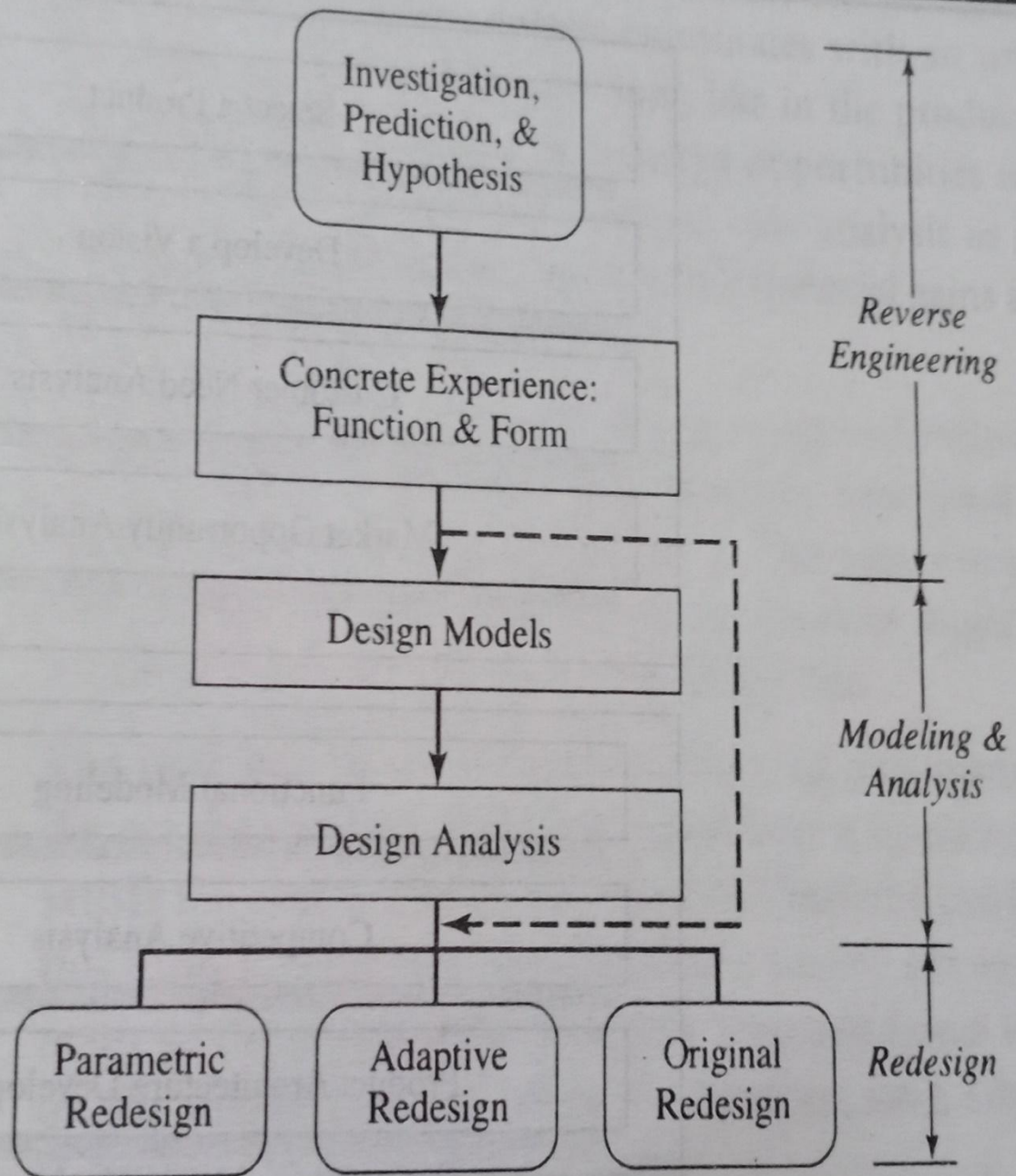
Figure 1.1 Concept of product development.

Kolb's Model of learning



Kolb's four learning styles are Diverging (feeling and watching), Assimilating (watching and thinking), Converging (doing and thinking), and Accommodating (doing and feeling).

Reverse Engineering



Reverse Engineer

Select a Product

Develop a Vision

Customer Need Analysis

Market Opportunity Analysis

Develop a Redesign

Functional Modeling

Competitive Analysis

Product Architecture Development

Concept Engineering

Implement a Redesign

Embodiment Engineering

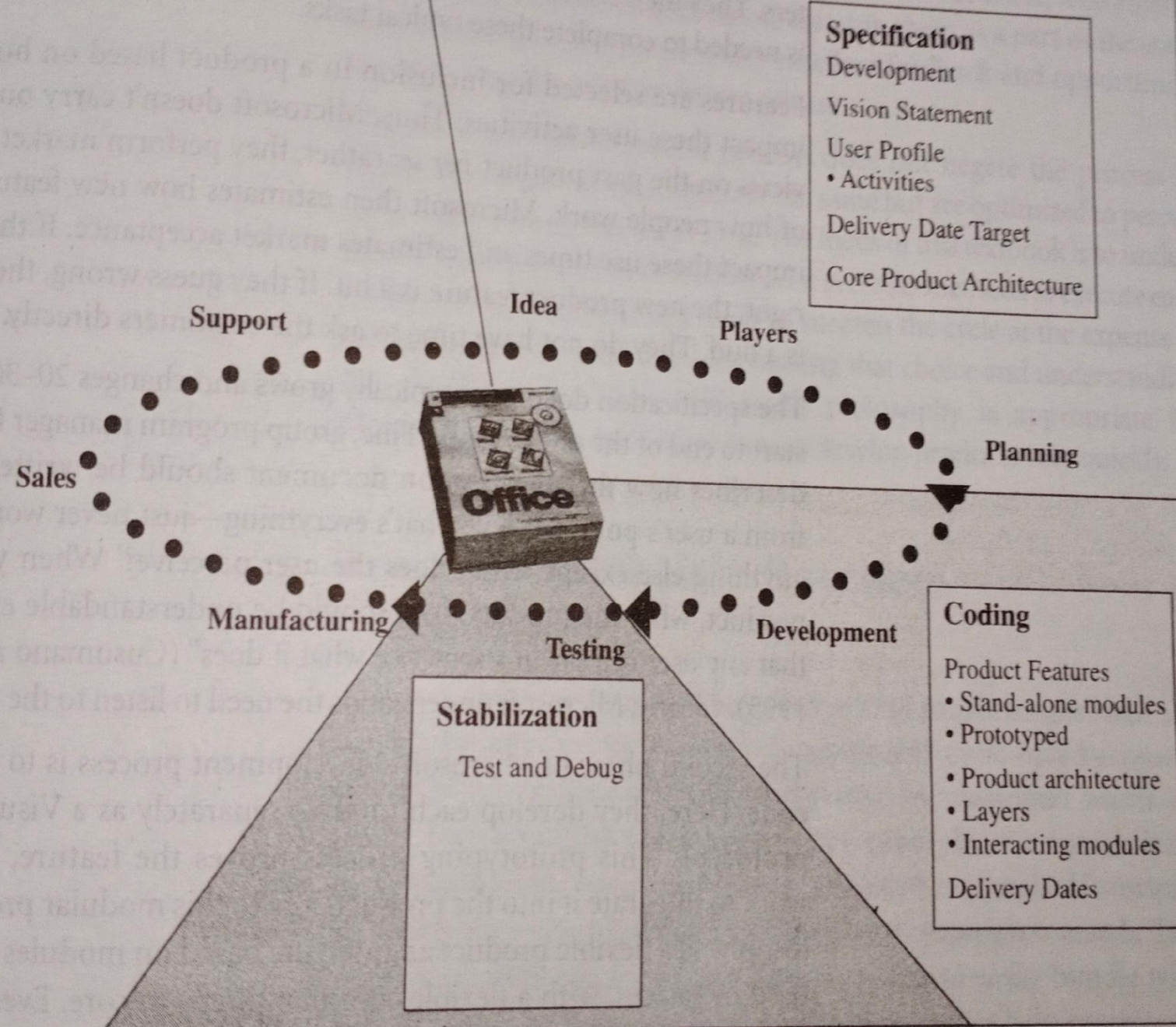
Physical and Analytical Modeling

Design for X

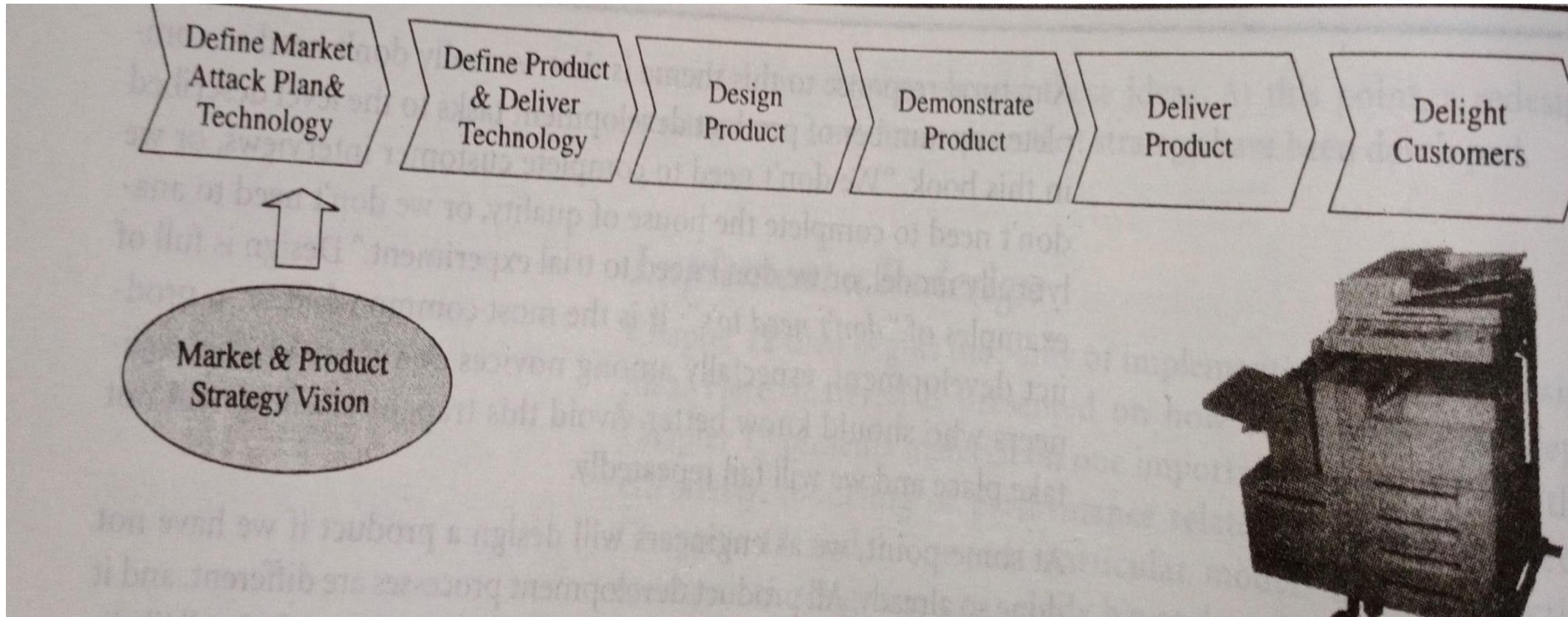
Robust Design

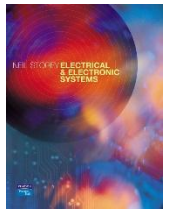
Reverse Engineering and ReDesign Development Process

Microsoft Product Development Process



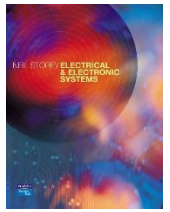
Xerox Product Development Process





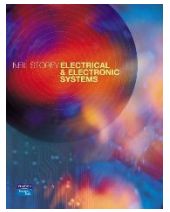
Data Acquisition and Conversion

- Introduction
- Sampling
- Signal Reconstruction
- Data Converters
- Sample and Hold Gates
- Multiplexing



Introduction

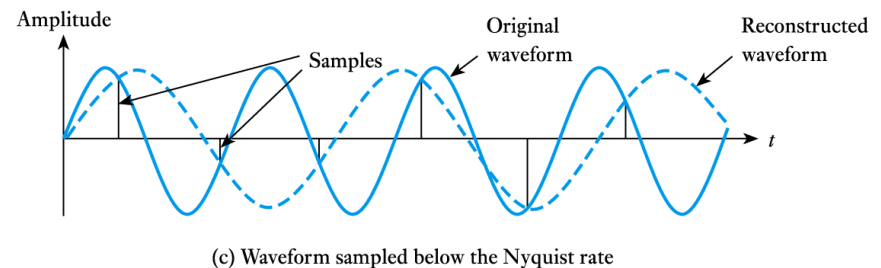
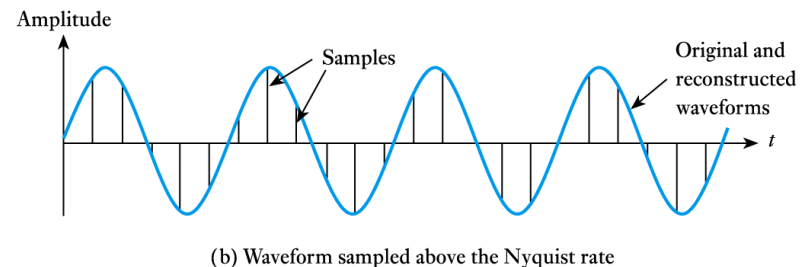
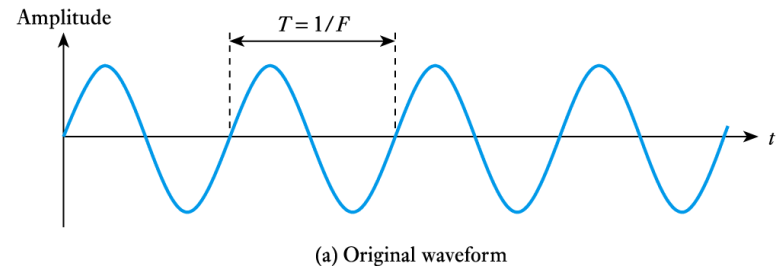
- Digital techniques have several advantages over analogue methods:
 - they are less affected by noise
 - processing, transmission and storage is often easier
- However, we often produce or use analogue signals
- Therefore, we often have the need to translate between analogue and digital representations



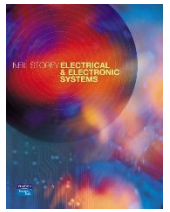
Sampling

- In order to obtain a picture of a varying quantity we need to take regular measurements
 - this process is called **sampling**
 - but how often do we need to sample?
- The answer is given by the **Nyquist sampling theorem** which says that:
 - the sampling rate must be greater than twice the highest frequency present in the signal being sampled*
 - this minimum sampling rate is the **Nyquist rate**

- The effects of sampling rate are illustrated here:
 - (a) shows the original signal
 - (b) shows the effects of sampling at a rate *above* the Nyquist rate
 - (c) shows the effects of sampling at a rate *below* the Nyquist rate

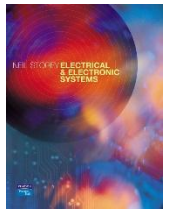


-
- **Note:** the sampling rate is determined by the highest frequency present in the signal, *not* the highest frequency of interest
 - If a signal contains unwanted high frequency components these should be removed before sampling
 - this is done using a low-pass filter
 - such a filter is called an **anti-aliasing filter**
 - It is common to sample at about 20% above the Nyquist rate to allow for imperfect filtering



Signal Reconstruction

- In many cases it is necessary to reconstruct an analogue signal from a series of samples
 - typically after they have been processed, transmitted or stored
- This requires the removal of the step transitions in the sampled waveform
- Reconstruction is achieved using a low-pass filter to remove these unwanted frequencies
 - this filter is called a **reconstruction filter**



Data Converters

- *Sampling* involves taking a series of instantaneous measurements of a signal and converting these into a digital form
- *Reconstruction* involves taking a series of digital readings and converting these back into their analogue equivalents
- These two operations are performed by **data converters** which can be of two basic types:
 - **analogue-to-digital converters (ADCs)**
 - **digital-to-analogue converters (DACs)**

■ Resolution of data converters

- a range of converters is available, each providing conversion to a particular **resolution**
- this determines the number of **quantisation levels** used
- an n -bit converter uses 2^n discrete steps
 - e.g. an 8-bit converter uses 2^8 or 256 levels
 - a 10-bit converter uses 2^{10} or 1024 levels
- an 8-bit converter gives a resolution of about 0.25%
- where greater accuracy is required converters with up to 20-bit resolution or more are available

■ Speed of conversion

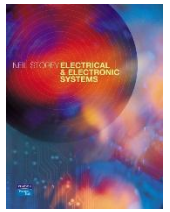
- conversions of either form take a finite time
- this is referred to as the **conversion time** or **settling time** of the converter
- the time taken depends on the converter
- DACs are usually faster than ADCs

■ Digital-to-analogue converters (DACs)

- available with a wide range of resolutions and in general conversion time increases with resolution
- a typical general-purpose 8-bit DAC would have a settling time of between 100 ns and 1 μ s
- a typical 16-bit converter would have a settling time of a few milliseconds
- for specialist applications high-speed converters have settling times of a few nanoseconds
- a video DAC might have a resolution of 8 bits and a maximum sampling rate of 100 MHz

■ Analogue-to-digital converters (ADCs)

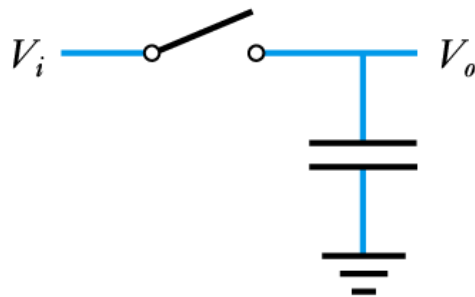
- again available in a range of resolutions and speeds
- most general-purpose devices use a **successive approximation** approach
- a typical 8-bit converter might have a settling time of between 1 and 10 μs
- a typical 12-bit converter might have a settling time of 10 to 100 μs
- high speed **flash converters** can exceed 150 million samples per second



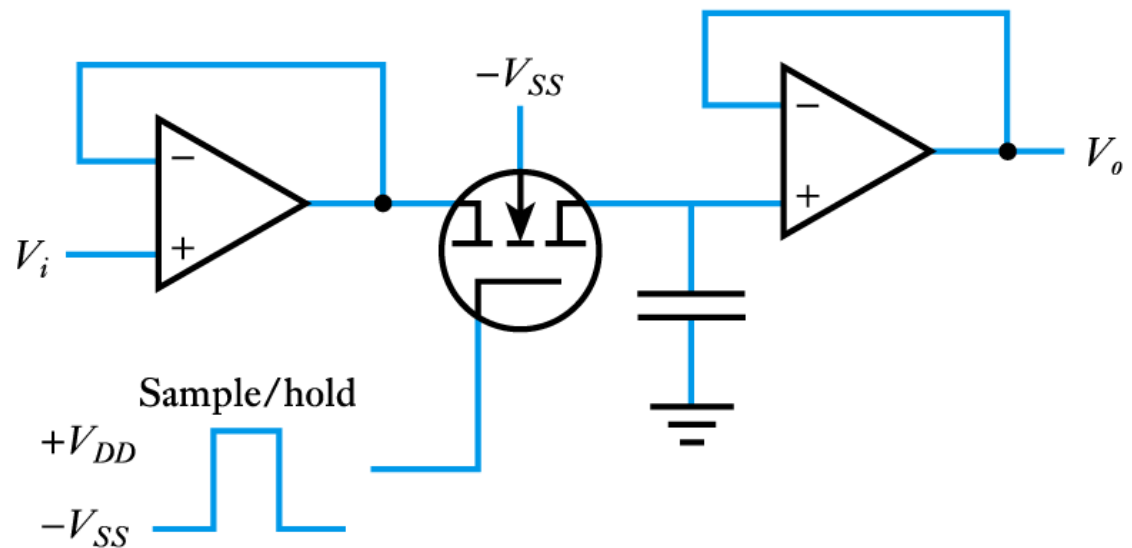
Sample and Hold Gates

- It is often useful to be able to *sample* a signal and then *hold* its value constant
 - this is useful when performing analogue-to-digital conversion so that the signal does not change during conversion
 - it is also useful when doing digital-to-analogue conversion to maintain the output voltage constant between conversions
- This task is performed by a **sample and hold gate**

■ Sample and hold gate circuits

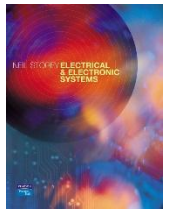


(a) Basic arrangement



(b) A typical circuit

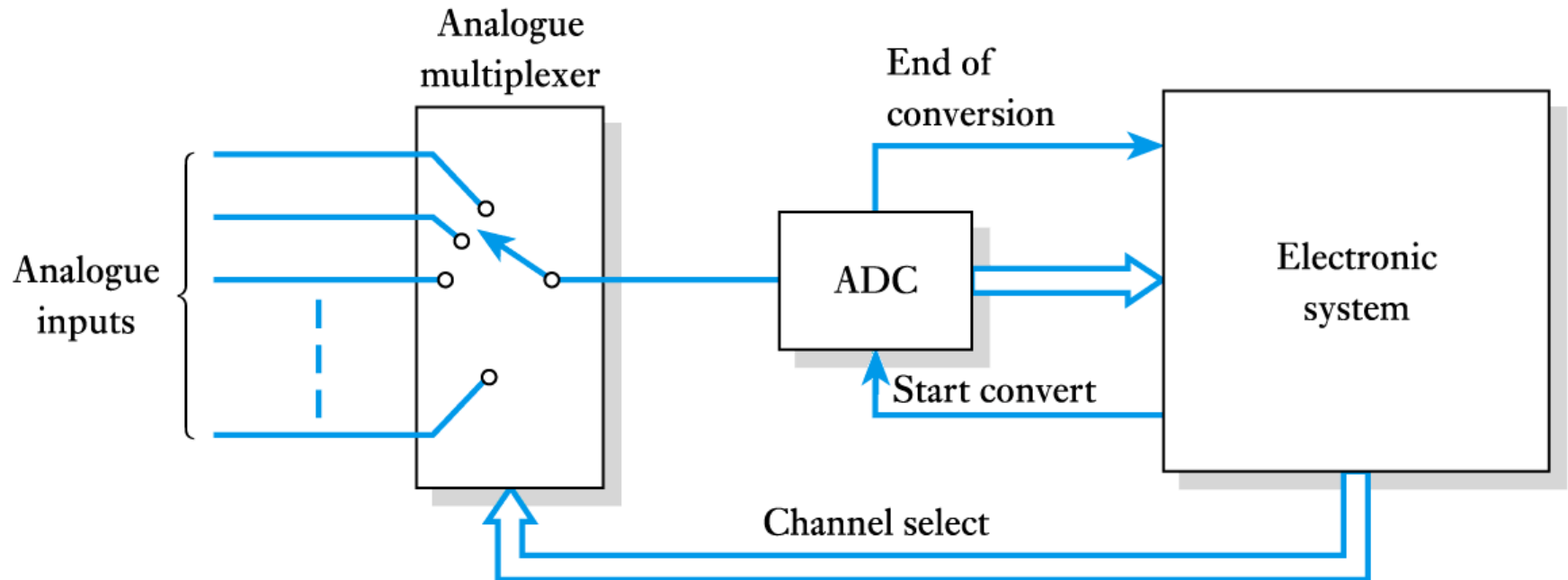
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- Most sample and hold gates are constructed using integrated circuits
 - Typical devices require a few microseconds to sample the incoming waveform, which then decays (or **droops**) at a rate of a few millivolts per millisecond
 - High speed devices, such as those used for video applications, can sample an input in a few nanoseconds, but may experience a droop of a few millivolts per microsecond



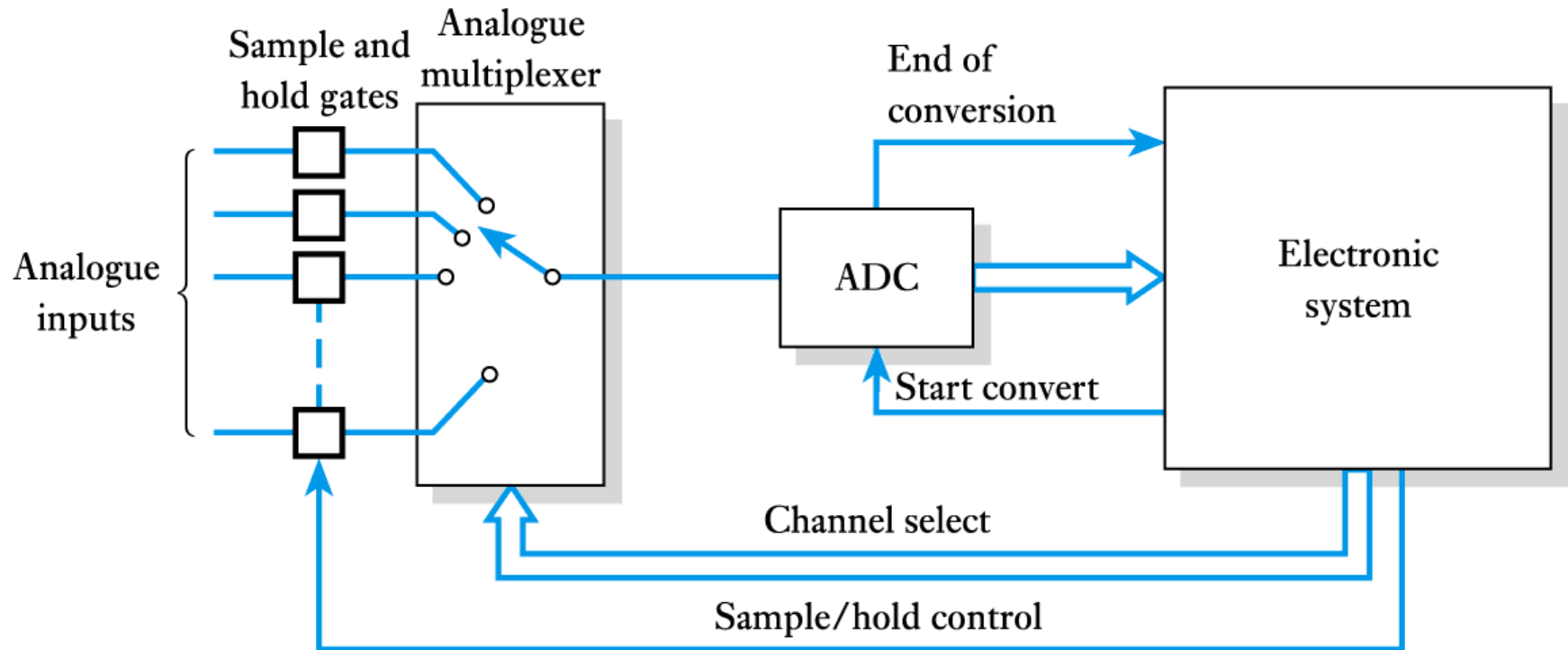
Multiplexing

- While some systems have a single input and a single output, often there are multiple inputs and outputs
- Rather than have separate converters for each input and output, we often use **multiplexing**
 - multiplexers make use of **electrically operated switches** to control the routing of signals
 - these can be used at the input or output of a system
 - normally separate anti-aliasing and/or reconstruction filters would be used with each input

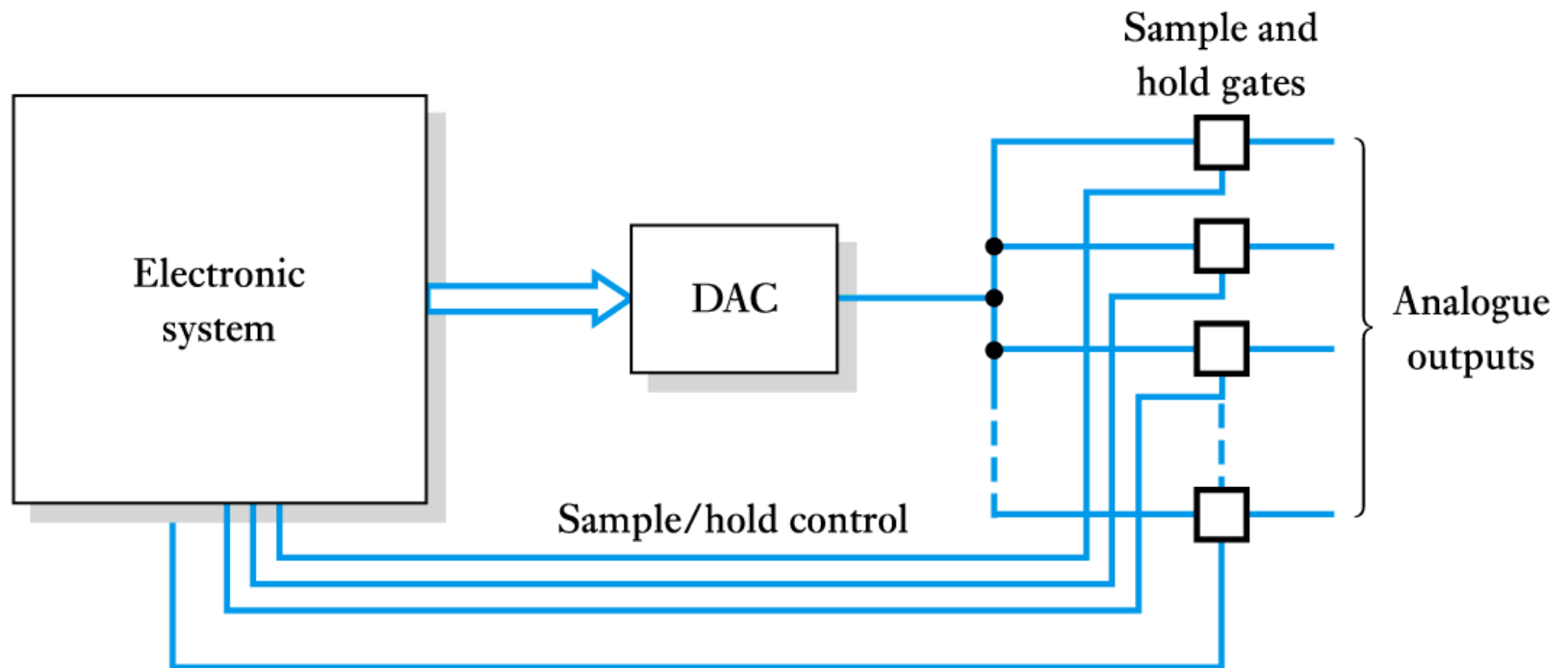
■ Input multiplexing



■ Input multiplexing with sample and hold gates



■ Output multiplexing



■ **Single-chip data acquisition systems**

- these are a combination of an ADC and a multiplexer within a single integrated circuit
- usually designed for use with microprocessor-based systems, and provide all the control lines necessary for simple interfacing

Key Points

- Converting an analogue signal to a digital form is achieved by sampling the waveform and then performing analogue to digital conversion
- As long as the sampling rate is above the Nyquist rate, no information is lost as a result of sampling
- When sampling broad spectrum signals we make use of anti-aliasing filters to remove unwanted components
- When reconstructing signals, filters are used to remove the effects of the sampling
- A wide range of ADCs and DACs is available
- Sample and hold gates may be useful at the input or output
- Multiplexers can reduce the number of converters required

Digital Data Acquisition System

#OE

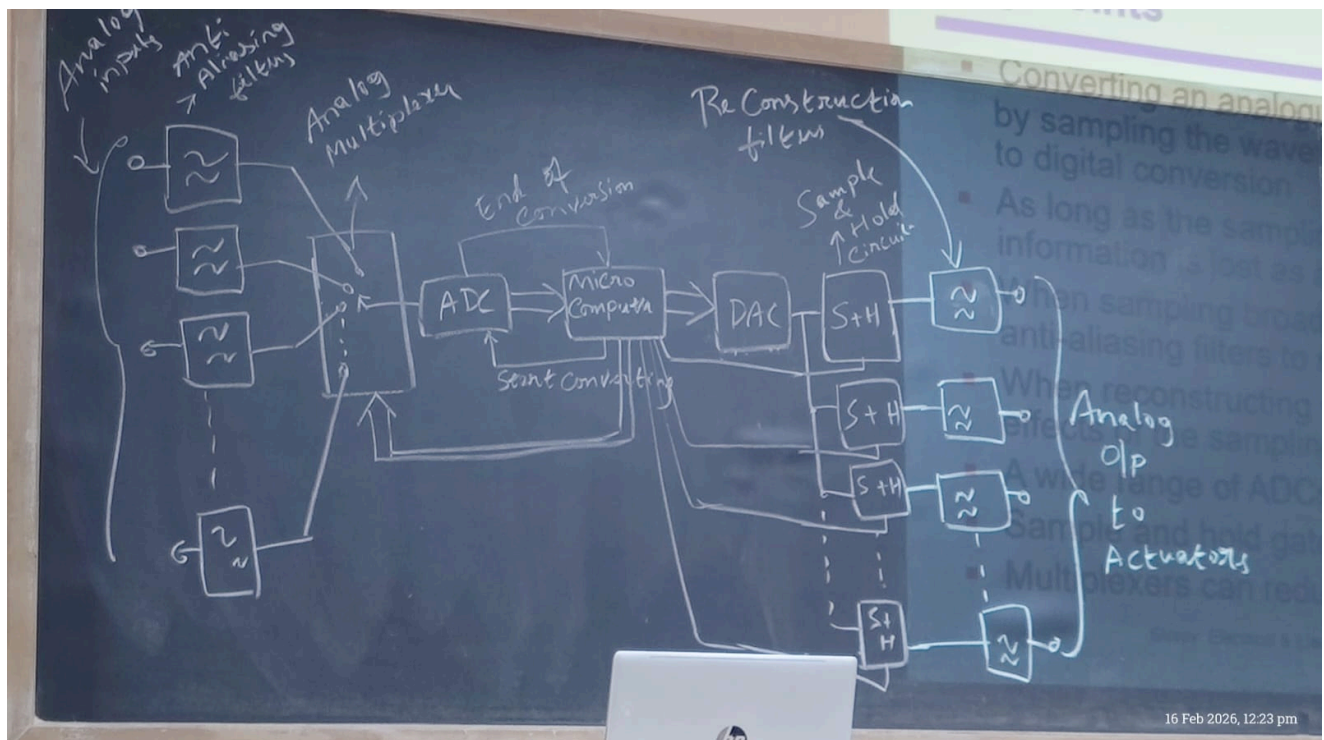
sampling

nyquit

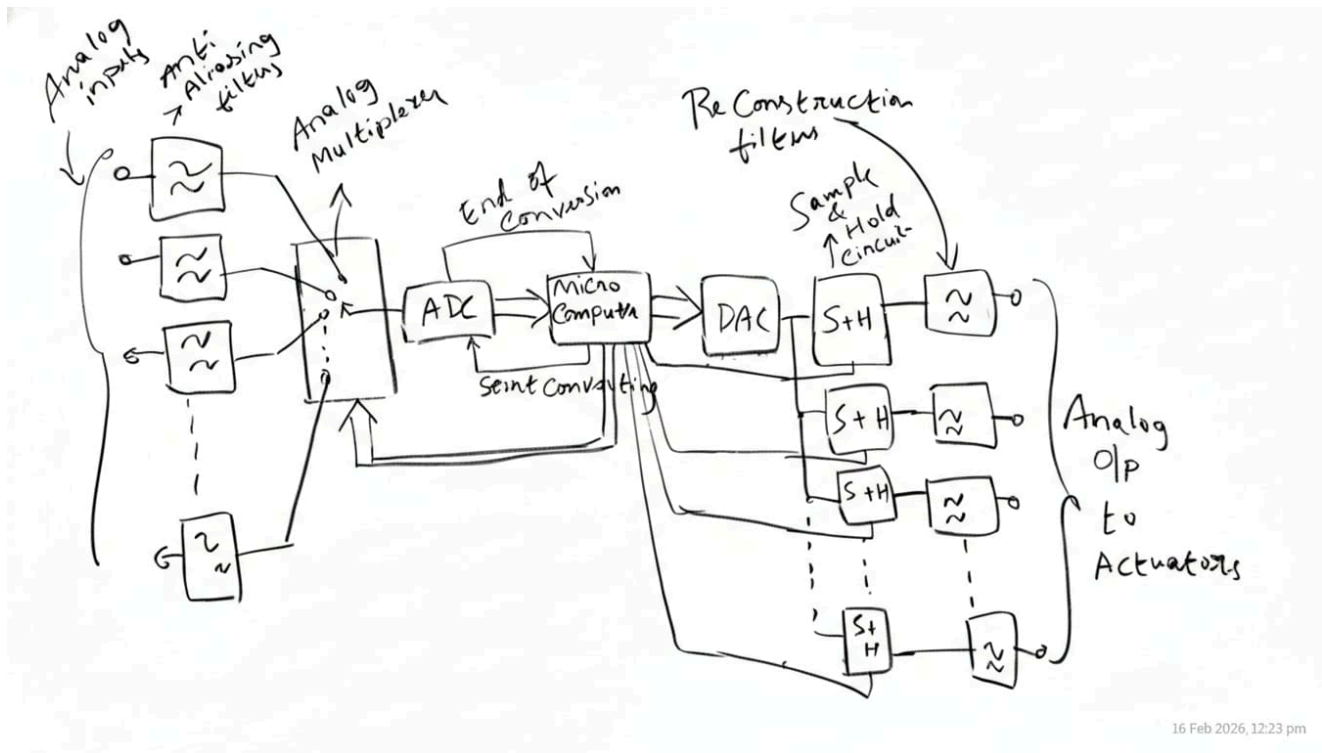
Important!!!!!! might come for midsem

Design A Microcomputer Based System to sense 8 analog input signals and produce 8 analog outputs. The inputs come from sensors that produce useful signals with a bandwidth of 1k Hz but are known to pick up higher frequency noise these signals are to be measured, to an accuracy of at least 1%. The output signals are to drive actuators with a maximum operating bandwidth of 100 Hz but which are affected by higher frequency signals. The actuators require signals to an accuracy of at least 1%. Draw the suitable block diagram for the above case.

solution



16 Feb 2026, 12:23 pm



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As the sensors have a useful signal bandwidth of 1Khz the converter will need to sample each channel with a sampling rate of at least twice the frequency. (2kHz) however the presence of high frequency noise necessitates the use of anti aliasing filters to remove the noise. Assuming that a slight attenuation near 1 kHz is acceptable, it would be appropriate to use low pass filters with a cutoff frequency of 1 kHz. 6th order butterworth filters would be a typical choice. as these are not ideal filters it is necessary to sample somewhat above the nyquist rate. an increase of 20% gives a sampling rate of 2.4kHz. if each channel is sampled at this rate, the ADC must be capable to sample $8 \times 2.4 \text{ kHz} = 19.2 \text{ kHz}$ which corresponds to a conversion time of 52 microseconds. in order to achieve an accuracy of 1% 7bit resolution is required. Infact most general purpose ADCs and DACs have a resolution of at least 8 bits. so it would be sensible to use such devices for the input and output the eight output channels are obtained from a single DAC using a series of sample and hold gates, which are individually controlled by lines from the processor , the outputs from the sample and hold gates would have step changes of voltage as the outputs were updated. as with the input filters, this reconstruction filters would typically be 6th order ButterWorth filter with a cutoff frequency of 100 Hz.



Consumer Electronics

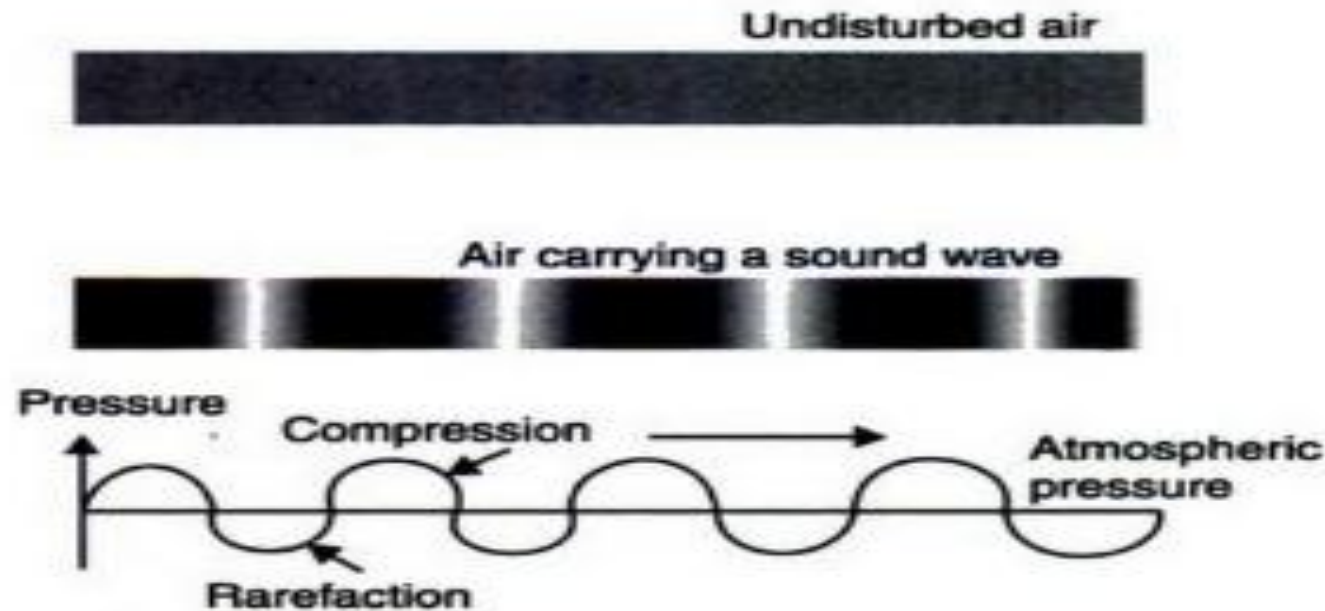
Consumer electronics include **devices** used for entertainment, communications and recreation.

Wave motion

Chapter 1

Wave Motion

Energy and momentum can be transported from one place to another through the motion of particles. Wave motion provides an alternative way for *transfer of energy and momentum without the movement of the material particles.*



1 Along the path of a sound wave in air, the longitudinal to and fro movement of particles produces alternate compressions and rarefactions.

Table 1.1 *The Speed of Sound*

	Medium	Speed (m/s)
Gases	Air (0°C)	331
	Air (20°C)	343
	Helium	965
	Hydrogen	1284
Liquids	Water (0°C)	1402
	Water (20°C)	1482
	Sea water	1522
Solids	Steel	5941
	Granite	6000
	Aluminium	6420

Longitudinal and Transverse waves

Waves in which the vibration of particles is in the same direction in which the wave travels are called *longitudinal waves*. Waves in which the movement of particles is at *right angles* to the direction in which the wave travels are called *transverse waves*.

Longitudinal waves do not require shearing stress and hence may be transmitted through an *elastic medium*. The derivation of the relation between the speed of longitudinal waves and the properties of the medium is somewhat involved, only the end result is given.

$$v = \sqrt{\frac{E}{d}} \quad \dots(1.1)$$

where E is the appropriate modulus and d is the density. For a rod, E is the **Young's modulus**; for a fluid, E is the **bulk modulus** B .

The properties of sound waves can be summed up as follows.

- (i) Sound waves are *longitudinal waves*.
- (ii) Sound travels through *all states of matter*.
- (iii) The velocity of sound is *maximum* in solids and *least* in gases.
- (iv) Sounds which can be heard by human beings are known as *audible sounds*. Their frequency range is from 20 to 20,000 Hz.
- (v) Sounds of frequencies greater than 20,000 Hz are called *ultrasonic sounds*. Ultrasonic sounds can be received and emitted by bats.
- (vi) Sounds of frequencies less than 20 Hz are called *infrasonic sounds*. Infrasonic sounds can be received by snakes.
- (vii) Sound waves can be *reflected*. In the phenomenon of reflection, the speed, frequency and wavelength of the sound waves do not change.
- (viii) Sound waves can be *refracted*. In the phenomenon of refraction, the frequency of the sound waves does not change but their velocity and wavelength change.
- (ix) Sound waves exhibit the phenomenon of *diffraction*.
- (xi) The *normal level* of sound is 60 dB to 120 dB.
- (xii) Sound waves cannot travel through *vacuum*.

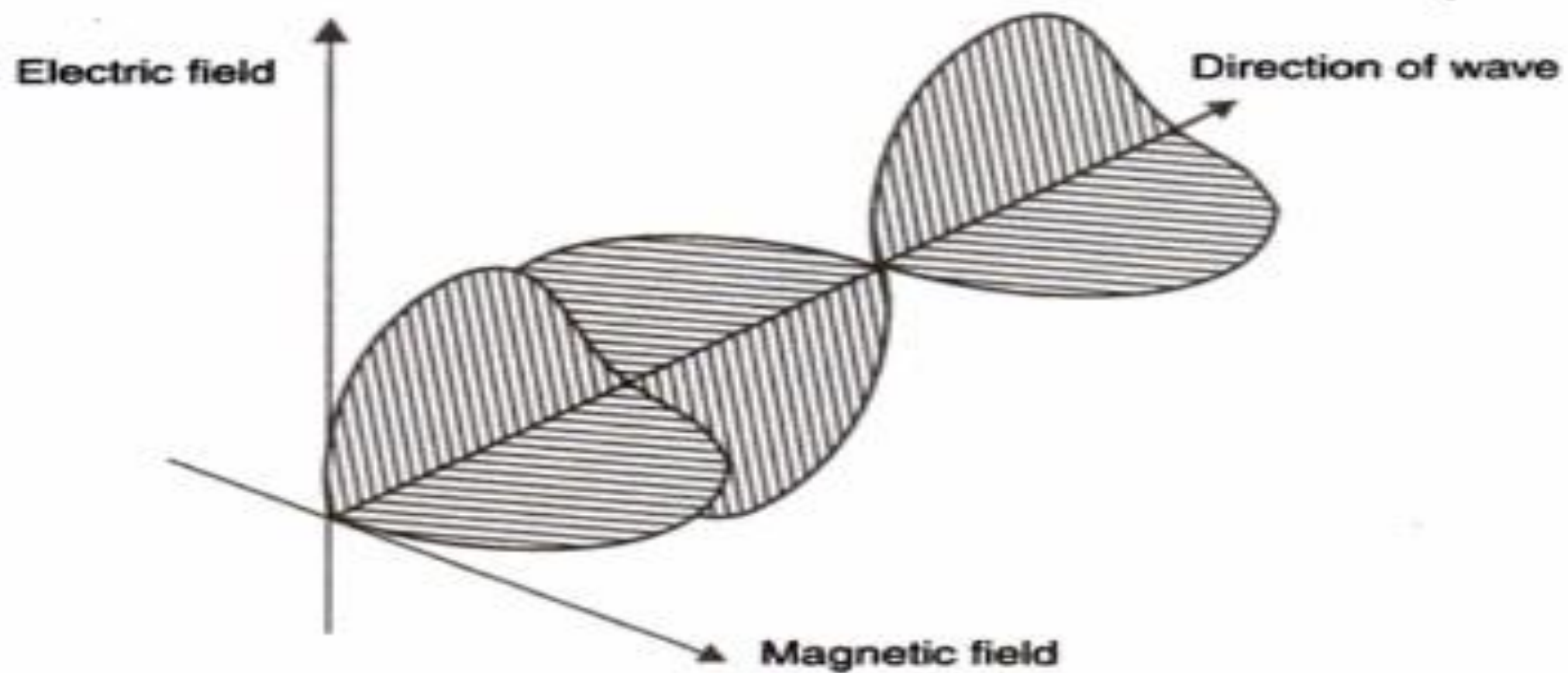


Fig. 1.12 *Field variations along an electromagnetic wave*

Wave motions

Example 1.1 Find the velocity of a compressional wave in an iron whose density is $7.7 \times 10^3 \text{ kg/m}^3$ and whose Young's modulus is $2.0 \times 10^{11} \text{ N/m}^2$.

Solution

$$v = \sqrt{\frac{2.0 \times 10^{11}}{7.7 \times 10^3}} = 5096.4 \text{ m/s}$$

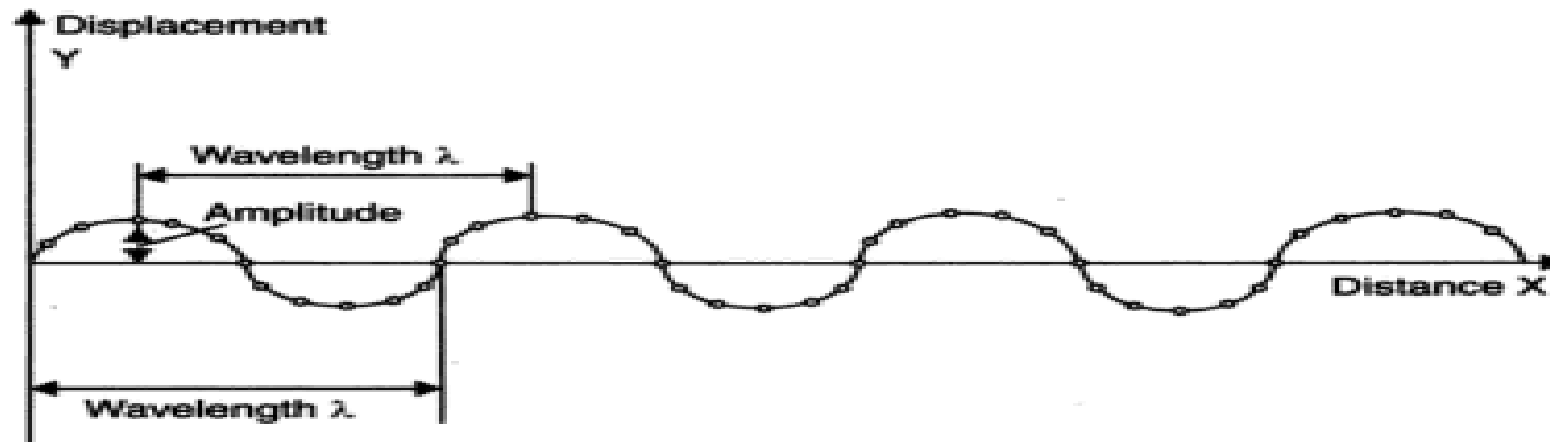
Example 1.2 Find the velocity of a longitudinal wave in water. $B = 2.0 \times 10^9 \text{ N/m}^2$ and $d = 1.0 \times 10^3 \text{ kg/m}^3$.

Solution

$$v = \sqrt{\frac{2.0 \times 10^9}{1.0 \times 10^3}} = 1414 \text{ m/s}$$

Wave Motions

$$\begin{aligned}\text{Velocity of wave} &= \frac{\text{distance travelled by wave}}{\text{time taken}} \\ &= \frac{\text{number of waves} \times \text{wave length}}{\text{time}} \\ \text{velocity} &= \text{frequency} \times \text{wavelength} \\ v &= n\lambda\end{aligned}$$



Example 1.3 One end of a rope is vibrated at 8 Hz and the waves generated measure 60 cm from crest to crest. How fast do they travel and what is the period of vibration of any part of the rope?

Solution

$$\begin{aligned}v &= n \lambda = 8 \times 0.6 \\&= 4.8 \text{ m/s}\end{aligned}$$

$$\begin{aligned}\text{Period} &\longrightarrow = \frac{1}{8} \text{ sec} \\&= 0.125 \text{ s}\end{aligned}$$

If v_t and v_0 be the velocities of sound at $t^\circ\text{C}$ and 0°C respectively, then

$$\frac{v_t}{v_0} = \sqrt{\frac{T_t}{T_0}} \quad \dots(1.3)$$

where T_t and T_0 are absolute temperatures.

Thus the velocity of sound is directly proportional to the square root of the absolute temperature.

Example 1.4 Calculate the increases in the velocity of sound for 1°C rise in temperature, if the velocity of sound at 0°C is 332 m/s .

Solution

$$\frac{v_t}{v_0} = \sqrt{\frac{T_1}{T_0}} = \sqrt{\frac{273+t}{273}}$$

$$v_t = v_0 \left(1 + \frac{t}{273} \right)^{\frac{1}{2}} = v_0 \left(1 + \frac{t}{2 \times 273} \right)$$

$$= v_0 \left(1 + \frac{t}{546} \right) = v_0 + v_0 \left(\frac{t}{546} \right)$$

$$v_t - v_0 = v_0 \left(\frac{t}{546} \right) = \frac{332 \times 1}{546}$$

$$= 0.61\text{ m/s}$$

Intensity of the wave : The intensity I or rate of flow of energy through unit area of the medium in a direction perpendicular to the direction of propagation of the wave is given by

$$I = 2\pi^2 f^2 a^2 v d \quad \dots(1.4)$$

where

f = frequency

a = amplitude

v = velocity of the wave

d = density of the medium.

If P is the power of an isotropic source of sound at a distance r , the intensity or *loudness* of sound.

$$I = \frac{P}{4\pi r^2} \quad \dots(1.5)$$

Since

$$P \propto a^2 \quad \therefore \quad I \propto \frac{a^2}{r^2} \quad \dots(1.6)$$

Example 1.6 Spherical sound waves are emitted uniformly in all directions from a point source, the radiated power P being 25 W. What are the intensity and the sound level of the sound wave at a distance of $r = 2.5$ m from the source?

Solution All of the radiated power must pass through a sphere of radius r centered on the source. Thus

$$I = \frac{P}{4\pi r^2}$$

We see that the *intensity* of the sound drops off as the *inverse square* of the distance from the source.

Thus

$$I = \frac{25}{(4\pi)(2.5)^2} = 0.32 \text{ W/m}^2$$

and

$$\begin{aligned} \text{SL} &= 10 \log \frac{I}{I_0} \\ &= 10 \log \frac{0.32}{10^{-12}} = 115 \text{ dB} \end{aligned}$$

Resonance wave

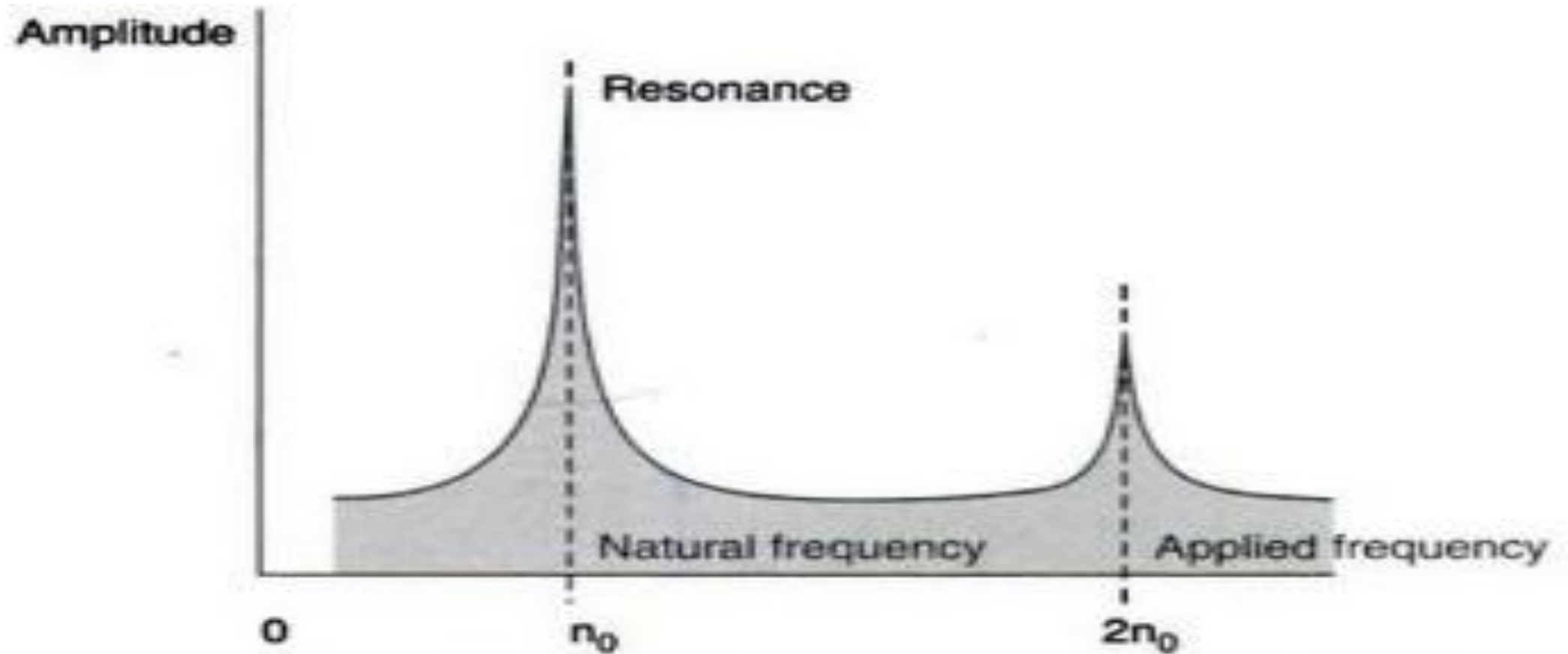
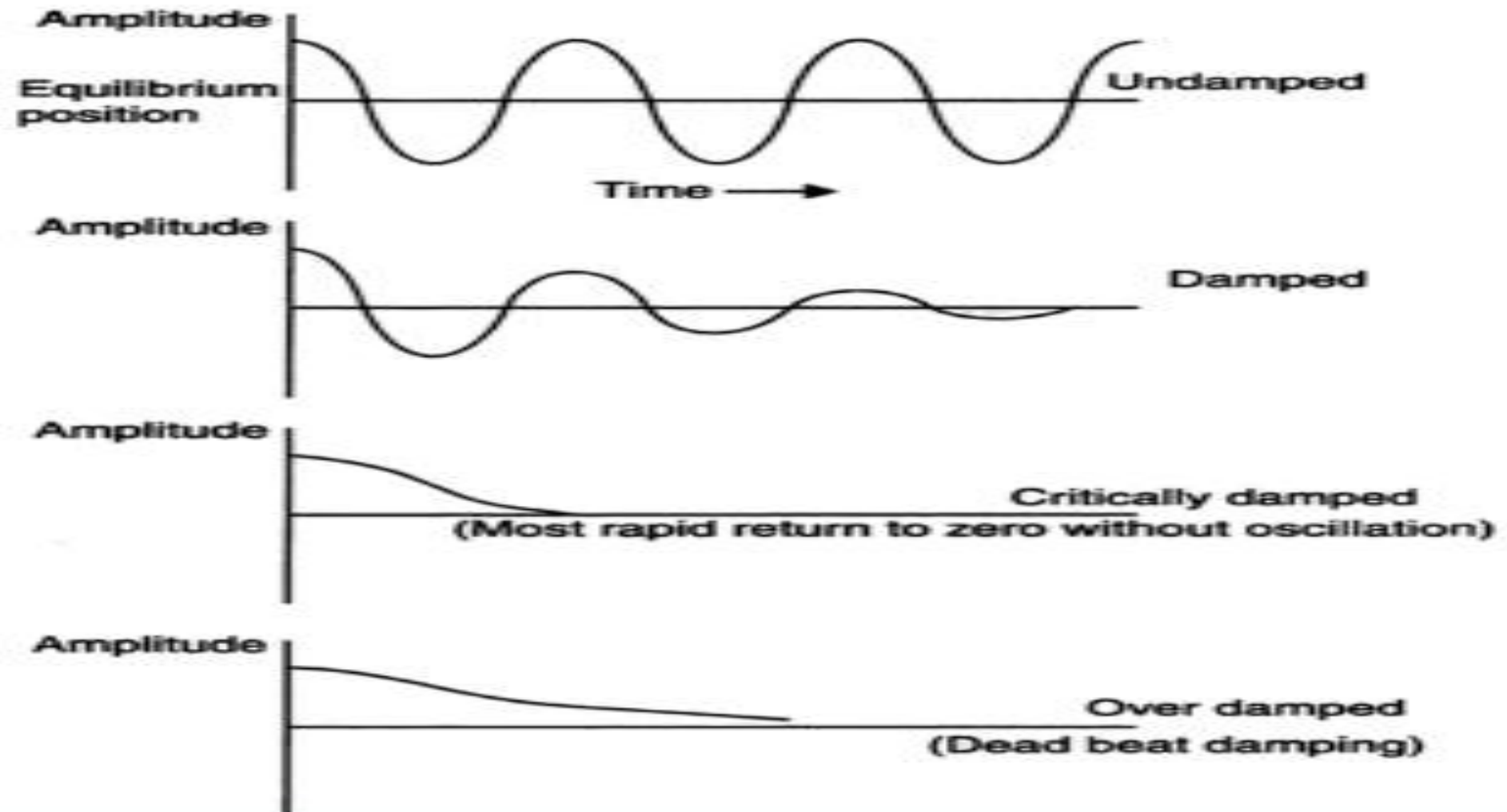


Fig. 1.23 *Effect of the frequency of the forced vibration on amplitude*

Echoes



Chapter 1

Microphones

KMN

A Microphone is a Transducer:

A transducer is a device, usually electrical or electronic, that converts one type of energy to another for the purpose of measurement or information transfer.

Most transducers are either sensors or actuators.

Common examples include microphones, loudspeakers,.

In a microphone, the diaphragm moves in response to changing air pressure (sound), creating an electrical current.

.

Micro

- 



images from RCA, Shure Brothers, Inc., Electrovoice, Audio Technica, Gibson and AKG

Carbon Button Microphone— Historical Design

- Thomas Alva Edison and Sir Emile Berliner filled patents for Carbon Button Microphone in 1877.
- Edison's design became the accepted design and patent.
- This is a popular microphone for telephones as well as early radio and speeches.

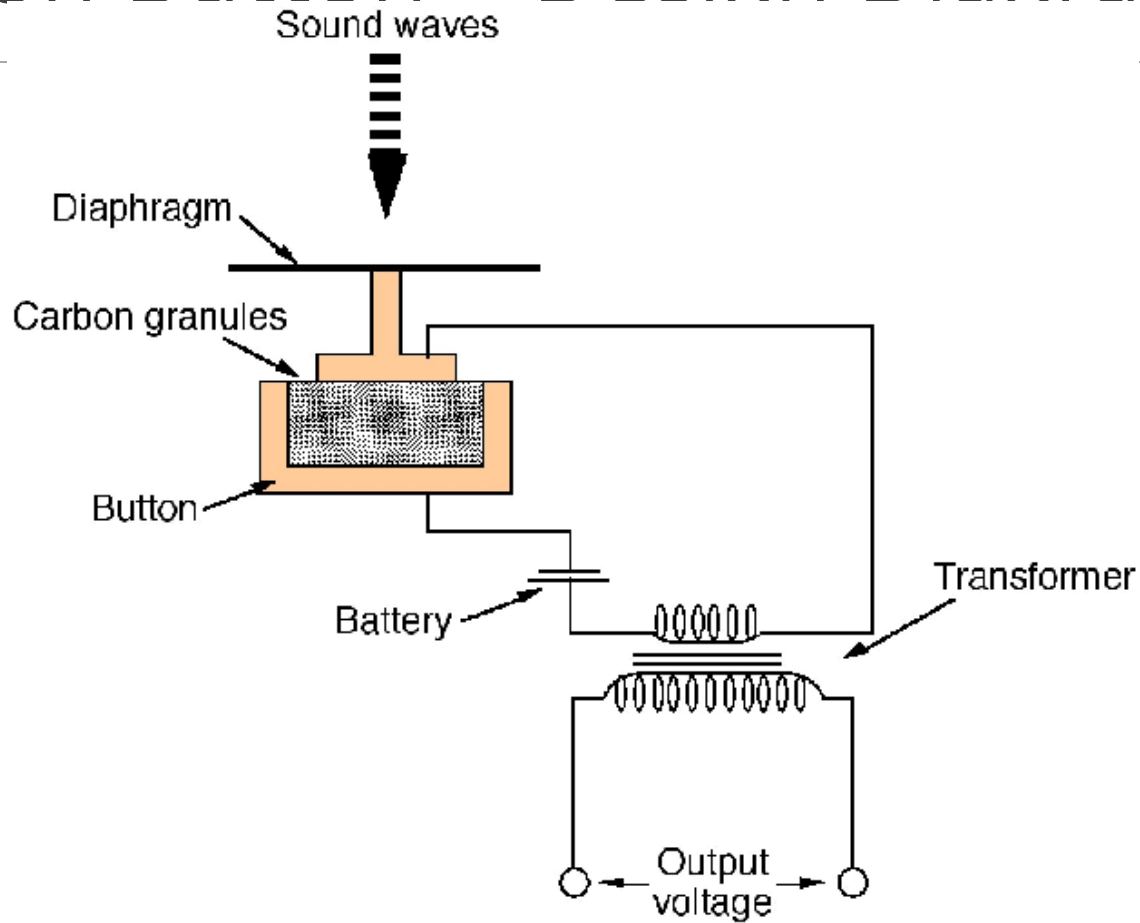


Carbon Button Microphone

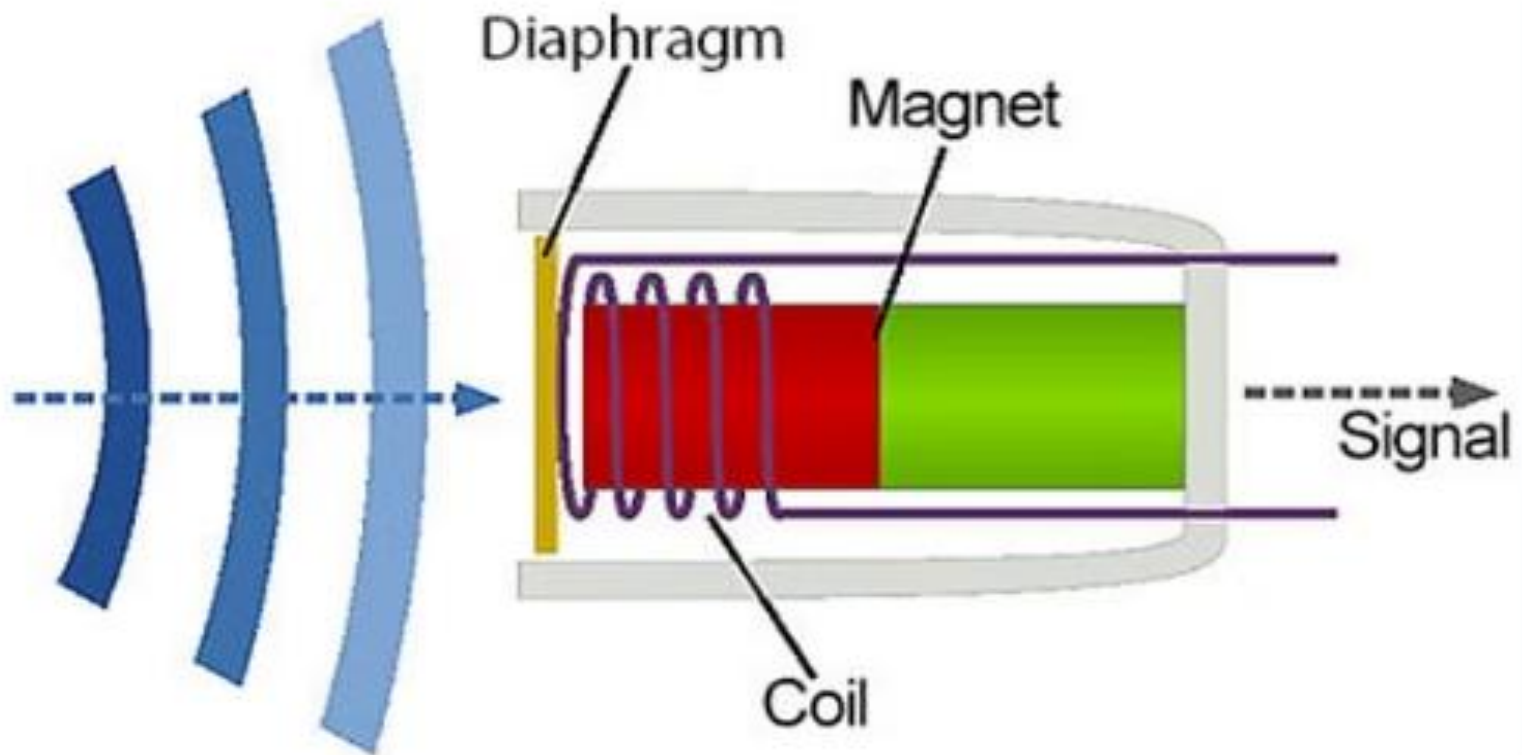
- Diaphragm: Thin Metal Plate

- The diaphragm is connected to a button full of carbon.
- A voltage is applied across the button.
- Sound pressure changes the resistance through the button by compressing/decompressing the carbon by pushing the plate.

Carbon Button—Design Diagram



Dynamic Microphone



Dynamic Microphone:

- a coil of wire attached to a diaphragm suspended in a magnetic field.
- Sound waves vibrate diaphragm, coil vibrates in magnetic field and generates a electrical signal similar to sound wave

Microphones: Dynamic

Tends to have rougher response

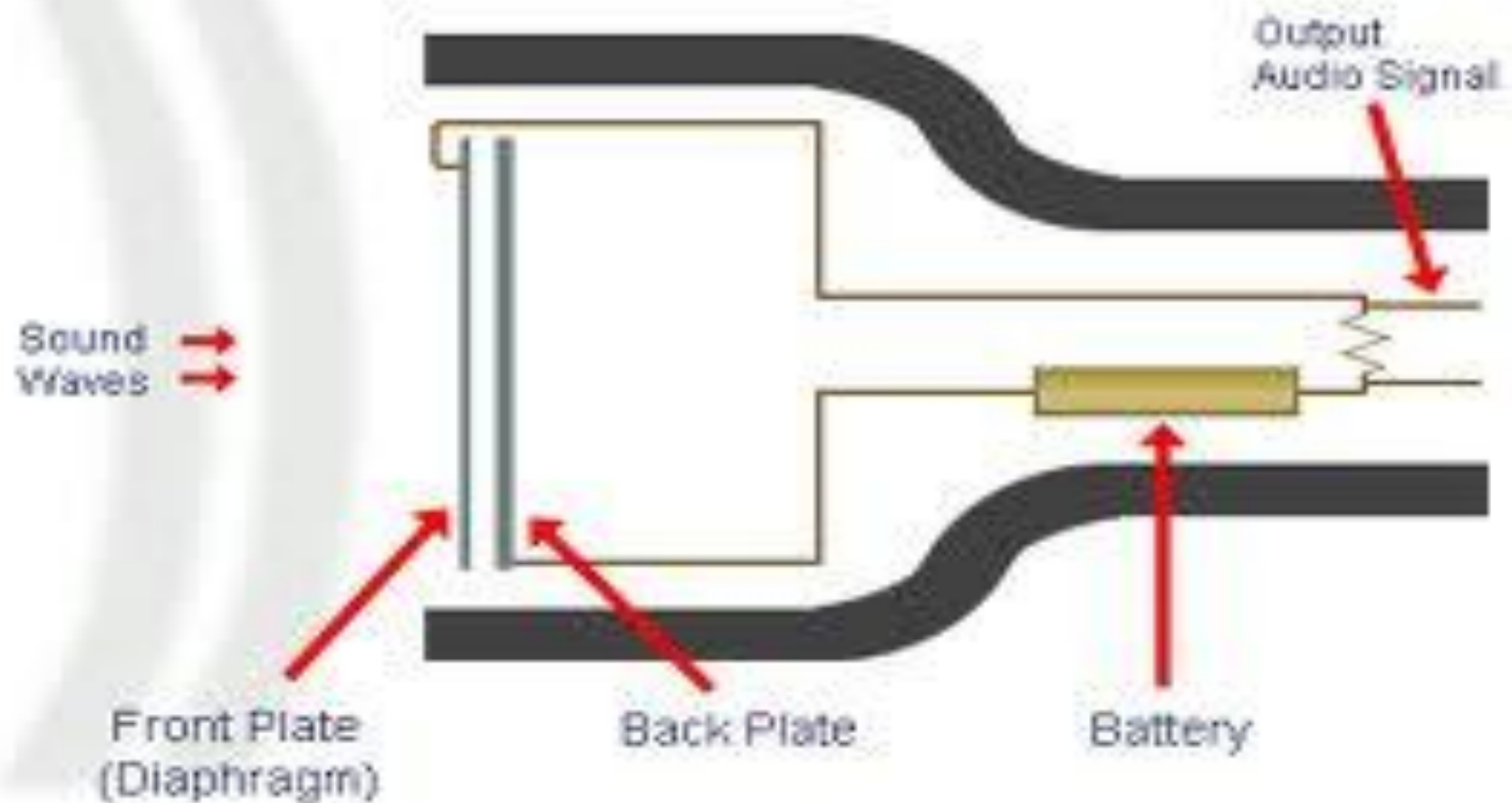
Rugged and reliable

Handles heat, cold, and high humidity

Preferred for high signal power: guitar amps, drums

flat response

Condensor Microphone



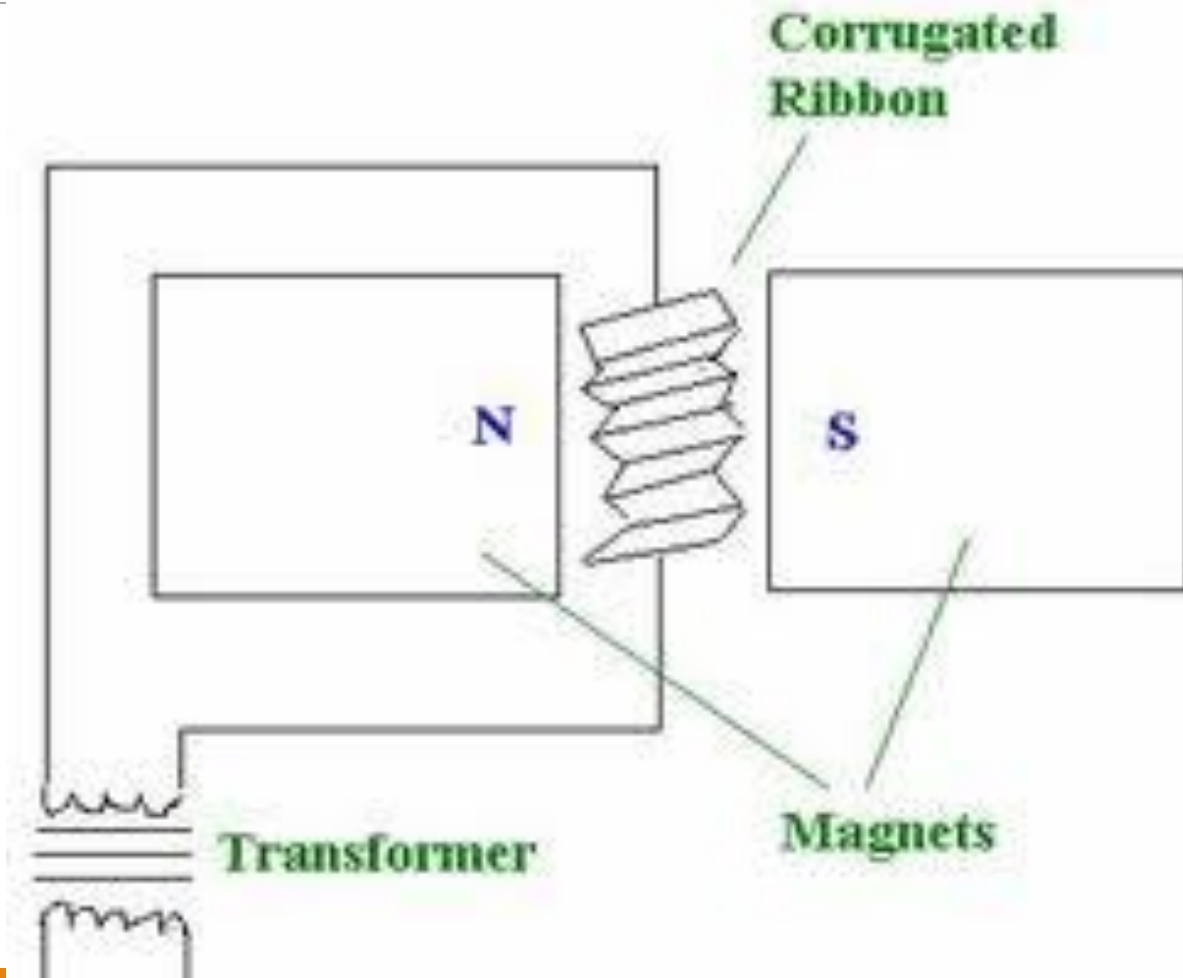
Condenser Microphone

- Sometimes called capacitor mic
- 2 parts: conductive diaphragm and metal backplate, spaced very close together
- Both charged with static electricity
- When sound waves strike diaphragm, it vibrates, varying the spacing between the plates
- This variation generates a similar signal to incoming sound wave
- Diaphragm mass is lower, responds faster to rapidly changing sound waves.

Condenser Microphone

- Condenser mics need external power supply (battery) 12 to 48 volts, Wide, smooth frequency response
- Detailed sound, Extended Highs
- Omni type has excellent low-frequency response
- Preferred for acoustic instruments, studio vocals
- Can be miniaturized

Ribbon Microphone



Ribbon Microphone

- Thin metal foil or ribbon is suspended in a magnetic field
- Sound waves vibrate the ribbon in the field and generate a electrical signal.
- warm, smooth tone quality, Expensive
- Delicate, damaged by temperature and humidity changes
- Complements digital recording

Microphones: Polar Patterns

- Polar pattern refers to the way a microphone responds from different directions
- Graph is based on its sensitivity (measured in decibels).

Microphones: Omni-directional

All around pickup

Not much isolation

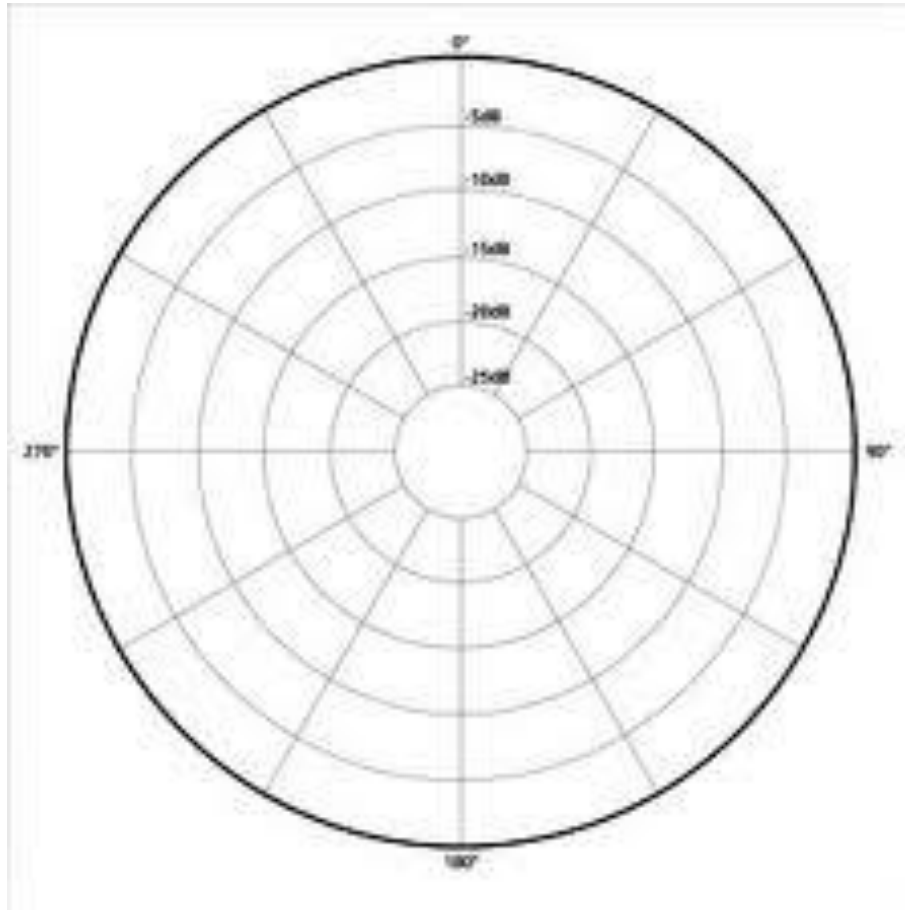
Low sensitivity to pops

Low handling noise

Extended low frequency response

Lower cost

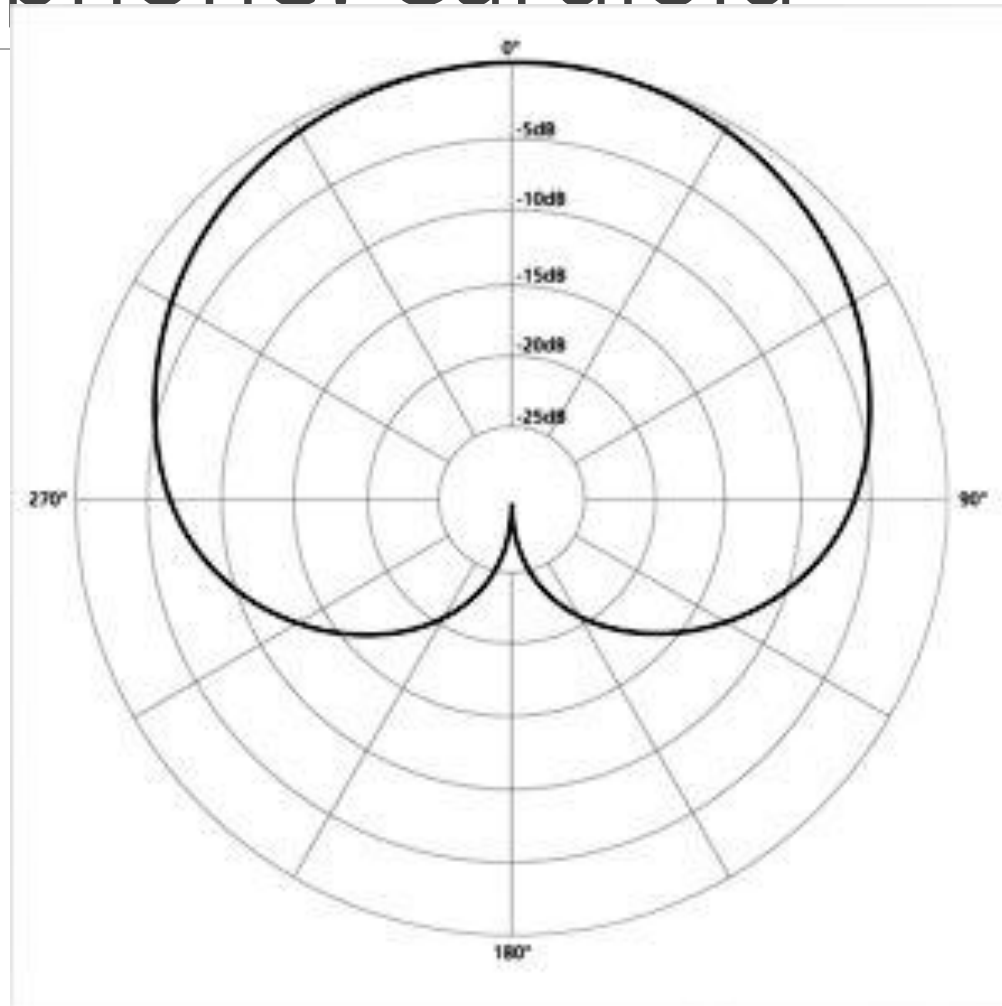
Microphone: Omni Pattern



Microphones: Unidirectional (cardioid, supercardioid, hypercardioid)

- Selective pickup
- Rejection of room acoustics, background noise, and leakage, Broad-angle pickup of sources in front of mic
- Maximum rejection of sound approaching the rear of the mic
- Most popular pattern

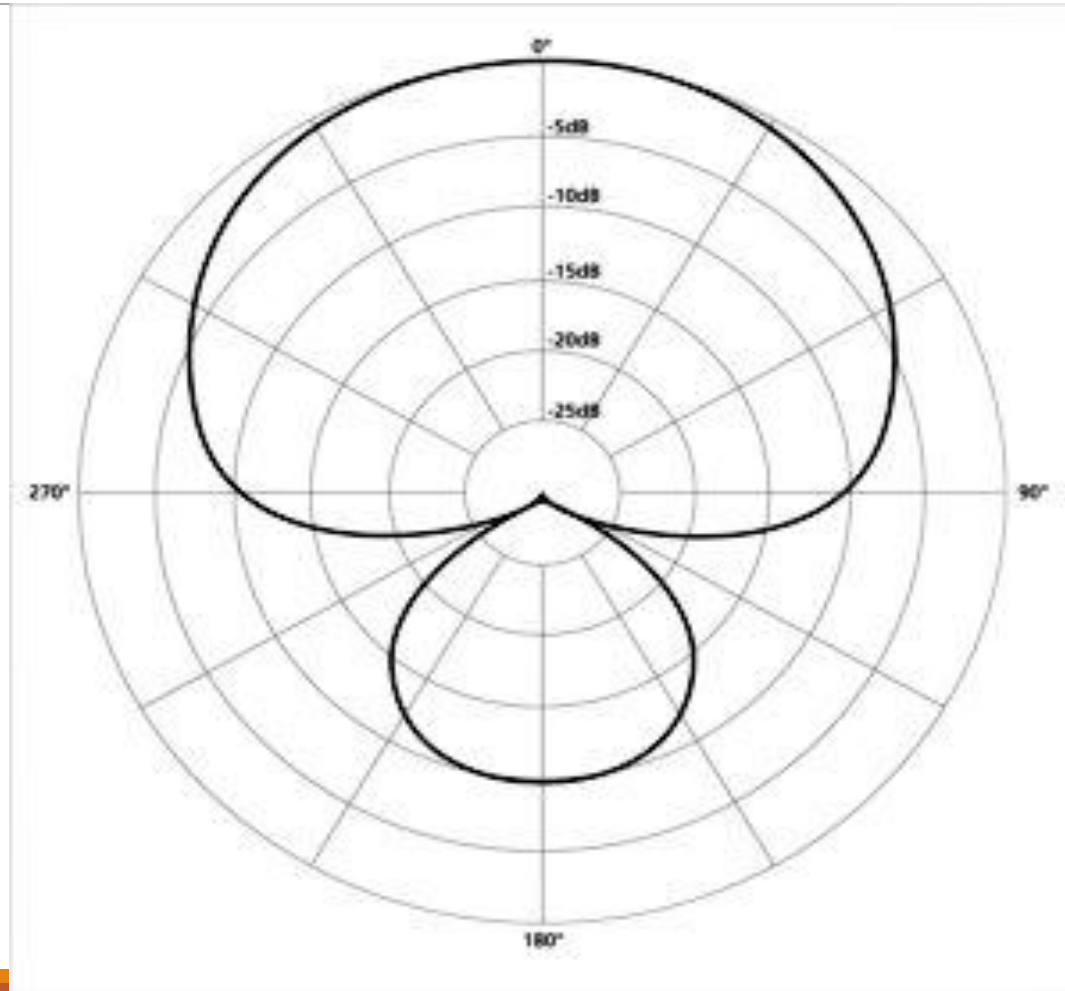
Microphone: Cardioid



Microphones: Supercardioid

- Maximum difference between front hemisphere and rear hemisphere (good for stage-floor miking)
- More isolation than cardioid
- Less reverb than cardioid

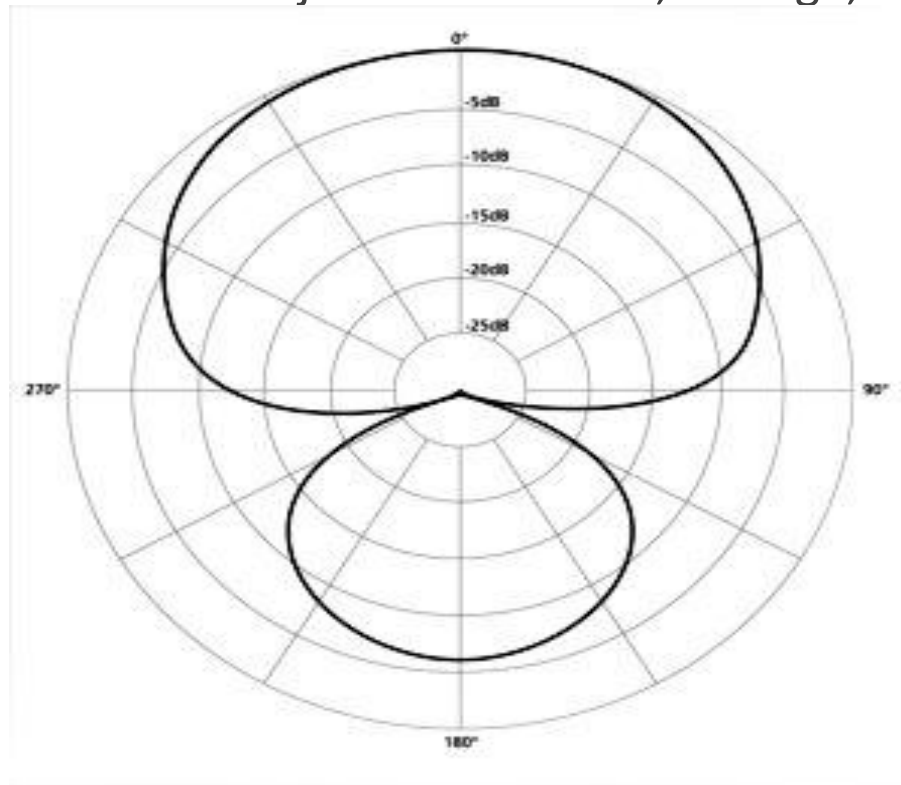
Microphone: Supercardioid



Microphone: Hyper-cardioid

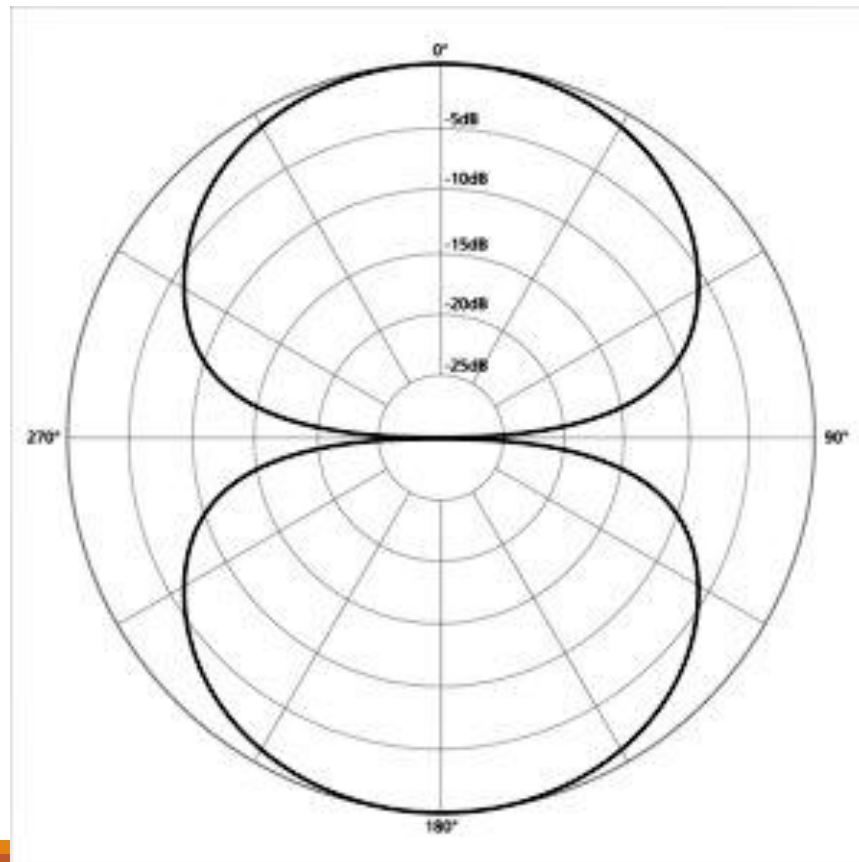
Maximum side rejection

Maximum isolation – maximum rejection of reverb, leakage, and background noise



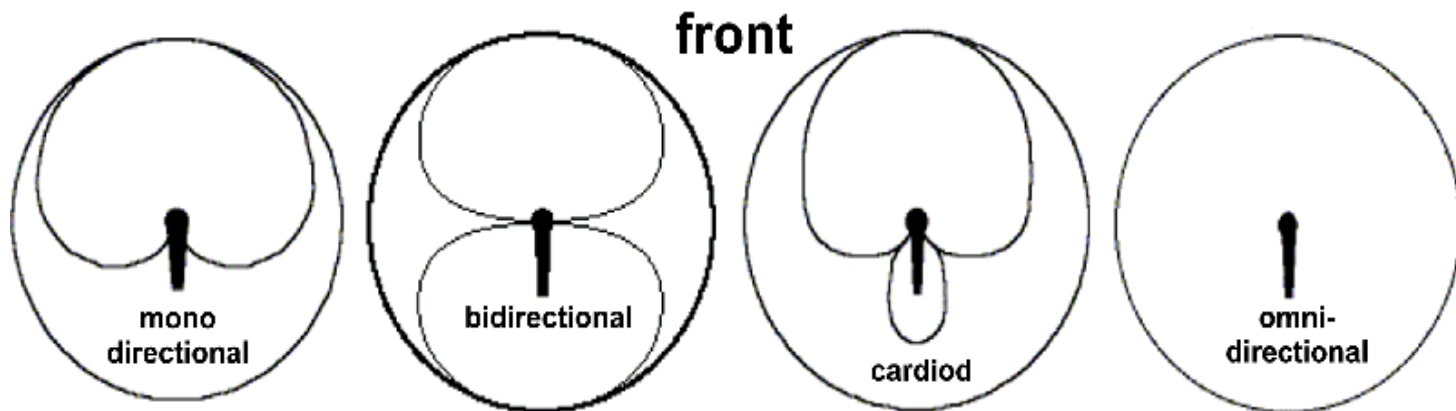
Microphone: Bi-directional

■ Front and rear pickup, with sides rejected



Microphone Pickup Patterns

- Pickup patterns help to reduce unwanted signal from getting pickups
- Popular patterns: Monodirectional, Bidirectional, Cardioid, and Omnidirectional



Microphones: Frequency Response

Range of frequencies a mic will reproduce at an equal level (within a tolerance like $\pm 3\text{db}$)

- Most instruments: 80Hz – 15kHz
- Bass instruments: 40Hz - 9kHz
- Brass and voice: 80 Hz – 12kHz
- Piano: 40 Hz – 12 kHz
- Orchestra or symphonic band: 40 Hz – 15kHz

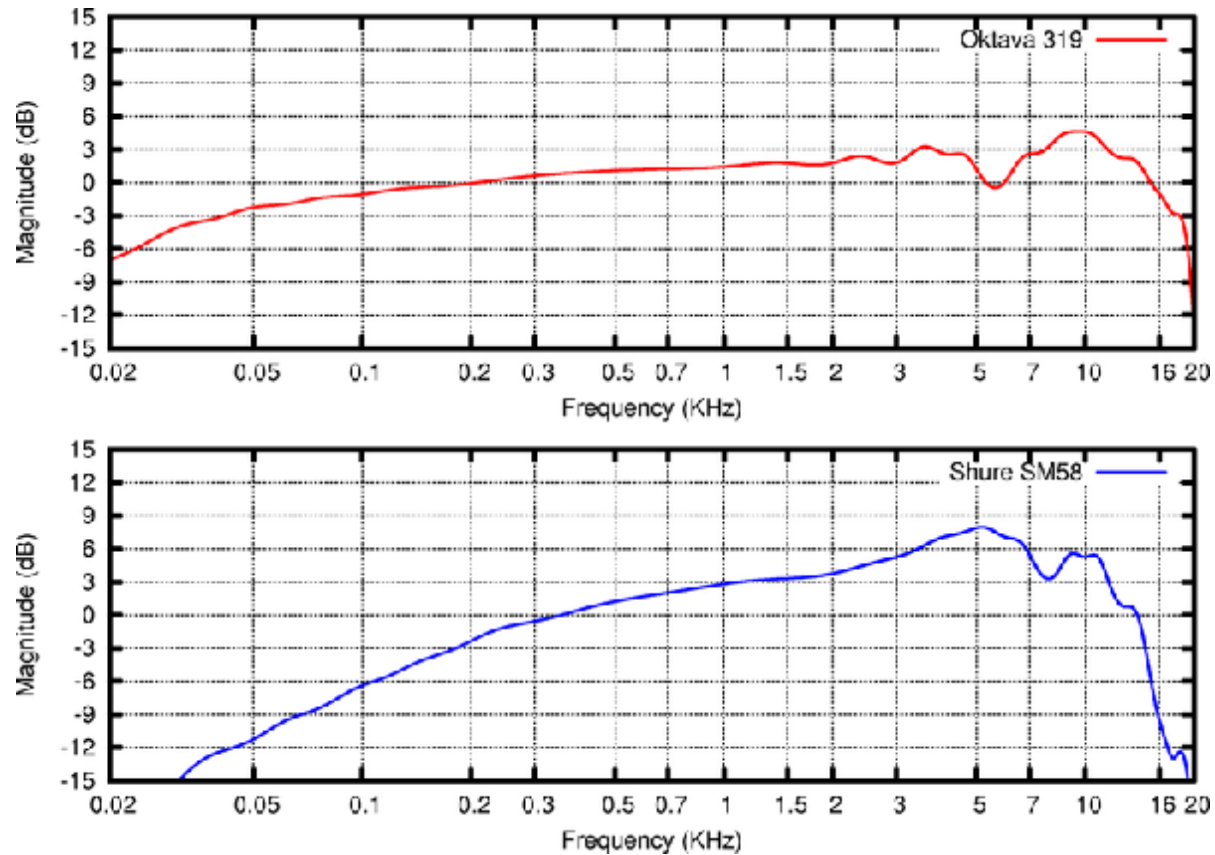
Microphones: Frequency Response

Roll-off switches: attenuates frequencies below a point below fundamental frequency of instrument you're recording (guitar does not play below 80 Hz)

Presence peak: rising high end around 5 – 10kHz makes sounds more crisp and articulate (crunchy)

Proximity effect: a bass boost tendency of a mic when placed too close.

Frequency Response



THANK YOU

Problems and solution

MTTF problems

- **Problem**
- A company designs a new type of LED street light.
From accelerated life testing:
- Out of **500 lights tested**,
- **25 lights** failed during a **2,000-hour** continuous operation test.
- **Find the Mean Time To Failure (MTTF)** for this LED light model.

$$\text{MTTF} = \frac{\text{Total operating time of all units}}{\text{Total number of failures}}$$

All 500 lights ran for 2,000 hours:

$$\text{Total operating time} = 500 \times 2000 = 1,000,000 \text{ hours}$$

$$\text{MTTF} = \frac{1,000,000 \text{ hours}}{25 \text{ failures}}$$

$$\text{MTTF} = 40,000 \text{ hours}$$

If the street lights run 12 hours/day,

$$\frac{40,000}{12} \approx 3,333 \text{ days } (\approx 9.1 \text{ years})$$

MTBF problems

A repairable conveyor system is being validated for a factory line.

During a trial, the system accumulated **3,600 hours of actual operating time** (uptime).

Across this period, the system experienced **10 failures**.

Each failure took, on average, **4 hours** to repair (MTTR)

Find:

- MTBF
- Failure rate λ (per hour)
- Steady-state availability estimate

Solution

1) MTBF

For repairable systems,

$$\text{MTBF} = \frac{\text{Total operating time (uptime)}}{\text{Number of failures}} = \frac{3600 \text{ h}}{10} = \boxed{360 \text{ hours}}$$

2) Failure rate

Assuming a constant failure rate (exponential model),

$$\lambda = \frac{1}{\text{MTBF}} = \frac{1}{360} \text{ h}^{-1} \approx \boxed{0.00278 \text{ h}^{-1}}$$

3) Availability (steady-state)

For an exponential repair/arrival model,

$$A \approx \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} = \frac{360}{360 + 4} = \frac{360}{364} = \frac{90}{91} \approx \boxed{98.90\%}$$

HEADPHONE



Outline

- ▣ History
- ▣ Transducer Technology
- ▣ Electrical characteristics
- ▣ Types & construction

History



Brandes radio headphones, circa 1920



Old telephone earpiece

Headphones originated from the earpiece, and were the only way to listen to electrical audio signals before amplifiers were developed. The first truly successful set was developed by Nathaniel Baldwin, who made them by hand in his kitchen and sold them to the United States Navy.

Transducer Technology

- ▣ Moving-coil
- ▣ Electrostatic
- ▣ Balanced armature

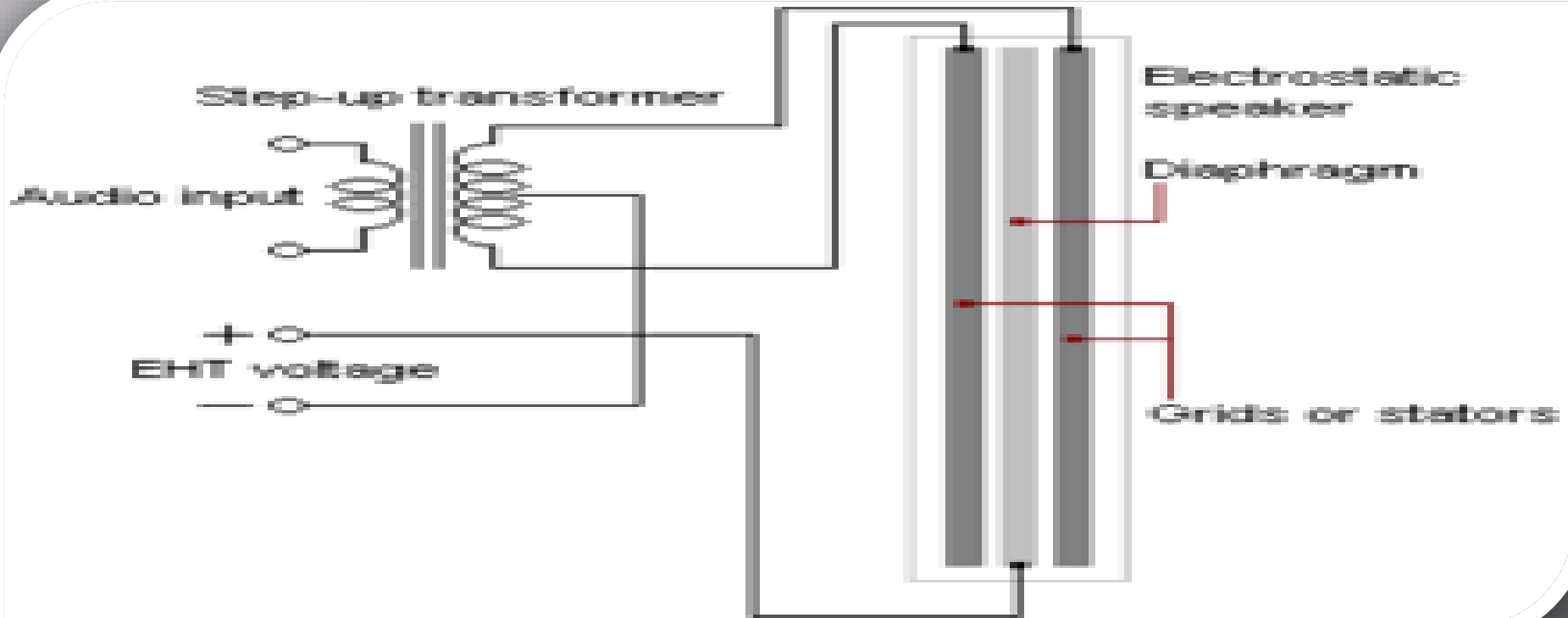


Moving-coil



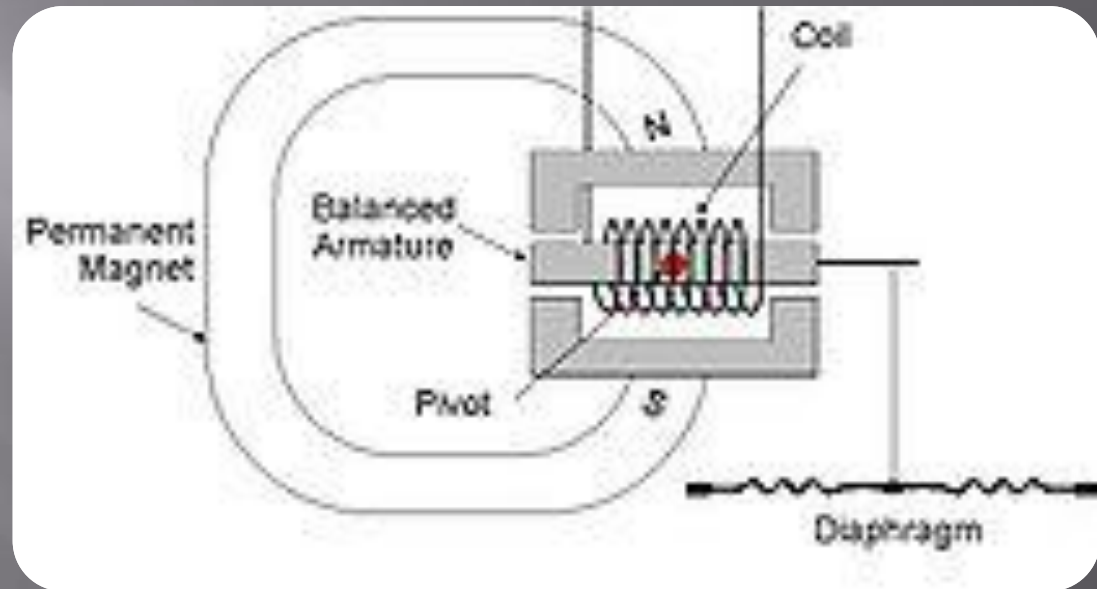
The moving coil driver, more commonly referred to as a "dynamic" driver is the most common type used in headphones. The operating principle consists of a stationary magnetic element affixed to the frame of the headphone which sets up a static magnetic field

Electrostatic



Electrostatic drivers consist of a thin, electrically charged diaphragm, typically a coated PET film membrane, suspended between two perforated metal plates (electrodes).

Balanced armature



A balanced armature is a sound transducer design primarily intended to increase the electrical efficiency of the element by eliminating the stress on the diaphragm characteristic of many other magnetic transducer systems.

Impedance

Headphones are available with low or high impedance . Low-impedance headphones are in the range 16 to 32 ohms and high-impedance headphones are about 100-600 ohms. As the impedance of a pair of headphones increases, more voltage but less current is required to drive it, and the loudness of the headphones for a given voltage decreases



In recent years, impedance of newer headphones has generally decreased to accommodate lower voltages available on battery powered CMOS-based portable electronics. This results in headphones that can be more efficiently driven by battery powered electronics. Consequently, newer amplifiers are based on designs with relatively low output impedance.



Sensitivity

Sensitivity is a measure of how effectively an earpiece converts an incoming electrical signal into an audible sound. It thus indicates how loud the headphones will be for a given electrical drive level. It can be measured in decibels of sound pressure level per milliwatt, or dB SPL/mW, which may be abbreviated to dB/mW. The sensitivity of headphones is usually between about 80 and 125 dB/mW.



Types

- ▣ Circumaural
- ▣ Supra-aural
- ▣ Clip-on earphones
- ▣ Canalphones



Circumaural



Circumaural headphones (sometimes called full size headphones) have circular or ellipsoid earpads that encompass the ears. Because these headphones completely surround the ear, circumaural headphones can be designed to fully seal against the head to attenuate external noise.

Circumaural



Supra-aural



Supra-aural headphones have pads that press against the ears, rather than around them. They were commonly bundled with personal stereos during the 1980s. This type of headphone generally tends to be smaller and lighter

Supra-aural



Clip-on earphones



Earbuds and earphones refer to very small headphones that are fitted directly in the outer ear, facing but not inserted in the ear canal; they have no band or other arrangement to fit over the head.

Clip-on earphones



Canalphones



headphones like earbuds, are small and without headband, but are inserted in the ear canal itself. They are sometimes known as canalphones. Price and quality range from relatively inexpensive to very high.

Canalphones



Genral construction of Ear phones

WORKING OF EARPHONES

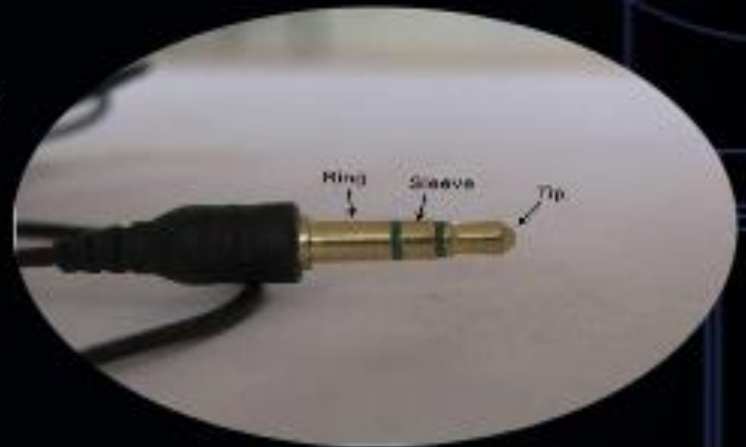
- Earphones are transducers : Converting Electrical Energy into Sound Energy.
- Earphones are available in many shape, size and colors with pricing depending on quality.
- The image shows a glimpse of commonly used earphones.



JACK

Components : Jack

- Jack : Part of the earphone which takes electrical signals from the audio output port of the devices like mobile phones, laptop, computers, music player etc.
- There are three contact areas known as the TIP, RING and the Sleeve as shown in the image and therefore this connector is also called as **TRS** connector.



Housing

Housing

- The housing is made up of rubber and is designed to give a good grip while connecting and disconnecting the connectors.
- The housing encloses the wires soldered to the contacts as shown in the above image.



Connecting Wires

- There are three wires each soldered to one of the metal contact of the jack. One wire is for ground, and the other two corresponding to the left and the right earphone.
- The below image shows the junction from where wires get distributed to the left and right ear piece.
- Each ear piece has two wires out of which one is common/ground. The other carries the electrical signal for the respective ear piece.



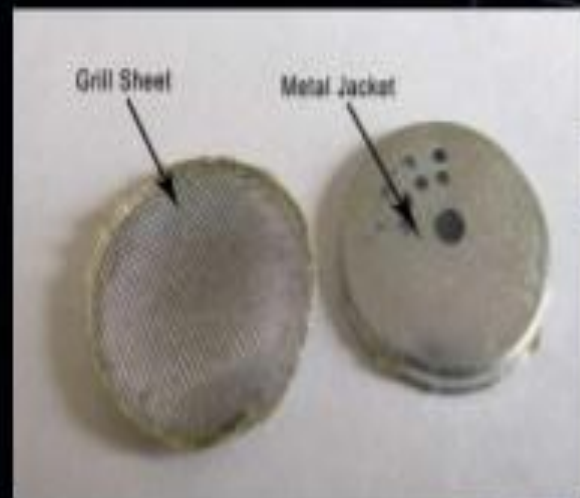
Filter

- The image shows the ear tip and the filter. Ear tip is that part of the earphone which makes contacts with the ear.
- The circular shaped tip is made up of silicone. It holds the earpiece in the ears and therefore needs to be soft and comfortable.
- The job of the filter is to protect the earwax and other dust particles from entering into the earpiece which in turn spoils the sound quality.



Structure of Filter

- The filter consists of two parts: a) metal jacket and b) metallic grill sheet.
- The metal jacket is the outermost part of the earphone from where the sound enters the ear.
- The design of the grill sheet is very important as it needs to provide a clear passage for the sound.
- Sound quality is highly dependent on the grill sheet.
- Both the metal jacket and the grill sheet serves the same purpose of protecting internal audio components from outside particles like ear wax and dust.



Driver Unit

- Removing the back casing we can see the wires soldered to the driver unit
- Driver unit of the earphone are connected to the voice coil.



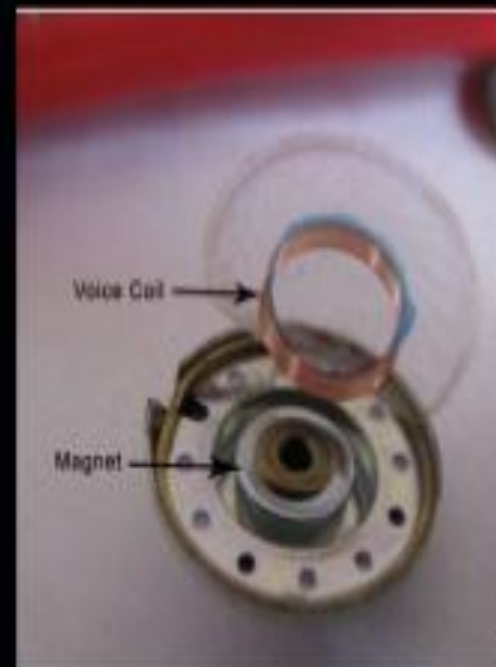
Diaphragm

- The diaphragm is made up of a thin film of plastic.
- It is the motion of the diaphragm which produces vibration of the surrounding air particles thereby producing sound.
- The vibration is dependent on the electrical signal from the device.



Voice Coil

- Diaphragm is connected to the voice coil and is placed in the slot surrounding the permanent magnet.
- The voice coil receives the analog audio signals and behaves as electromagnet.
- The resultant force produced by the two magnetic fields causes the diaphragm to vibrate, thus producing sound.

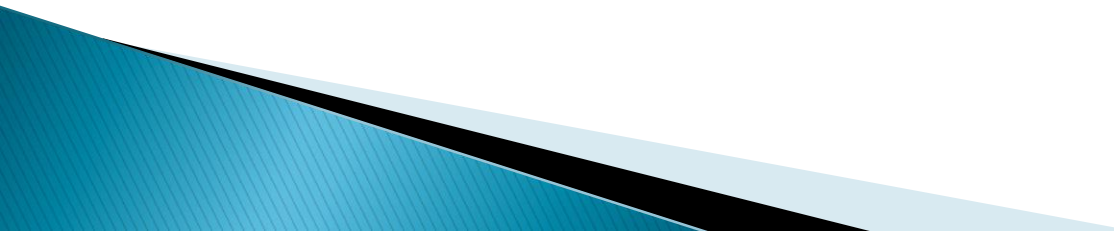


▣ THANK YOU

TYPES OF LOUDSPEAKER

BY
KMN

WHAT IS A LOUDSPEAKER

- ▶ A LOUDSPEAKER IS TRANSDUCER WHICH CONVERTS ELECTRICAL SIGNALS OF AUDIO FREQUENCY INTO SOUND WAVES OF THE SAME FREQUENCY.
 - ▶ IT IS ALSO CALLED AS OUTPUT TRANSDUCER OR REVERSE TRANSDUCER.
 - ▶ A loudspeaker must be able to reproduce a wide range of audio frequencies (i.e., 20 Hz to 20 kHz).
- 

POLE PIECES

These focus the magnetic field so that it is strongest around the voice coil.

MAGNET

This is the motor of a loudspeaker – it provides the energy that causes the voice coil to move.

BASKET

This is the chassis of the drive unit to which all elements are attached and which itself bolts into the cabinet.

SUSPENSION SPIDER & VOICE COIL

The spider holds the voice coil central within the magnet and acts as a spring to bring it back after each pulse. The speaker cone is attached to the voice coil which sits in a magnetic field and moves when a signal passes through it. Variations in the signal make the coil vibrate the drive unit in a pistonic motion which produces sound by resonating airwaves in the room much like a drum.

DRIVE CONE & SURROUND

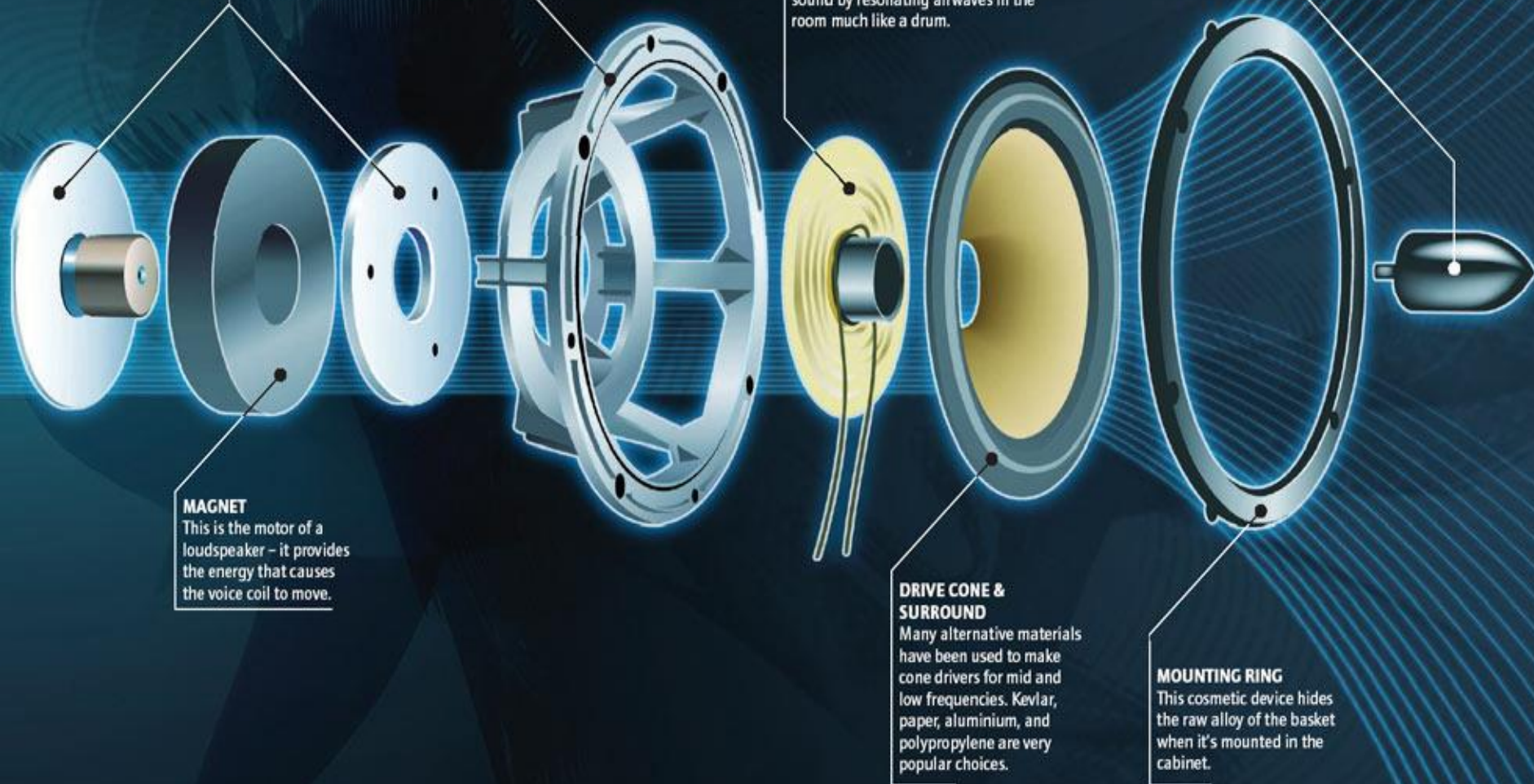
Many alternative materials have been used to make cone drivers for mid and low frequencies. Kevlar, paper, aluminium, and polypropylene are very popular choices.

PHASE PLUG

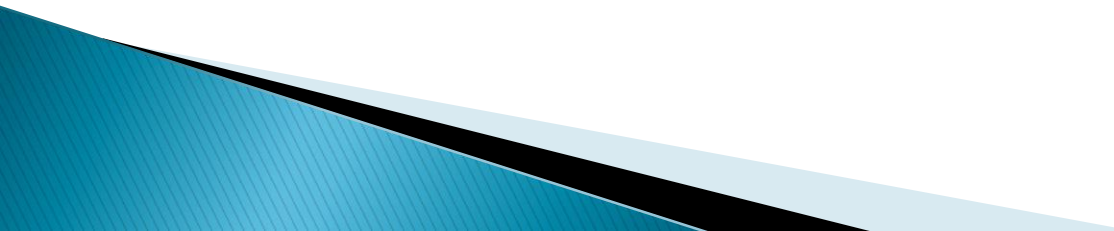
Not found on all drive units but designed to avoid phase changes. More commonly drive units feature a dust cap to stop detritus entering the system.

MOUNTING RING

This cosmetic device hides the raw alloy of the basket when it's mounted in the cabinet.

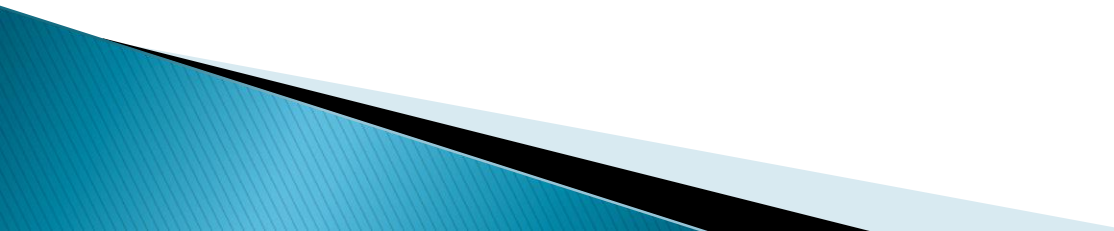


Types Of Loudspeaker

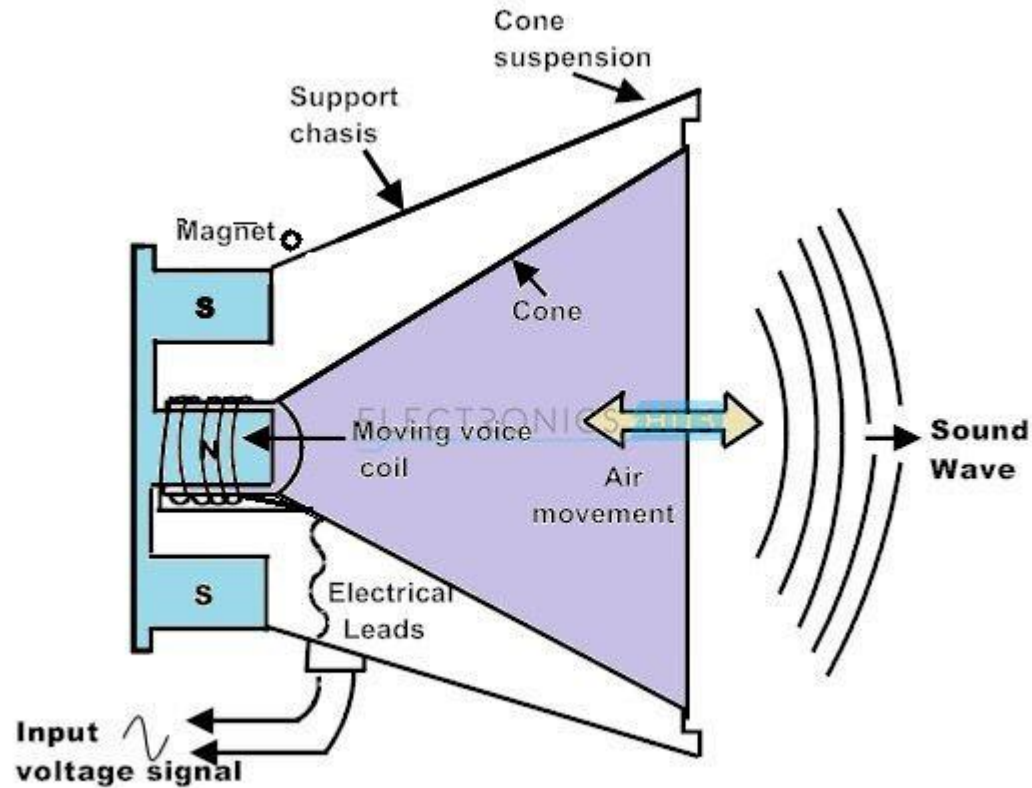
- ▶ Basically there are three types of loudspeaker, they are listed as follows.
 - ▶ MOVING COIL LOUDSPEAKER OR CONE TYPE LOUDSPEAKER
 - ▶ ELECTRODYNAMIC LOUDSPEAKER
 - ▶ HORN TYPE LOUDSPEAKER
- 

MOVING COIL LOUDSPEAKER

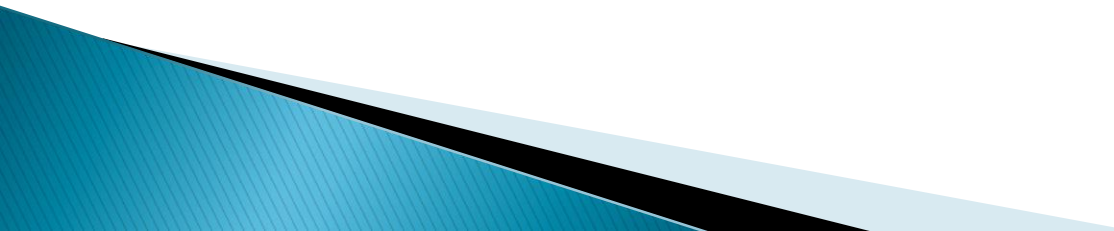
Principle–

- ▶ The moving coil loudspeaker works on the interaction between a magnetic field and current flowing through the coil.
 - ▶ A coil called voice coil, is placed in a uniform magnetic field. When the audio current passes through the voice coil, there is an interaction between the magnetic field and the current, resulting in a force working on the movable coil.
 - ▶ This force is proportional to the audio current, and hence causes vibratory motion in the coil, which makes a conical paper diaphragm to vibrate and produce pressure variations in air, resulting in sound waves.
- 

Diagram



Features

- ▶ Here are some characteristics given below
 - ▶ $\text{SNR} = 30 \text{ DB}$
 - ▶ $\text{FREQUENCY RESPONSE} = \text{MEDIUM FREQUENCY}$
 - ▶ $\text{DISTORTION} = 10\%$
 - ▶ $\text{DIRECTIVITY} = \text{OMNIDIRECTION}$
- 

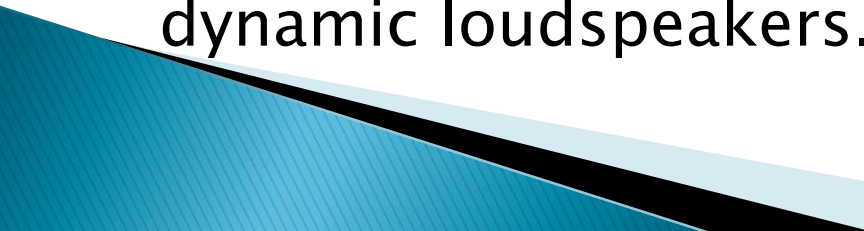
Construction

- ▶ The moving coil loudspeaker consist of a voice coil,wound on a cardboard
 - ▶ Audio current is fed to the loudspeaker through two terminals. The coil is placed in a magnetic field
 - ▶ The magnet is a permanent magnet
 - ▶ The coil is attached to the conical diaphragm,.
 - ▶ The cone is corrugated having circular corrugation to increase surface area for better efficiency.
 - ▶ A flexible strip of rubber round its periphery is used to support it.
 - ▶ The spider springs are used to support the complete diaphragm and also provides the required stiffness to restrain the motion.
 - ▶ The spider also keep the coil centered so that cone moves forward and backward only.
- 

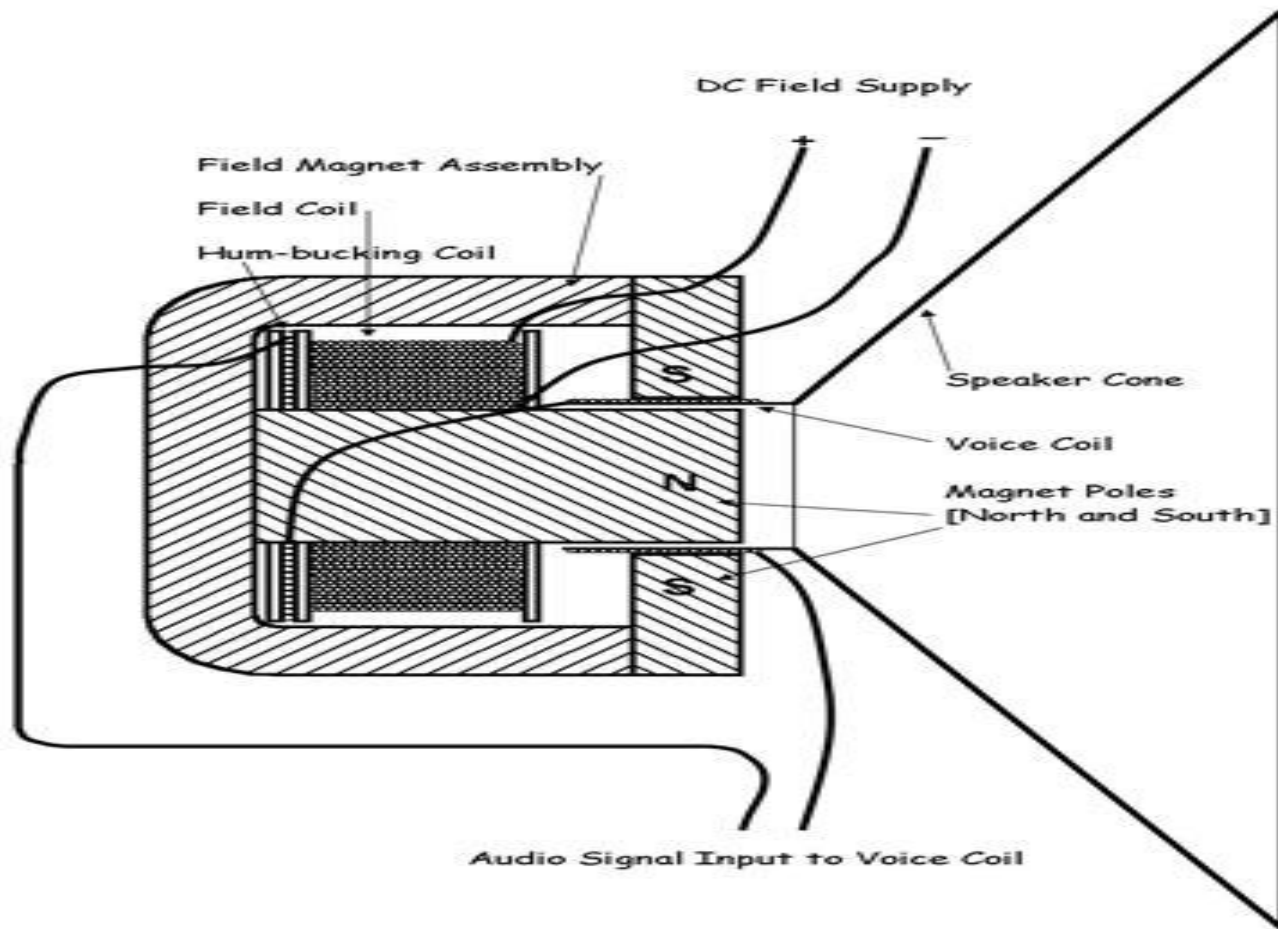
Working

- ▶ At the front of a loudspeaker, there is a fabric, plastic, paper, or lightweight metal **cone** called a **diaphragm**.
- ▶ The outer part of the cone is fastened to the outer part of the loudspeaker's circular metal rim. The inner part is fixed to an **iron** coil sometimes called the **voice coil**, that sits just in front of a permanent **magnet** sometimes called the field magnet.
- ▶ When you hook up the loudspeaker to a stereo, electrical signals feed through the speaker cables into the coil.
- ▶ This turns the coil into a temporary magnet or **electromagnet**.
- ▶ As the electricity flows back and forth in the cables, the electromagnet either attracts or repels the permanent magnet. This moves the coil back and forward, pulling and pushing the loudspeaker cone.

ELECTRODYNAMIC LOUDSPEAKER

- ▶ An electrodynamic loudspeaker or field coil loudspeaker is a dynamic loudspeaker in which the field is produced by an electromagnet rather than by a permanent magnet.
 - ▶ An electrodynamic loudspeaker therefore has two coils:
 - ▶ The voice coil common to all dynamic loudspeakers, positioned in the air gap between the pole pieces, the motion of which moves the loudspeaker cone.
 - ▶ A fixed field coil which together with its magnetic core replaces the permanent magnet of other dynamic loudspeakers.
- 

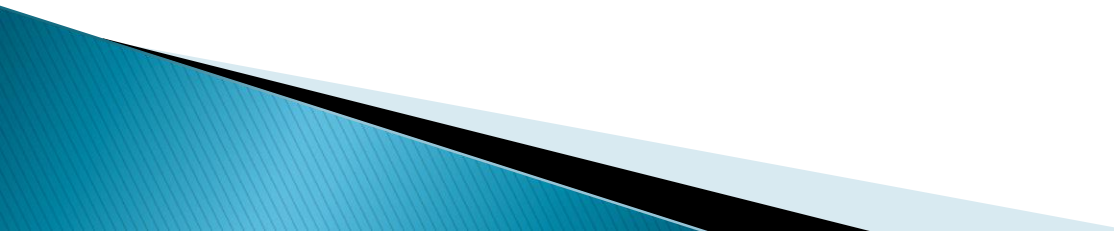
Diagram



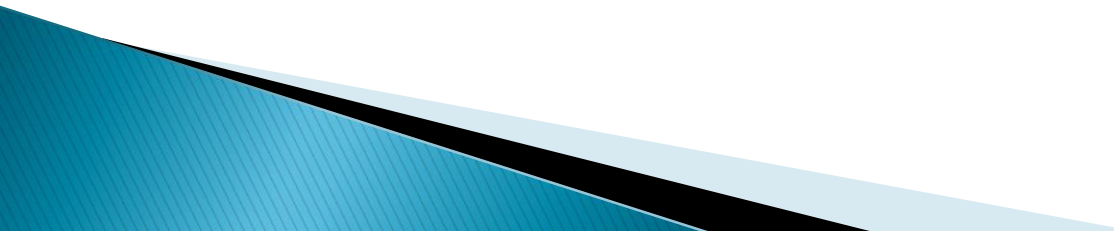
Features

- ▶ To provide strong magnetic field for high wattage speakers an electromagnet is used instead of permanent magnet.
- ▶ Some characteristics are given below
 - ▶ EFFICIENCY 30–50%
 - ▶ FREQUENCY RESPONSE – 30HZ–10KHZ
 - ▶ DISTORTION – LOW
 - ▶ SNR–30–40db

Construction

- ▶ The strong and steady magnetic field is produced by a large field coil wrapped around a core.
 - ▶ The shape of the magnet is pot type with the south pole in the centre and the north pole in the periphery.
 - ▶ The special shape of the core allows magnetic flux to remain concentrated in the angular gap between pole pieces.
- 

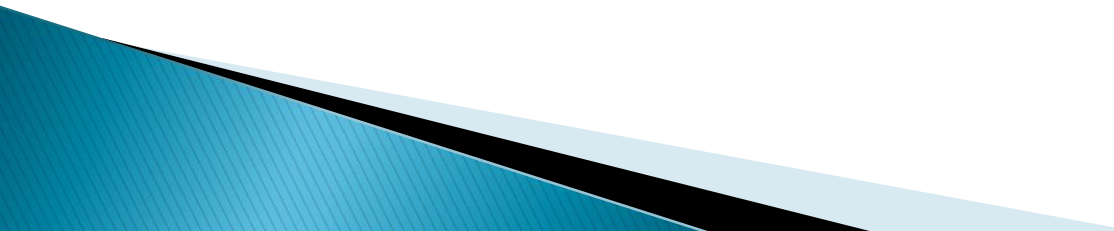
Working

- ▶ The voice coil is wound on fibre or aluminium it is placed in the angular gap. The audio signal from the amplifier's output transformer is applied to the voice coil.
 - ▶ This signal causes a varying magnetic field. The resultant interaction between the two magnetic field produces mechanical vibrations in the coil assembly, which corresponds to the audio signals.
- 

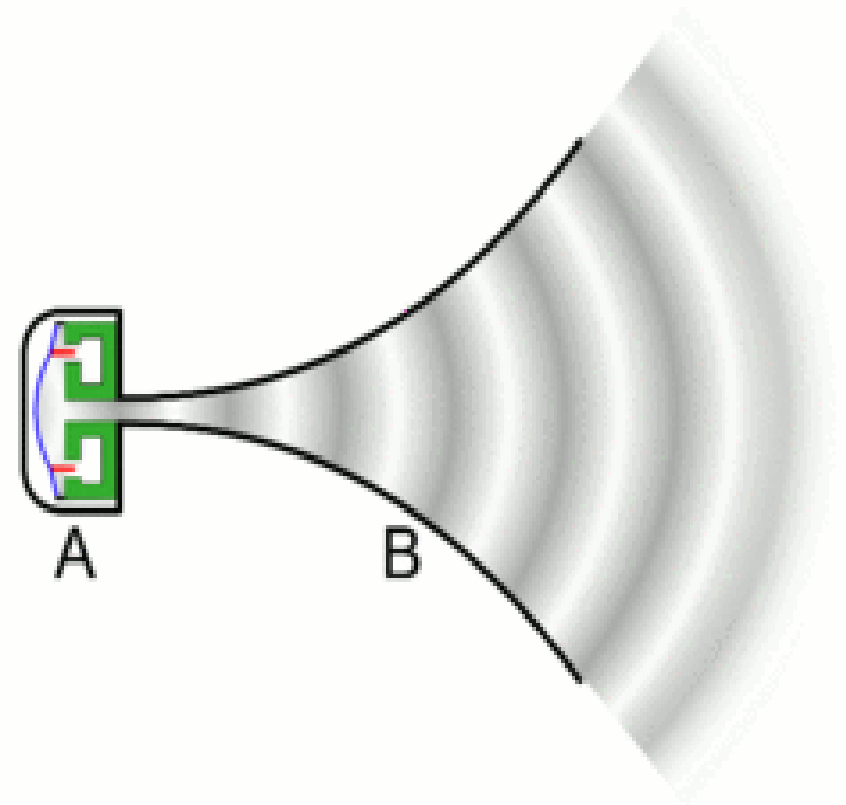
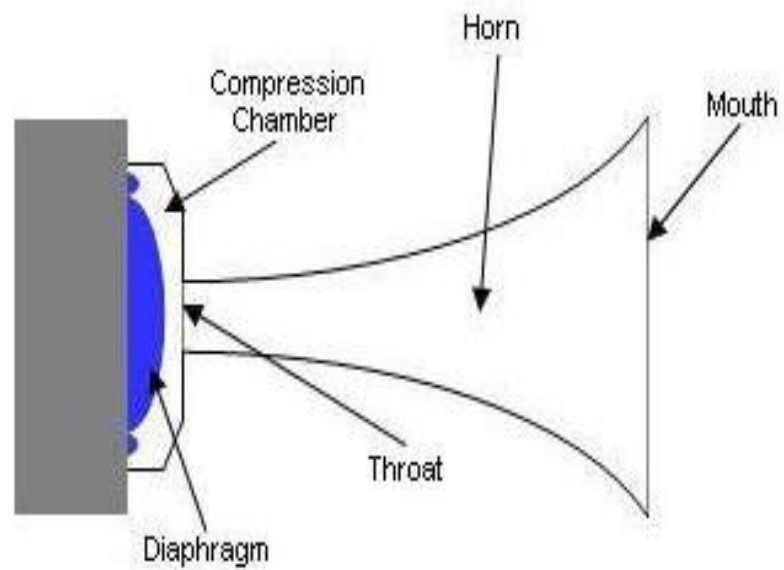
HORN-TYPE LOUDSPEAKER

- ▶ A horn loudspeaker is a loudspeaker or loudspeaker element which uses an acoustic horn to increase the overall efficiency of the driving element(s).
- ▶ It consists of a compression driver which produces sound waves with a small metal diaphragm vibrated by an electromagnet, attached to a horn to produce sound waves to the open air.

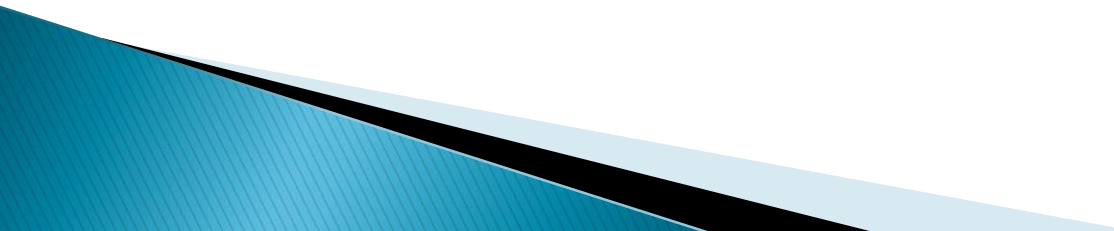
Some characteristics is given below:

- ▶ EFFICIENCY – 30 to 50 %
 - ▶ SNR – 40 db
 - ▶ FREQUENCY RESPONSE – 30–10000Hz
 - ▶ DISTORTION – 5%
- 

Diagram



Construction

- ▶ The horn is tapered enclosure whose diameter increases from the throat to the mouth
 - ▶ There is an air chamber trapped between throat and diaphragm.
 - ▶ The horn acts as a high pass filter.
- 

Working

- ▶ Horn type loudspeaker use moving coil place in magnetic field but instead of radiating acoustic and sound power direct to the open space of listener area.
- ▶ The power is first deliver to air chamber and than tapped to the horn and from their to the listener area.

Loudspeaker-system manufacturers

Manufacturer	Headquarters country
<u>Acapella Audio Arts</u>	Germany
<u>ADAM Audio</u>	Germany
<u>ADRAudio</u>	Slovenia
<u>Advance Acoustic</u>	France
<u>Advent</u>	United States
<u>Alesis</u>	United States
<u>Altec Lansing</u>	China and United States
<u>Amphion Loudspeakers</u>	Finland
<u>Apertura</u>	France
<u>Audison</u>	Italy
<u>Audiovox</u>	United States
<u>AVI Sound International</u>	Canada
<u>Bang & Olufsen</u>	Denmark

THANK YOU

Compact Disc (CD)

COMPACT
disc



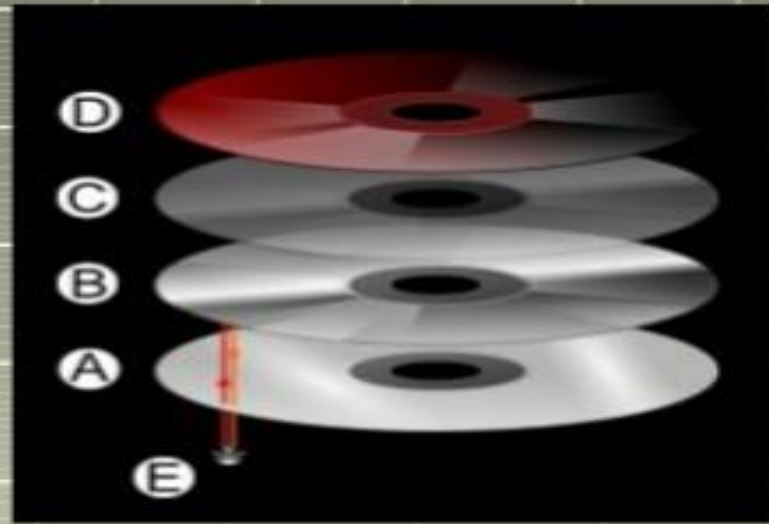
What is Compact Disc (CD)?

- *It is an optical disc used to store digital data.*
- *‘SONY’ first publicly demonstrated an Optical Digital Audio Disc in September 1976.*
- *It was originally developed to store & playback sound recordings exclusively, but later expanded to encompass-*

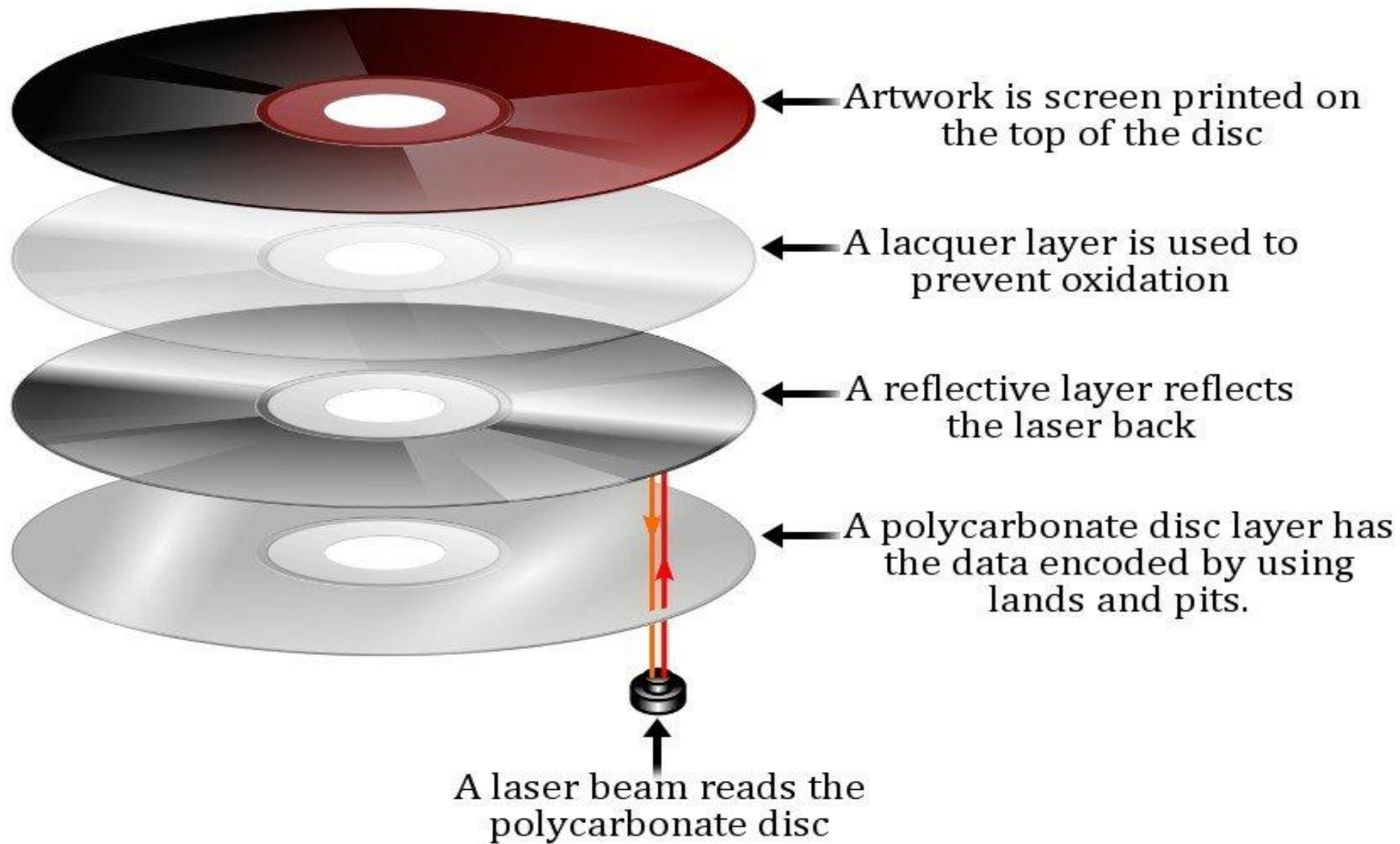
CD later expanded to encompass:-

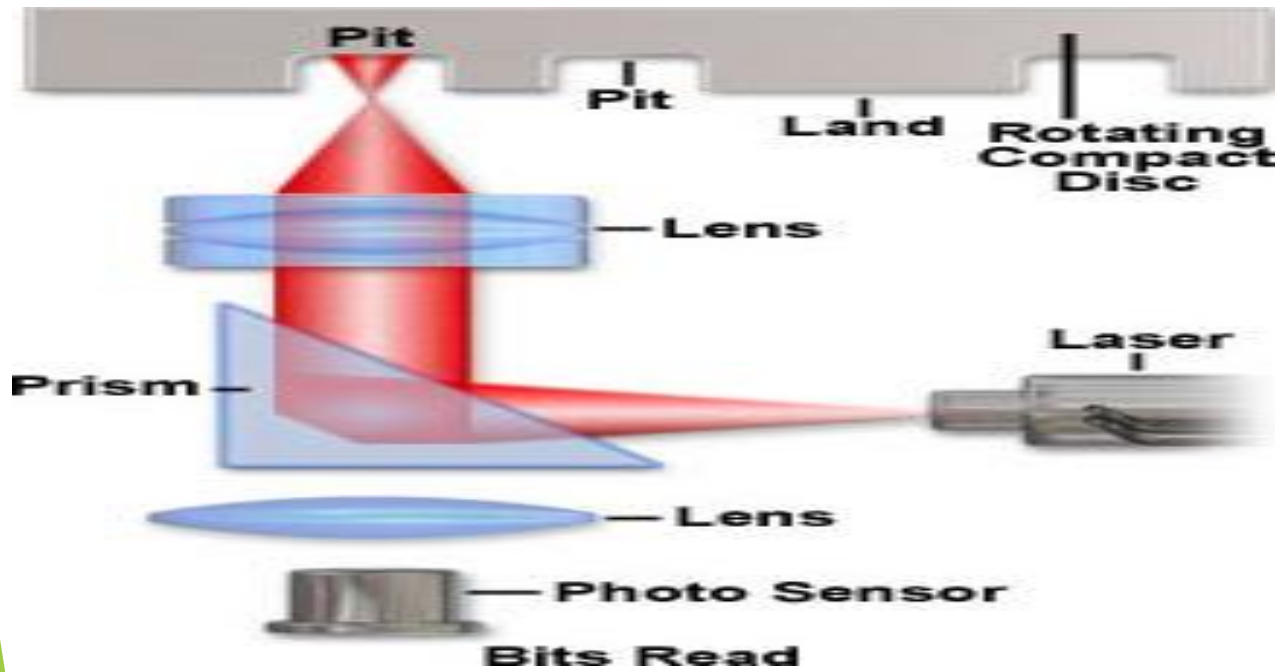
- *Data storage (CD-ROM)*
- *Write once audio & data storage (CD-R)*
- *Rewritable Media (CD-RW)*
- *Video compact Discs (VCD)*
- *Super Video Compact Discs (SVCD)*
- *Photo CD*
- *CD-i*
- *Enhanced CD*

Physical details of COMPACT DISCS:-



- A. A polycarbonate disc layer has the data encoded by using bumps.*
- B. A shiny layer reflects the laser.*
- C. A layer of lacquer protects the shiny layer.*
- D. Artwork is screen printed on the top of the disc.*
- E. A laser beam reads the CD & is reflected back to sensor, which converts it into electronic data.*





Disc Size and Capacity

Physical Size	Audio Capacity	CD-ROM data Capacity	Note
<i>120mm</i>	<i>74-99 min</i>	<i>650-870 MB</i>	<i>Standard size</i>
<i>80mm</i>	<i>21-24 min</i>	<i>185-210 MB</i>	<i>Mini-CD size</i>
<i>85x54mm- 86x64mm</i>	<i>~ 6 min</i>	<i>140-65 MB</i>	<i>"Business Card" size</i>

Digital Versatile/Video Discs (DVD)



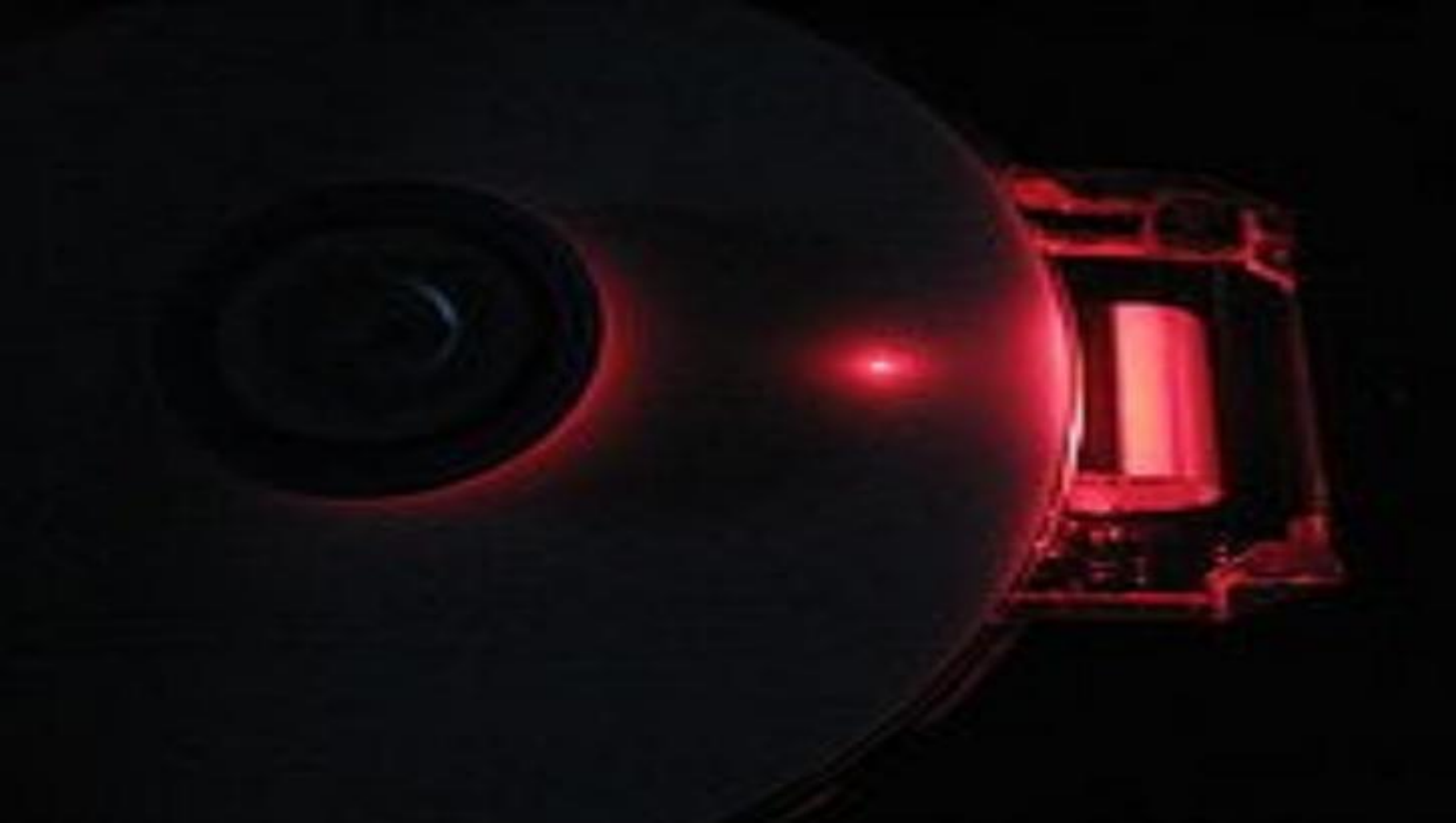
What is DVD?

- *DVD stands for Digital Versatile/Video Disc.*
- *It is an optical disc storage media format.*
- *Invented & Developed by Philips, Sony, Toshiba and Panasonic in 1995.*
- *It offers higher storage capacity than CD's while having the same dimensions.*

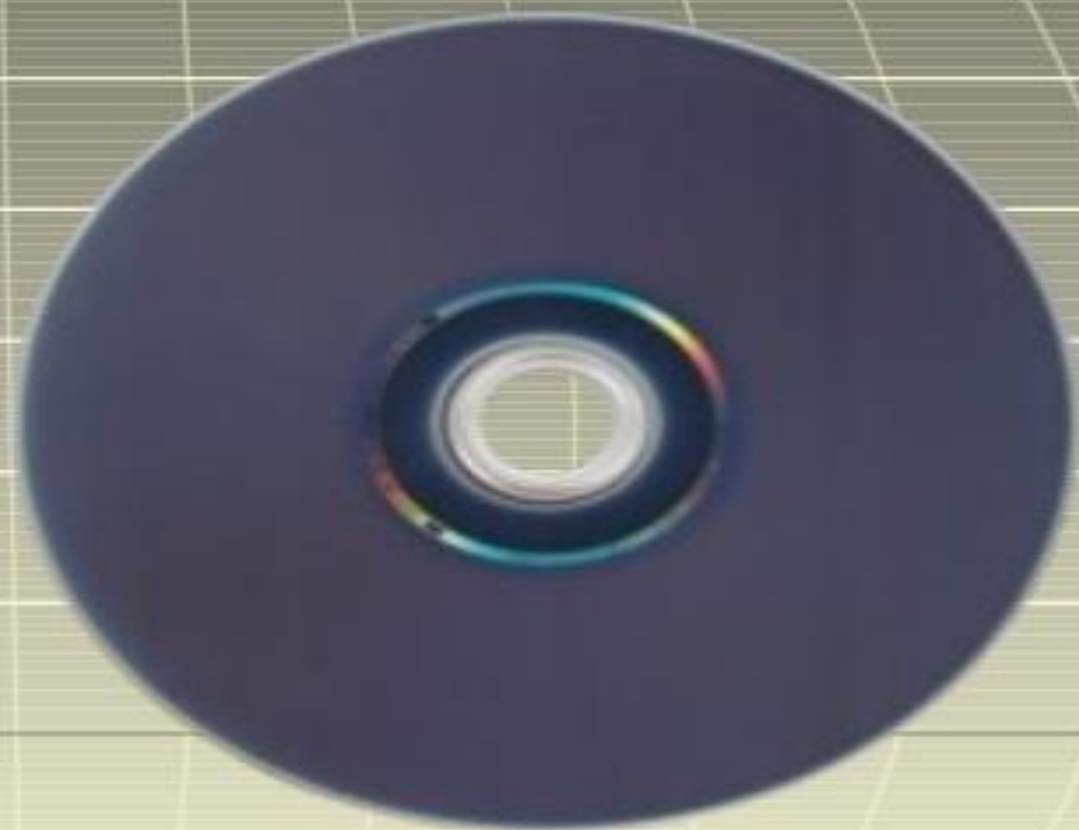
DVD sizes

- *There are basically 4 different DVD sizes.*

<i>DVD size (name)</i>	DVD-5	DVD-9	DVD-10	DVD-18
<i>Storage capacity</i>	4.7 GB	7.95 GB	8.75 GB	15.9 GB



BLU-RAY DISC

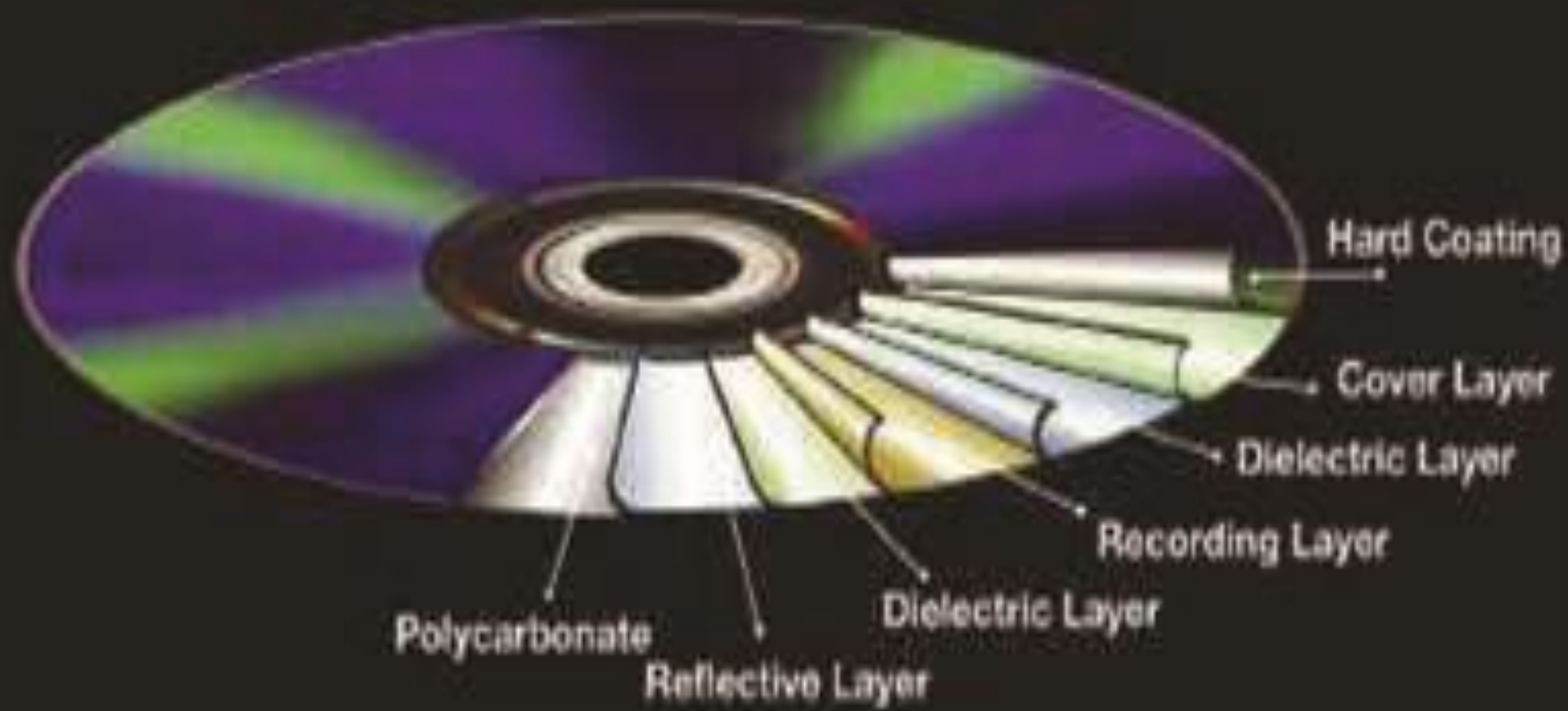


What is Blu-ray Disc?

- *It is a next-generation optical disc format meant for storage of high definition video & high density data.*
- *Jointly developed by a group of consumer electronics & PC companies called Blu-Ray disc Association (BDA).*



STRUCTURE OF A BLU-RAY DISC



How it works

- ▶ While current optical disc technologies such as DVD rely on a red laser to read and write data, the new format uses a blue-violet laser instead, hence it's name Blu-ray
- ▶ The discs come in single or double layer. Single layer can hold 25GB and double can hold 50GB of data compared to DVDs holding 4.7GB of data

Laser wavelength: Blu-ray & DVD



Lasers in blue on the visible spectrum can read more bits of data over a given area.

DVDs: because wavelengths of red light are longer, less data can be read over a given area.



Blu-ray vs. DVD capacity

Blu-ray vs. DVD Capacity

Single-layer DVD



4.7 GB of data



average two-hour standard-definition movie
with a few extra features

Single-layer Blu-ray disc



27 GB of data



more than 13 hours
of standard video



more than two hours of
high-definition video

Double-layer Blu-ray disc



54 GB of data



more than 20 hours of standard video



4.5 hours of high-definition digital video

0 10 20 30 40 50

How it works

- ▶ The benefit of using a blue-violet laser is that it has a shorter wavelength than a red laser, which makes it possible to focus the laser spot with even greater precision. This allows data to be packed more tightly and stored in less space, so it's possible to fit more data on the disc even though it's the same size as a CD/DVD.

Capacity & Usage :-

- ***Capacity = 25 GB (single-layer)
50 GB (dual-layer)
100/128 GB (BDXL)***

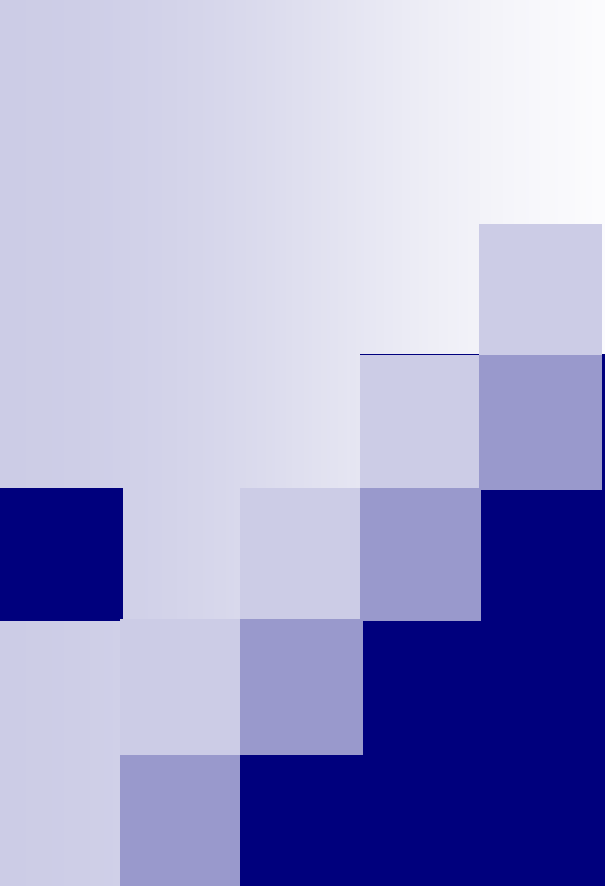
- ***Usage :-***

- 1. Data Storage***
- 2. High definition Video***
- 3. High definition Audio***
- 4. Stereoscopic 3D***
- 5. Playstation Games***

Size & Capacity:-

<i>Type</i>	<i>Diameter (cm)</i>	<i>Layers</i>	<i>Capacity</i>
			<i>GB</i>
<i>Standard disc size, single layer</i>	<i>12</i>	<i>1</i>	<i>25.0</i>
<i>Standard disc size, dual layer</i>	<i>12</i>	<i>2</i>	<i>50.1</i>
<i>Mini disc size, single layer</i>	<i>8</i>	<i>1</i>	<i>7.8</i>
<i>Mini disc size, dual layer</i>	<i>8</i>	<i>2</i>	<i>15.6</i>

THANK YOU



LED Technology

By: KMN



Background

- Organic Light Emitting Diode (OLED)
- Kodak scientist Dr. Ching (1970's)
- OLED materials reported in 1987
- Color improvements by 1989
- Becoming a major competitor with today's LCD/plasma displays

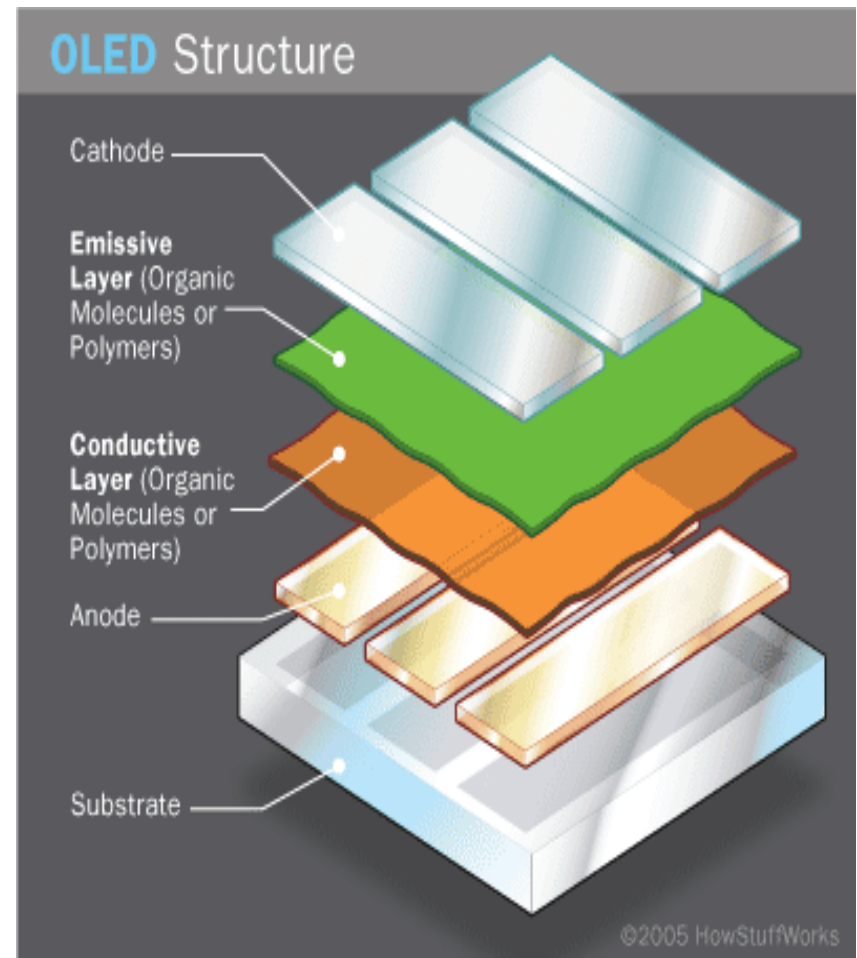


OLED/LED: What's the diff?

- Both OLED and LED use the same principle of electroluminescence- the optical and electrical phenomenon where certain materials emit light in response to an electric current passing through it.
- OLED
 - Lighter weight
 - Less power consumption
 - Organic based chemicals

How It Works

- Composed of two substrate layers on the outside
- Emissive and conductive layer lie between the cathode and the anode layers
- A current is applied across the LED, where electrons move from cathode to anode
- The cathode gives electrons to the emissive layer, where the anode withdraws these electrons from the conductive layer
- The emissive layer becomes rich in negative charge while the conductive layer becomes more positively charged
- The two charges recombine in the emissive layer, creating a drop in energy levels of the electrons
- The drop in energy levels results in radiation that is on the visible spectrum, emitting light



OLED's Today

- Various Mp3 players
 - Muzio JM-200
 - Sony NW-E015F Walkman
 - Arcos 204
- Kodak's LS633 3mpx digital camera
 - Priced at 399\$
 - Good battery life
 - 2.2" OLED screen



Continued...

- Mobile devices

- Apple iPhone



- Sony Ericsson W51S



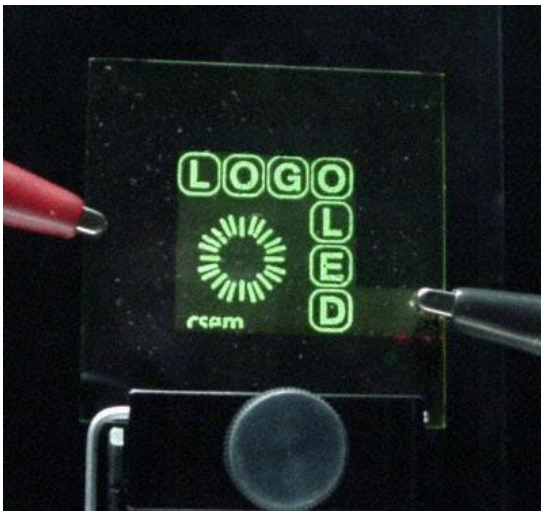
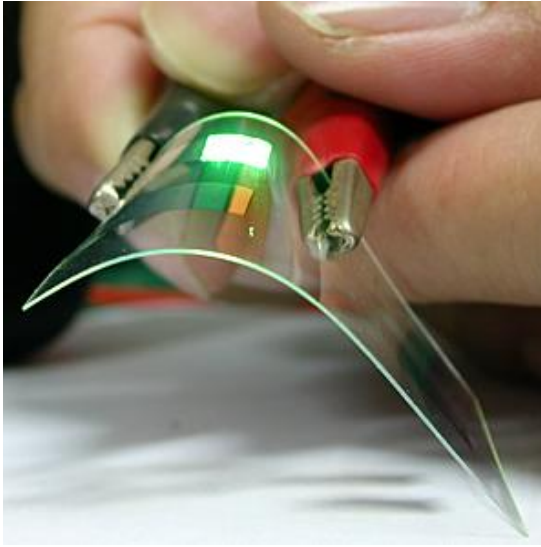
OLED vs. Plasma/LCD

■ Advantages

- ☐ Brighter, clearer picture
- ☐ More efficient viewing angle
- ☐ Thinner, lighter in weight
- ☐ Low powered
- ☐ Can be printed on various surfaces



...continued

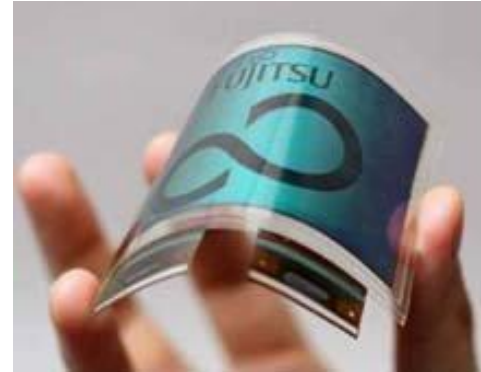


■ Disadvantages

- Short term battery life
- Expensive

Why so Expensive?

- Uses ink-jet printing technology
- Sprays conductive polymer substances, instead of ink.



Future Developments

- Smaller sized OLED TV's
- Middle-larger sized flexible semiconductor technology



Companies

- Samsungs
 - 12mm thick
 - 1600x1200 aspect ratio
- Sony intends to give OLED technology making OLED TV's.
- Toshiba and other



LED vs LCD

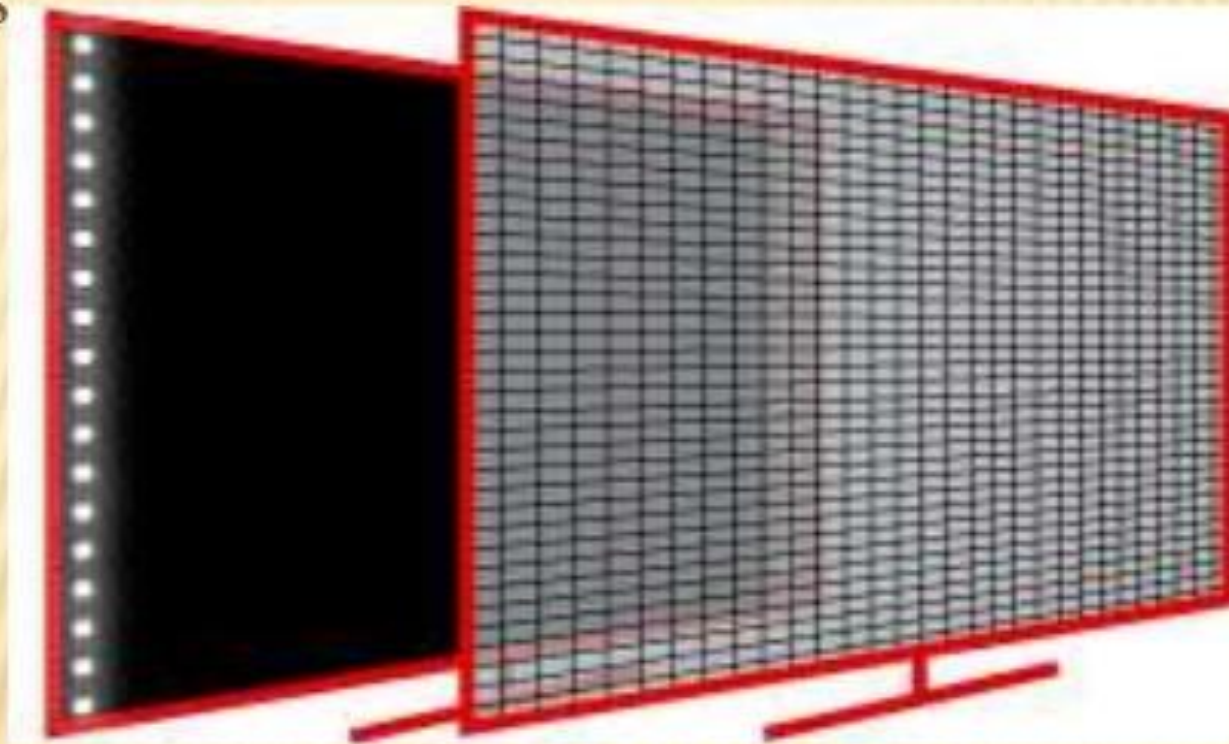
- The difference in lights and in lighting placement has generally meant that LED TVs can be thinner than LCDs, although this is starting to change.
- It has also meant that LED TVs run with greater energy efficiency and can provide a clearer, better picture than the general LCD TVs.
- Because of difference in lights and lightning placement there are two types

EDGE LIT

FULL ARRAY LIT

EDGE LIT

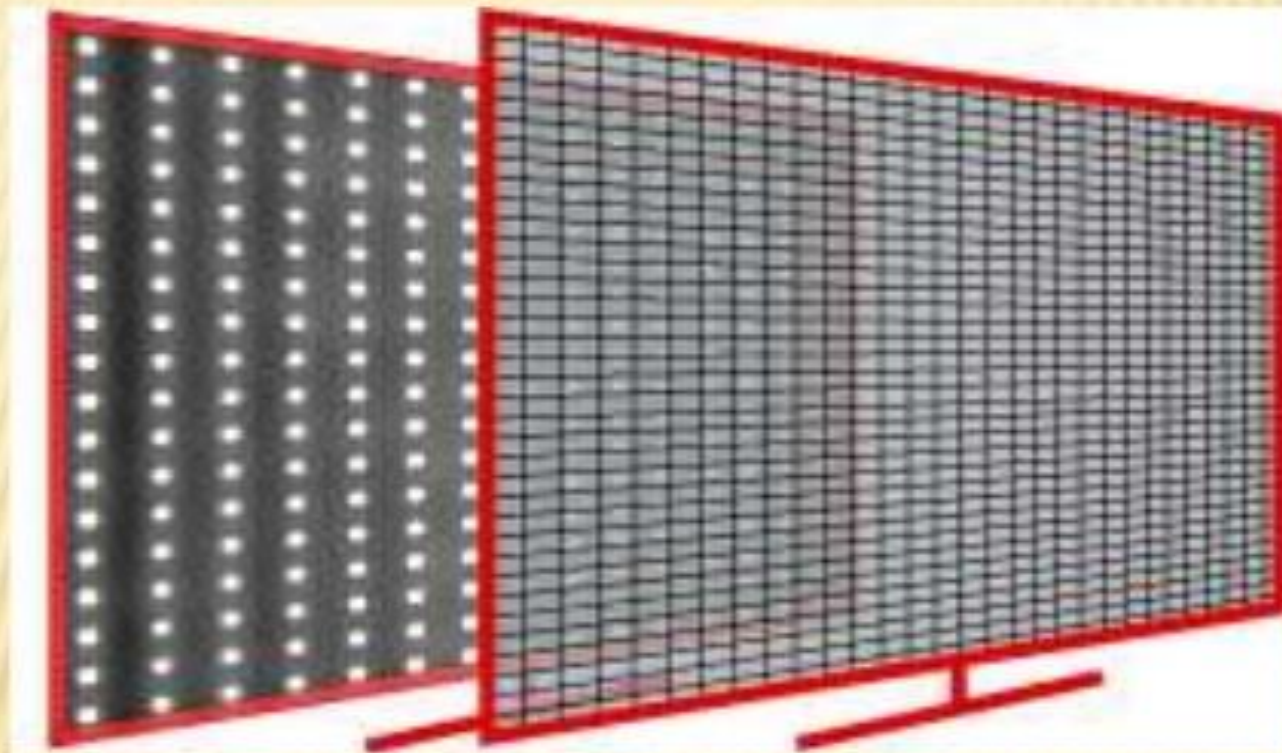
- This is the most common method for LED TVs



EDGE LIT

- In an edge lit LED screen, the LEDs are placed at the peripheral of the screen.
- Depending on the television, it can be all around the screen or only on the sides or the bottom.
- This allows the screen to be very thin. However, it can cause some spots on the screen to be brighter than others, like the edges.

FULL ARRAY



FULL ARRAY

- This method is considered the best LED backlight type, but can only be found on a very limited number of models.
- In a full array LED screen, the LEDs are distributed evenly behind the entire screen. This produces a more uniform backlight and it provides a more effective use of local dimming, where it can change the luminosity of only a specific part of the screen.

SPECIFICATION

- Cost : As per size of TV(almost 20k -70k)
- Size : available in 32",42",.....
- As per the market survey LG, SAMSUNG, SONY,TOSHIBA,PANASONIC are making LED TV.



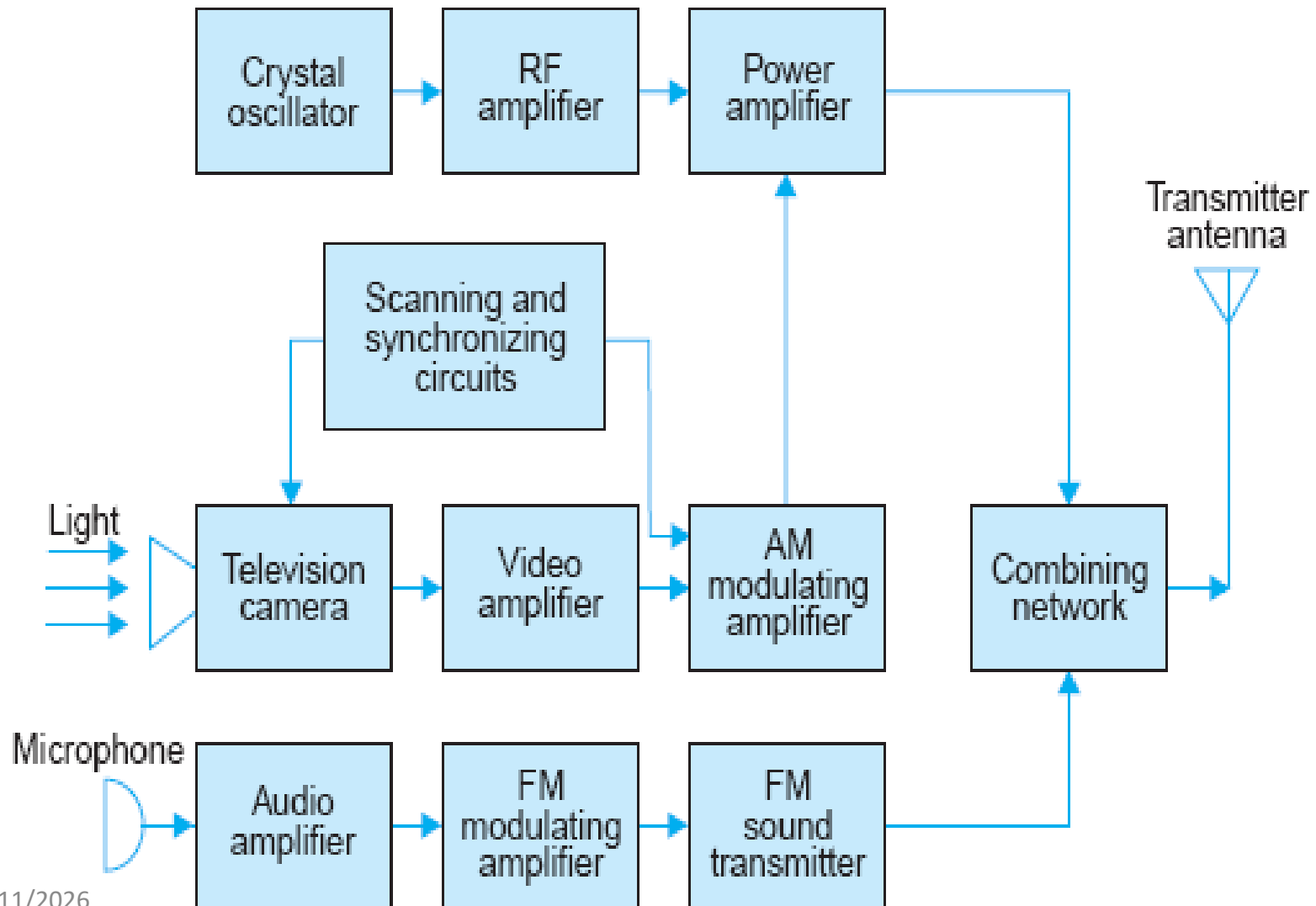
THANK
YOU

TELEVISION

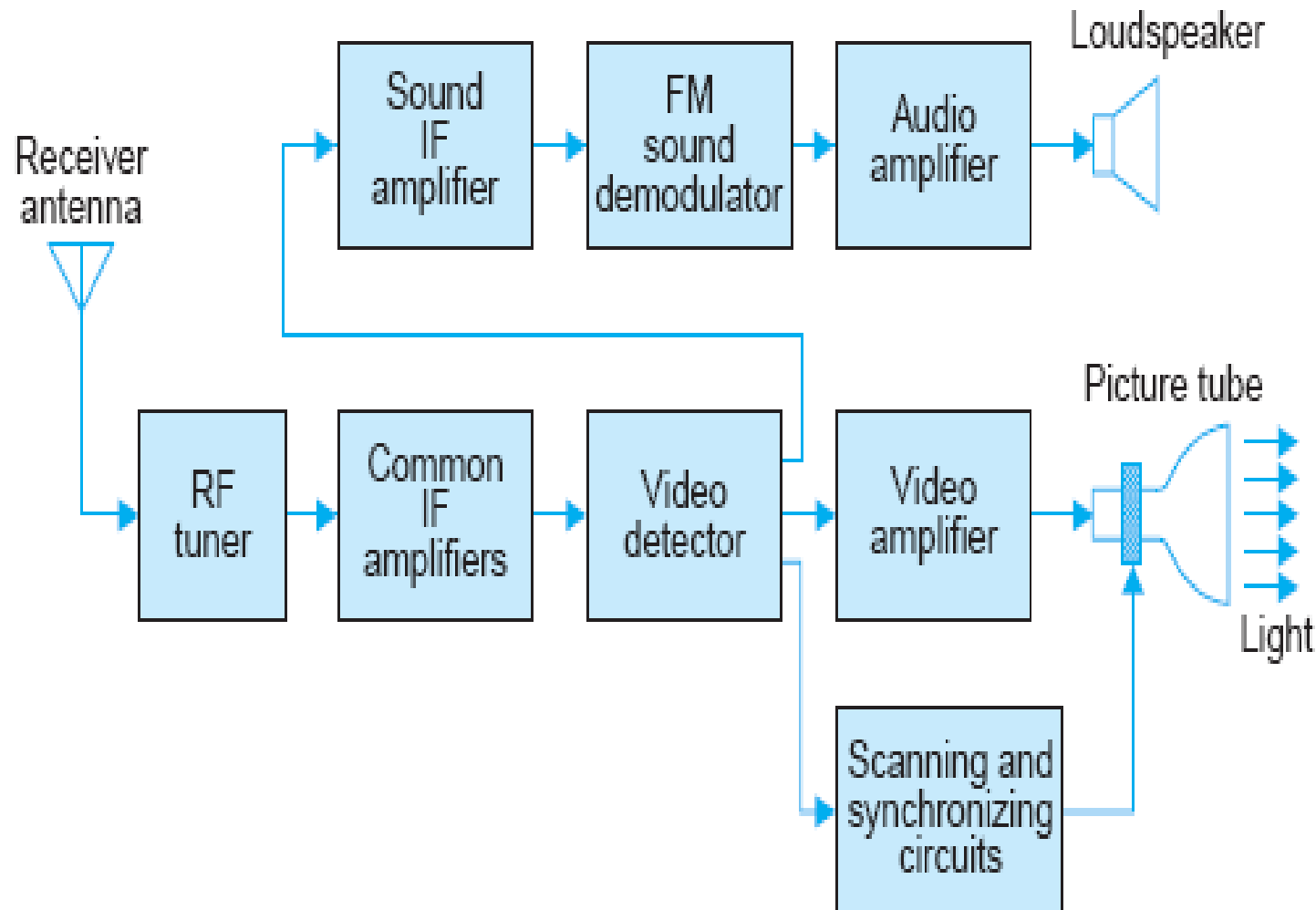
Elements of a T V System

- Picture Transmission
- Sound Transmission
- Picture Reception
- Sound Reception
- Synchronization
- Receiver Controls
- Colour Television.

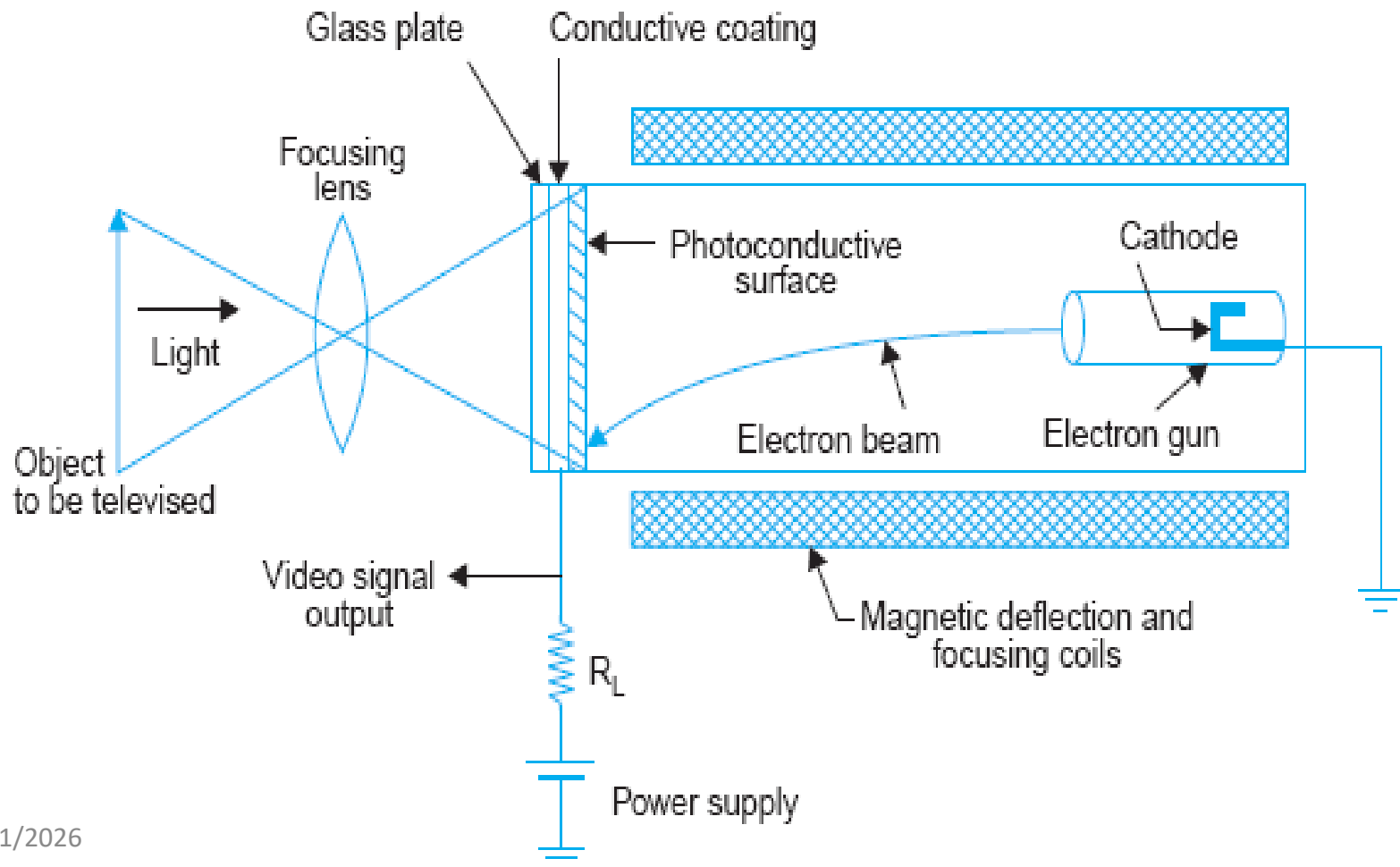
Basic monochrome television transmitter



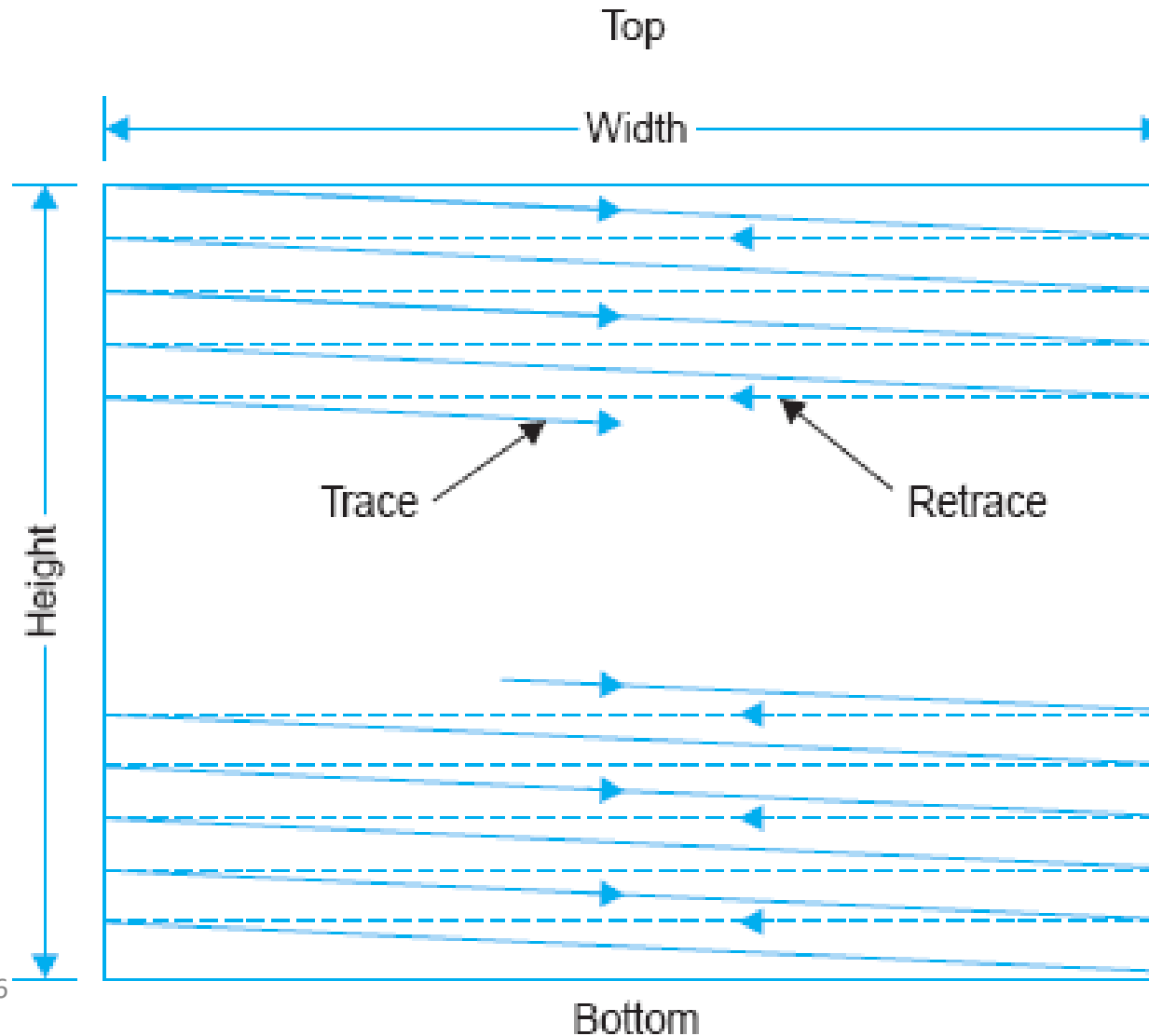
Basic monochrome television receiver



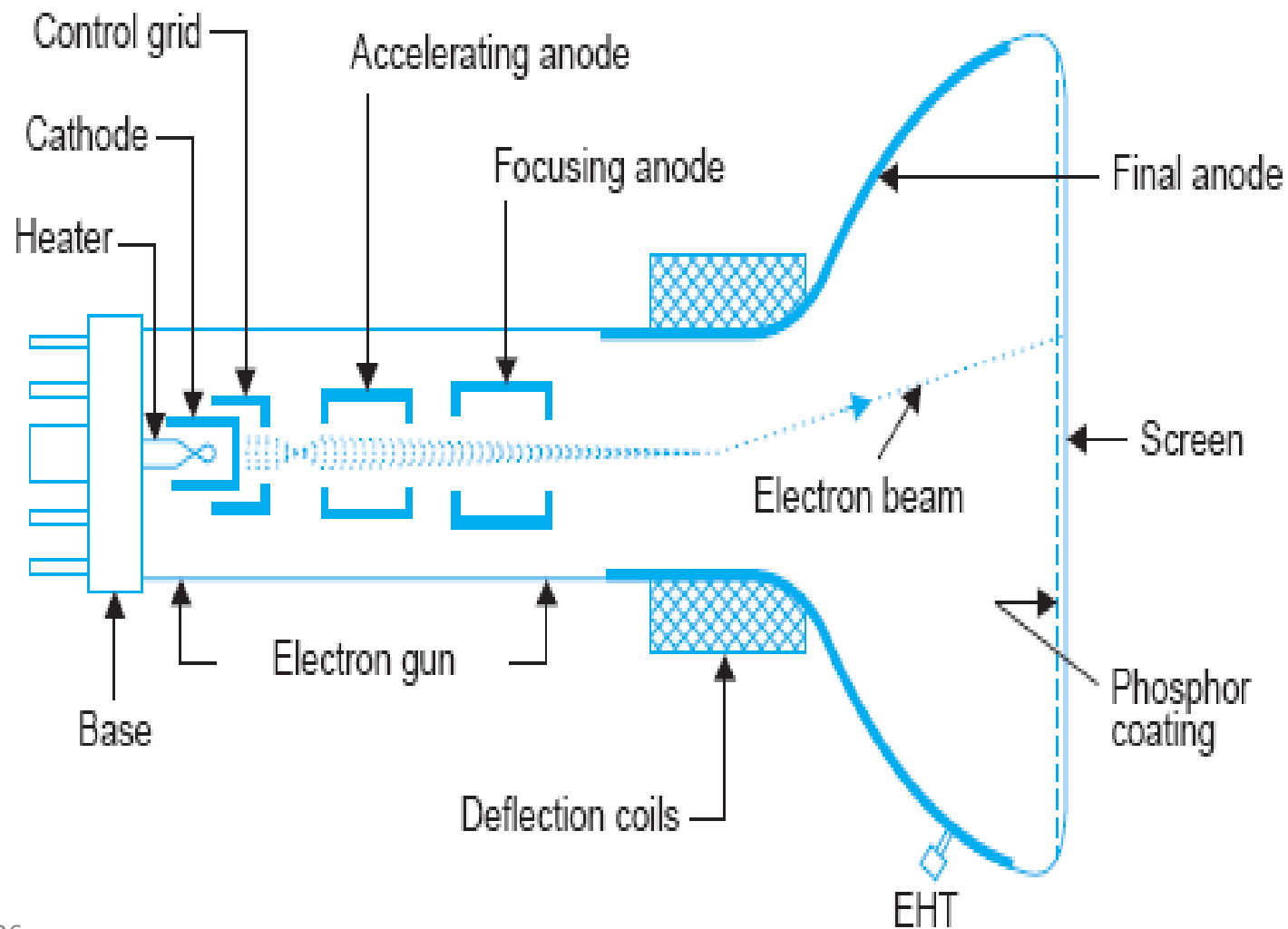
Simplified cross-sectional view of a Vidicon TV camera tube



Path of scanning beam in covering picture area

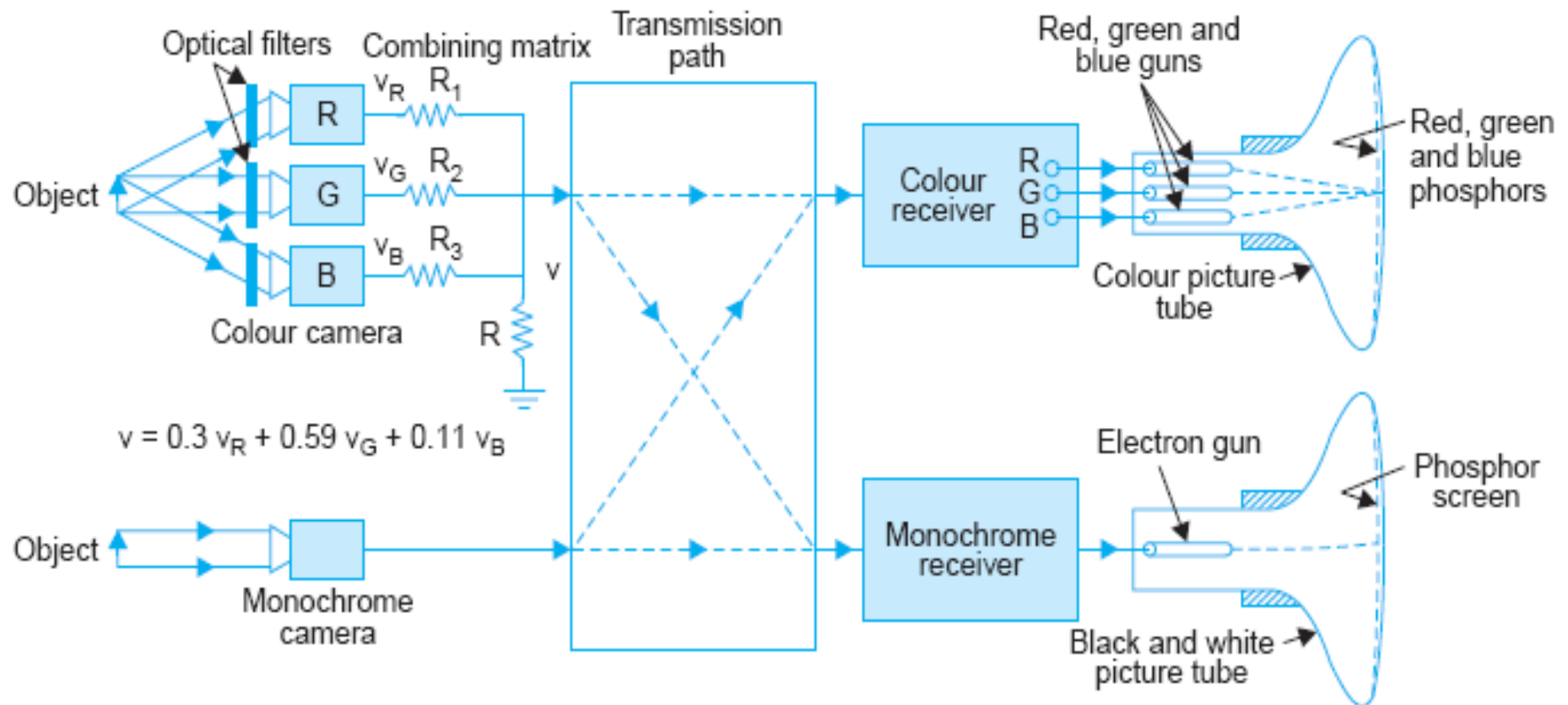


Elements of a picture tube



Signal transmission paths illustrating compatibility between colour and monochrome TV systems.

R, G and B represent three camera tubes which develop video signals corresponding to the red, green and blue contents of the scene being televised



Television System

NTSC Standard

(National Television Standards Committee)

- Standard for U.S., Japan, and Korea
- 4 X 3 Aspect Ratio
- 525 Lines
- 30 Frames Per Second
- Scanned in "Fields"

PAL System

(Phase Alternating Line)

- Another World Television Standard
- 4 X 3 Aspect Ratio
- 625 Lines
- 25 Frames Per Second
- Scanned in "Fields"
- There are Slight Variations: PAL-B, -G, -H, -N
- Used in Continental Europe and Parts of Africa, Middle East & So. America
- More Lines = Better Resolution
- Fewer Frames/Fields = More "Flicker"

SECAM System

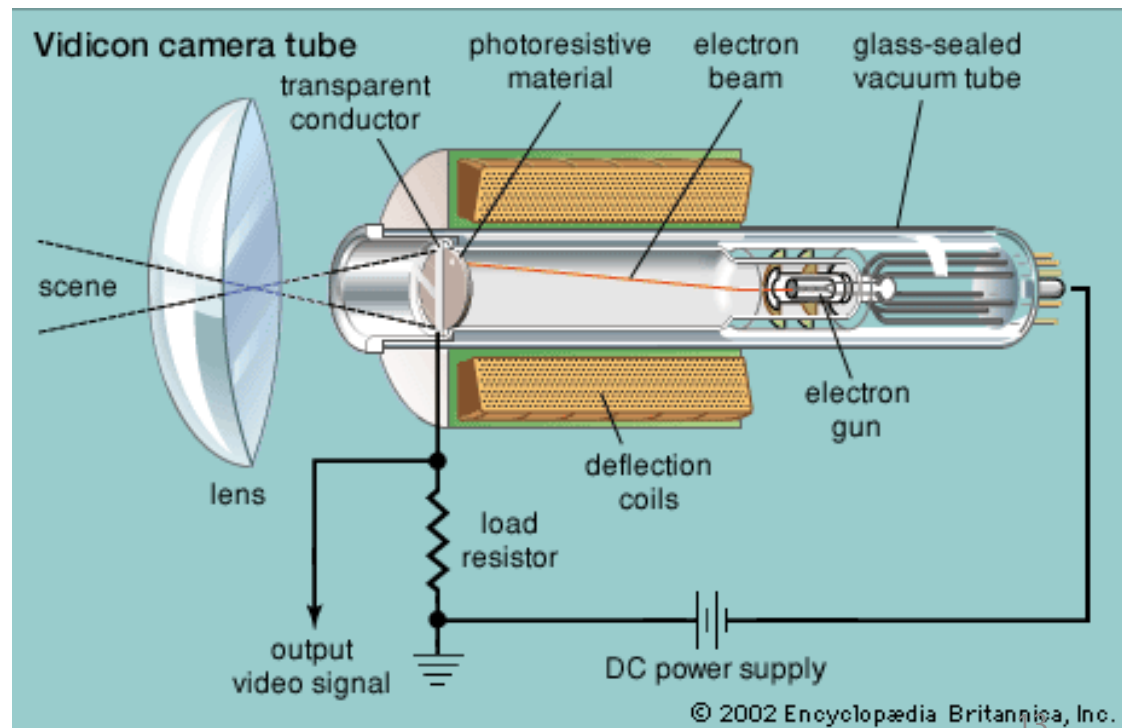
(Sequential Color and Memory)

- Another World Television Standard
- 4 X 3 Aspect Ratio
- 625 Lines
- 25 Frames Per Second
- Scanned in "Fields"
- Used in France, Eastern Europe and Parts of the Middle East & Africa
- More Lines = Better Resolution
- Fewer Frames/Fields = More "Flicker"

HDTV System (High Definition Television)

- 16 X 9 Aspect Ratio
- A **Digital** System
- Permits Several Levels of Picture Resolution Similar to that of High-Quality Computer Monitors, With 720 or 1080 Lines (1280 x 720 pixels or 1920 x 1080 pixels)
- Ranges from 24 to 60 frames per second, progressive or interlaced scan
- Uses MPEG-2 Compression to squeeze a 19 megabit-per-second data flow so that it can be accommodated by a standard broadcast TV channel of 6 MHz bandwidth
- 5.1 Channels of Dolby AC-3 Digital Surround-Sound Audio

Vidicon TV Pick-up Tube

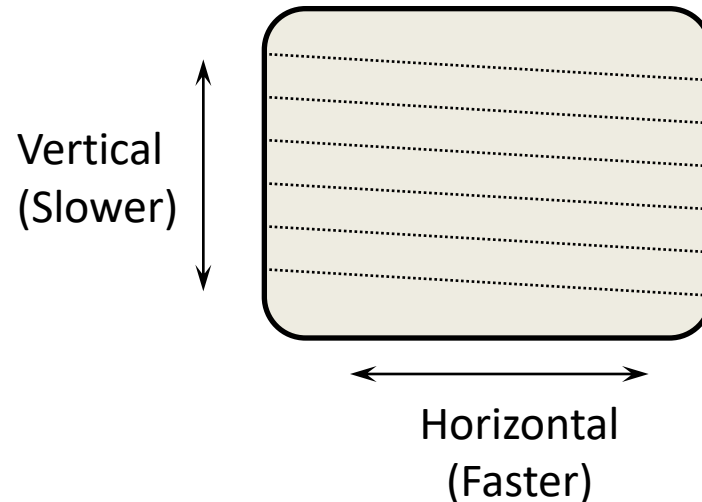
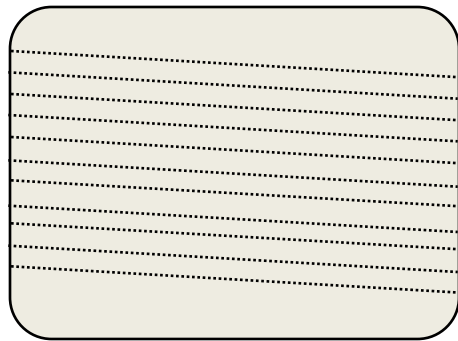


Basic Factors

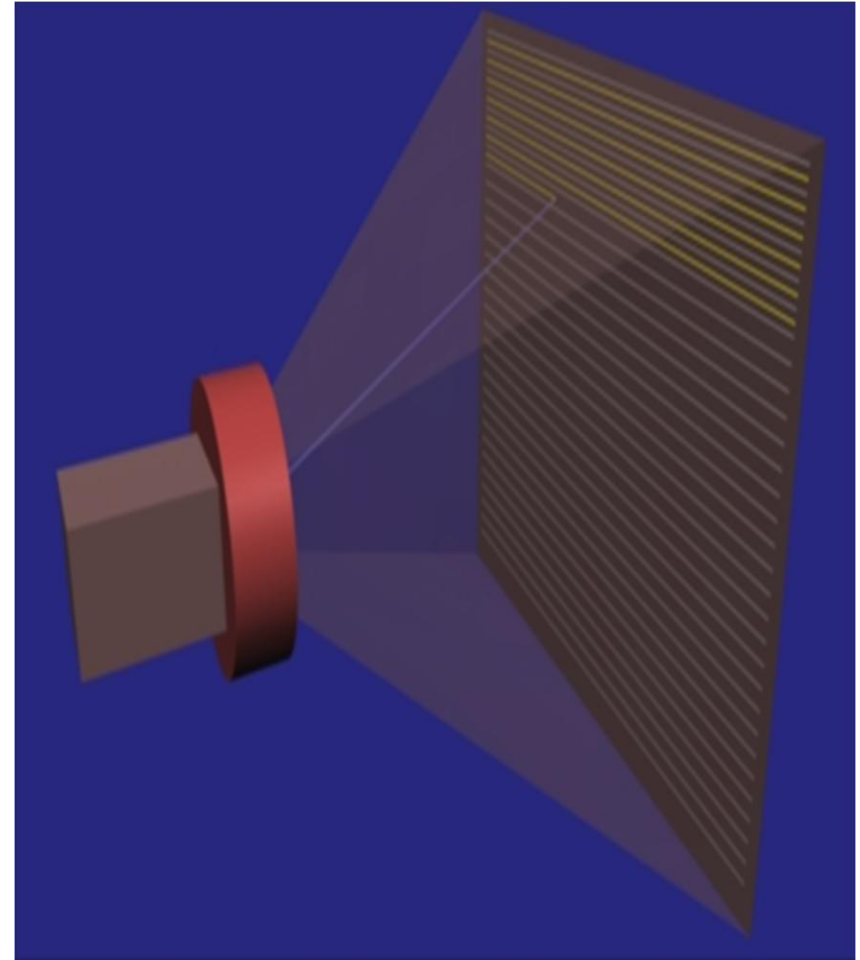
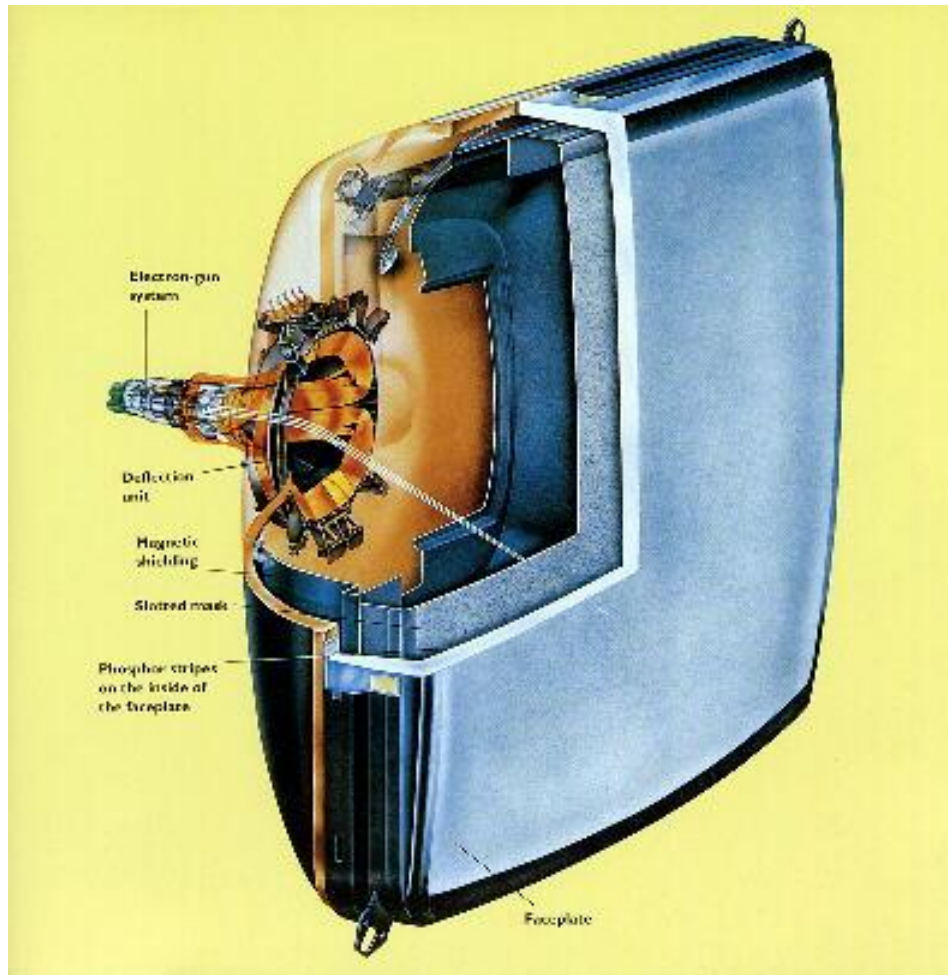
- Structure (Aspect ratio – 4:3)
- Continuity (25 FPS)
- Scanning lines (860 x 0.7)
- Flicker (Interlaced Scanning)
- Resolution (Vertical & Horizontal)
- Other factors.

Television Scanning

- beam scanning for standard TV
 - 625 lines in total image
 - 25 images (**frames**) scanned per second
- Oscillators
 - Vertical
 - Horizontal

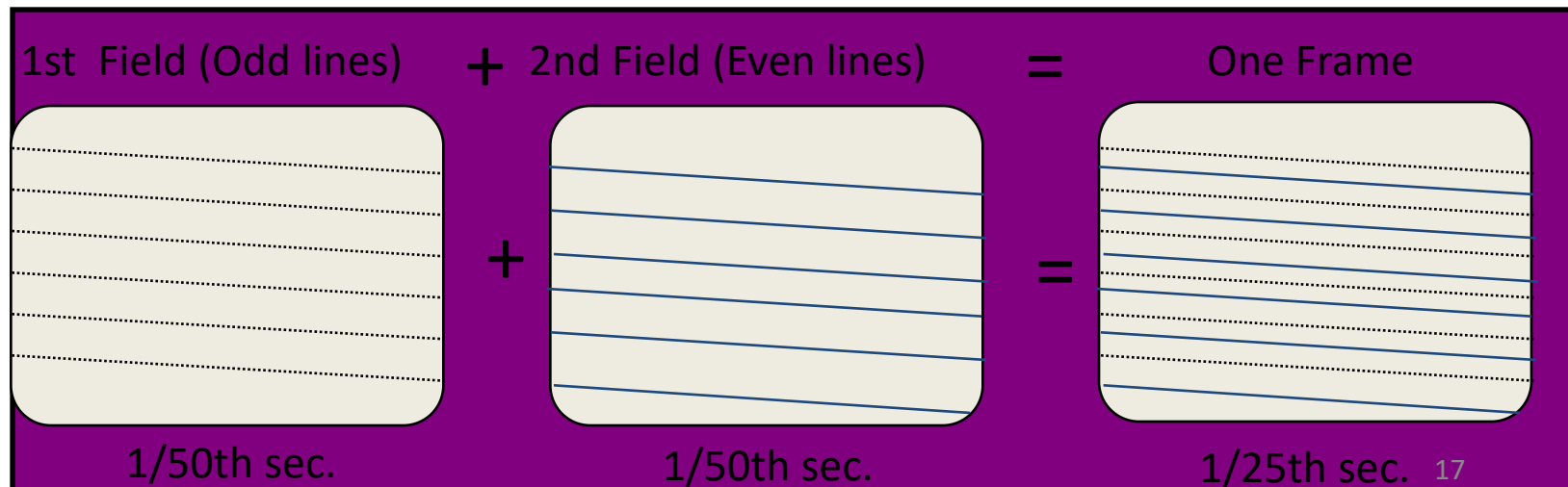


CRT



TV Interlacing

- **frame** scanned in two passes, first the odd lines then the even
 - One frame takes 1/25th second
 - avoids flicker
- each pass called a **field**
 - takes 1/50th second



Interlaced



Odd lines
Field 1

Even lines
Field 2

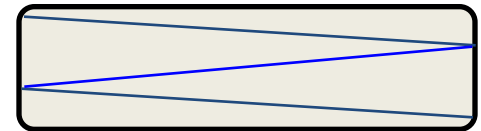


Field 1 + Field 2 = Frame (complete image)

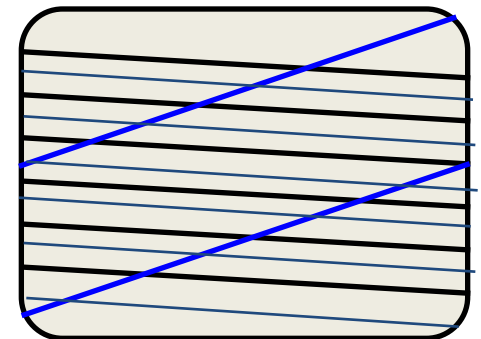
Display Rate: 50fields per second

Synchronization

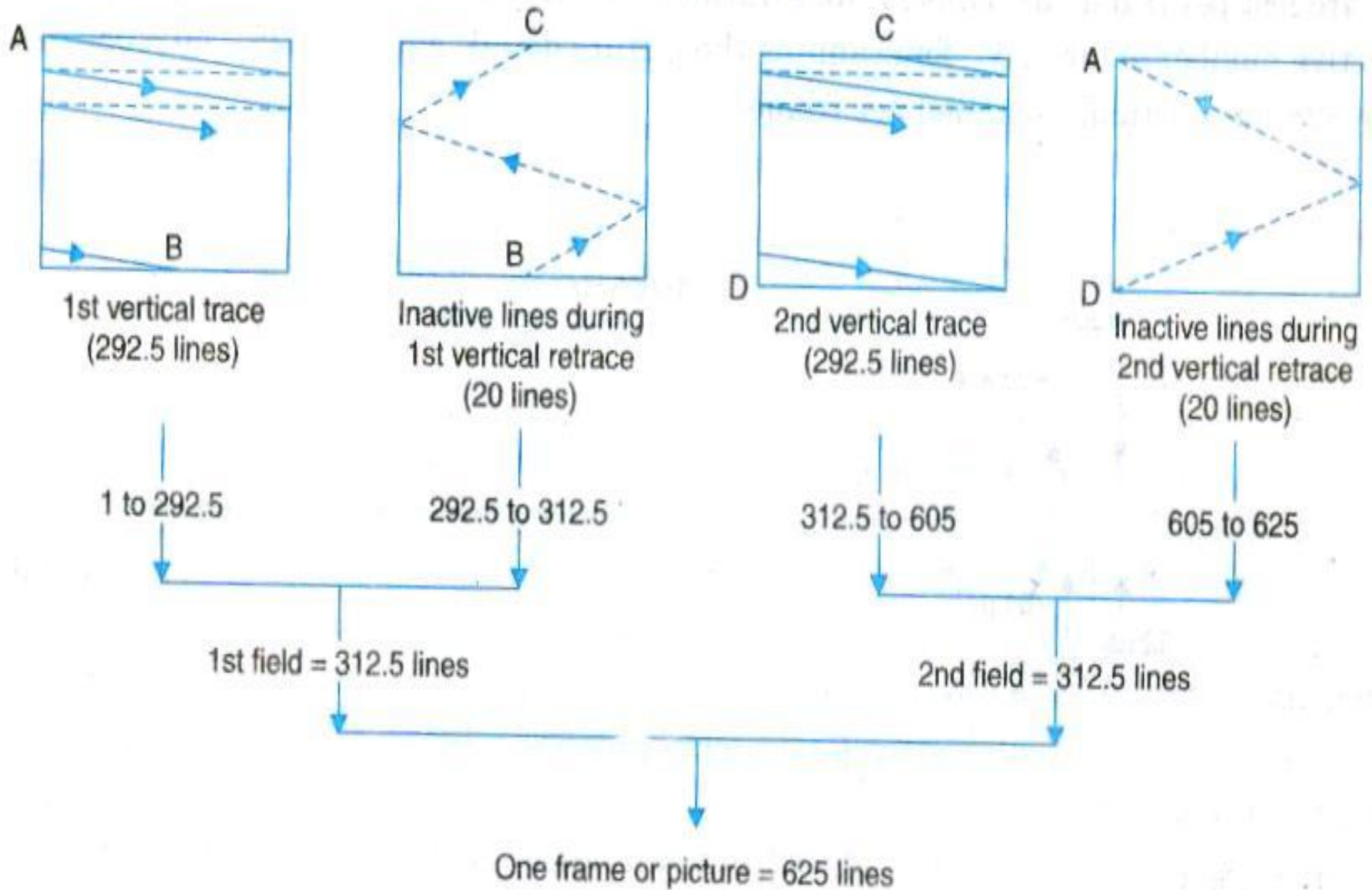
- TV Camera & Monitor must be synchronized
 - beam at same place on image
- special sync pulses sent at end of each horizontal line & vertical field
- retrace
 - horizontal retrace
 - beam returned to left side of screen
 - vertical retrace
 - beam returned to the top of screen
 - Turns off video during retrace

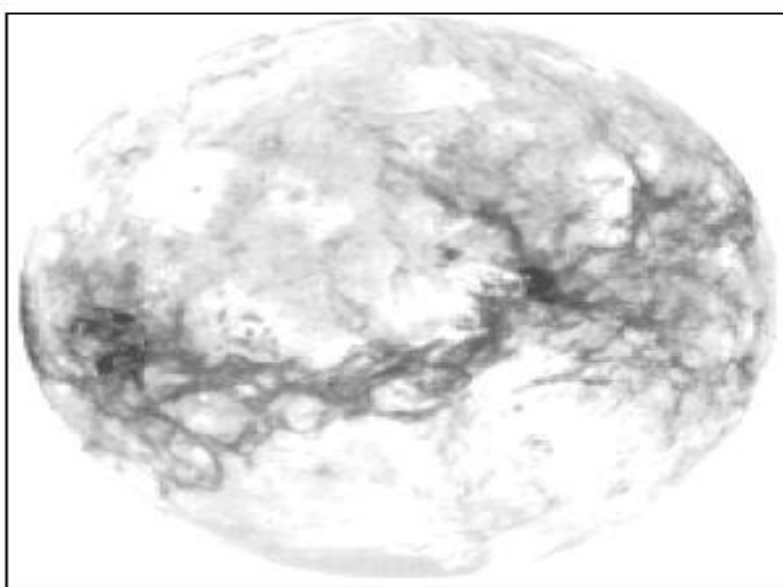


Horizontal Retrace

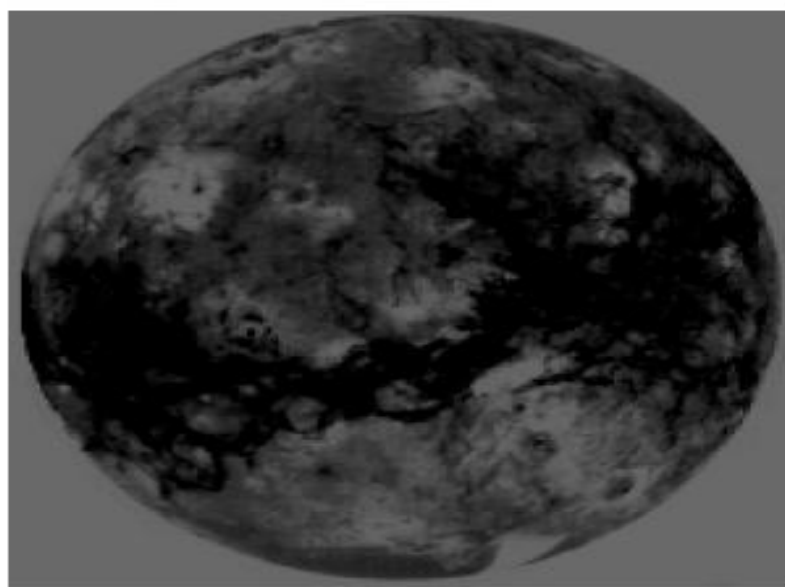


Vertical Retrace¹⁹

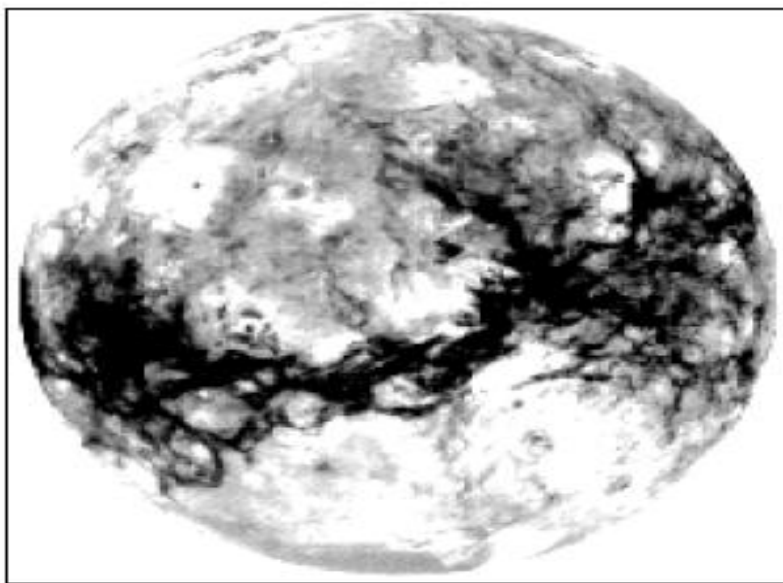




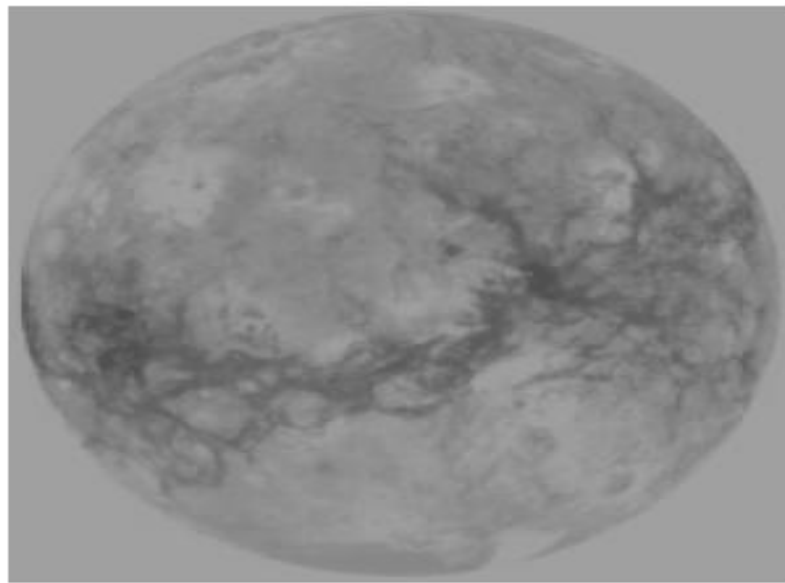
Brightness too high



Brightness too low



Contrast too high



Contrast too low

Brightness and contrast adjustments. Increasing the *brightness* makes every pixel in the image becomes lighter. In comparison, increasing the *contrast* makes the light areas become lighter, and the dark areas become darker. These images show the effect of misadjusting the brightness and contrast.

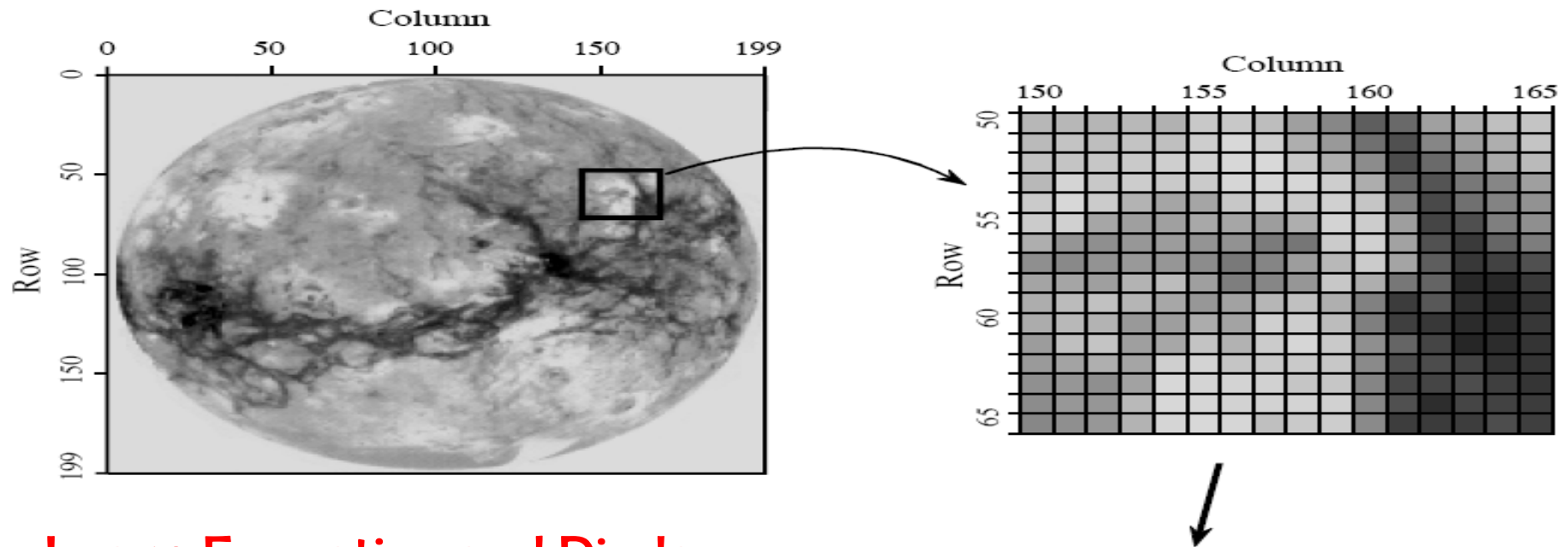
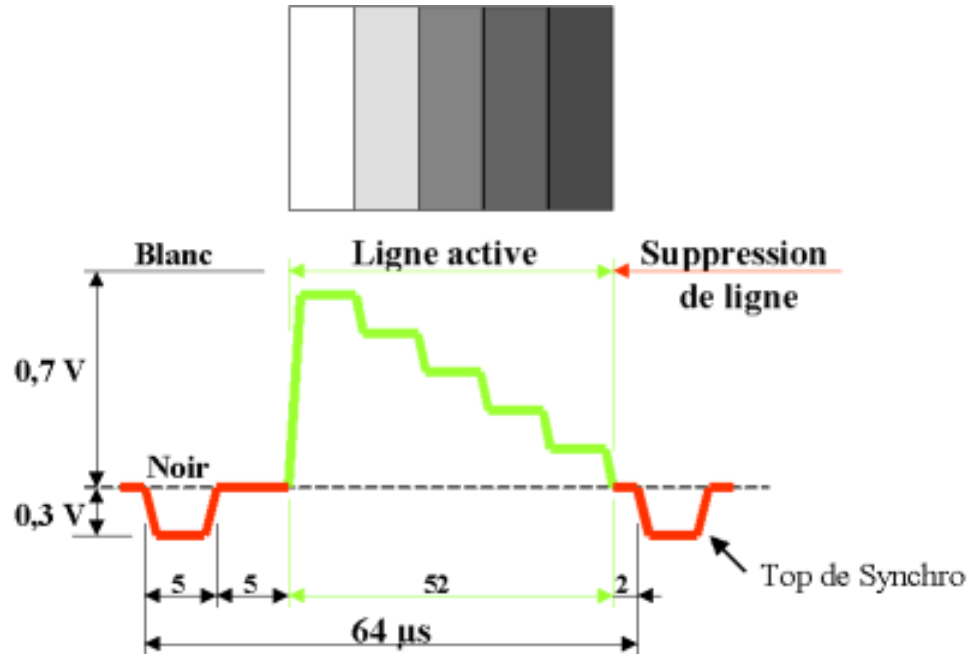


Image Formation and Display

Digital image structure. This example image is the planet Venus, as viewed in reflected microwaves. Digital images are represented by a two-dimensional array of numbers, each called a *pixel*. In this image, the array is 200 rows by 200 columns, with each pixel a number between 0 to 255. When this image was acquired, the value of each pixel corresponded to the level of reflected microwave energy. A *grayscale* image is formed by assigning each of the 0 to 255 values to varying shades of gray.

		Column															
		150				155				160				165			
Row	50	183	183	181	184	177	200	200	189	159	135	94	105	160	174	191	196
		186	195	190	195	191	205	216	206	174	153	112	80	134	157	174	196
		194	196	198	201	206	209	215	216	199	175	140	77	106	142	170	186
		184	212	200	204	201	202	214	214	214	205	173	102	84	120	134	159
		202	215	203	179	165	165	199	207	202	208	197	129	73	112	131	146
	55	203	208	166	159	160	168	166	157	174	211	204	158	69	79	127	143
		174	149	143	151	156	148	146	123	118	203	208	162	81	58	101	125
		143	137	147	153	150	140	121	133	157	184	203	164	94	56	66	80
		164	165	159	179	188	159	126	134	150	199	174	119	100	41	41	58
		173	187	193	181	167	151	162	182	192	175	129	60	88	47	37	50
	60	172	184	179	153	158	172	163	207	205	188	127	63	56	43	42	55
		156	191	196	159	167	195	178	203	214	201	143	101	69	38	44	52
		154	163	175	165	207	211	197	201	201	199	138	79	76	67	51	53
		144	150	143	162	215	212	211	209	197	198	133	71	69	77	63	53
		140	151	150	185	215	214	210	210	211	209	135	80	45	69	66	60
	65	135	143	151	179	213	216	214	191	201	205	138	61	59	61	77	63

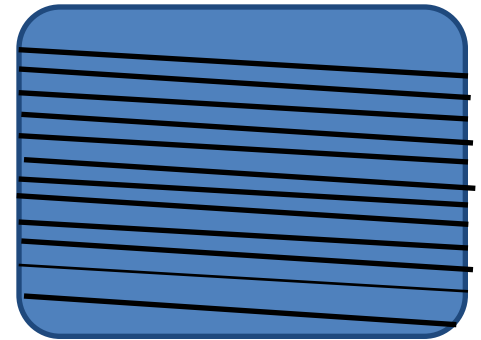
Video Signal



- Voltage level indicates brightness
- Blanking during non-video
 - retrace

Vertical Resolution

- proportional to # of vertical scan lines
- theoretic maximum
 - half # of visible scan lines
 - black lines alternate with white
 - $\text{max. line pairs} = \text{video lines} / 2$



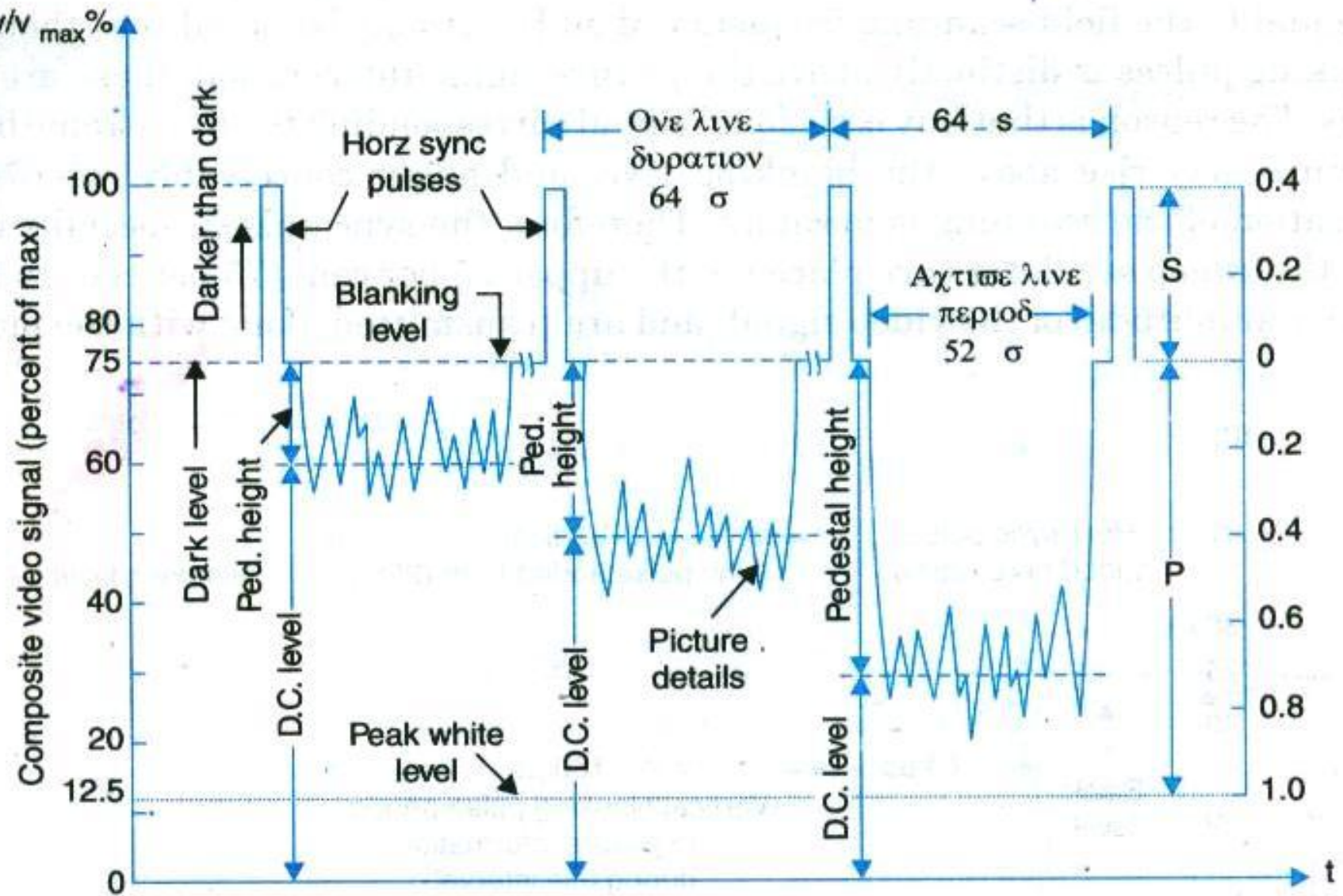


Fig. 3.1 Arbitrary picture signal details of three scanning lines with different average brightness levels. Note that picture to sync ratio $P/S = 10/4$.

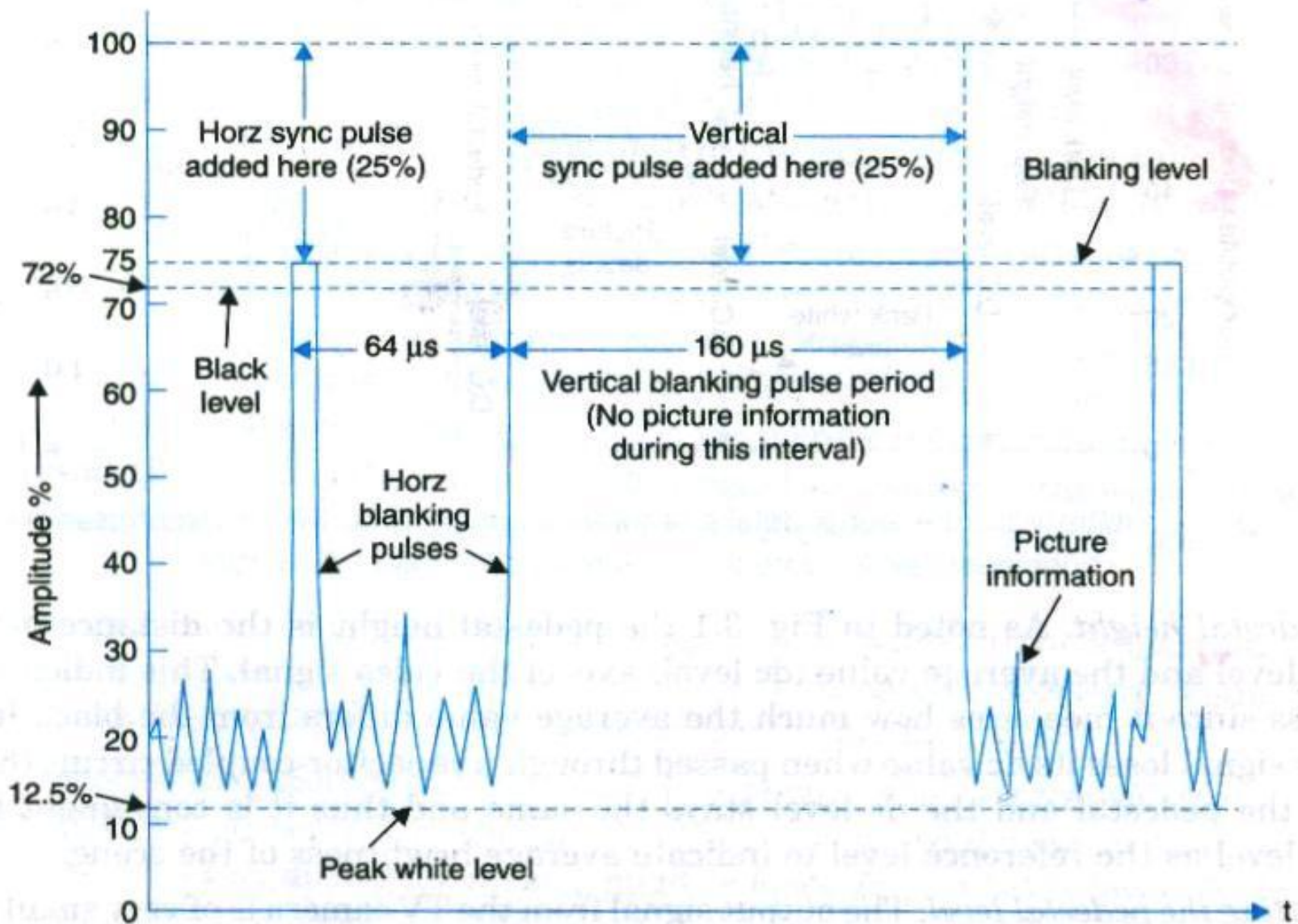


Fig. 3.2 Horizontal and vertical blanking pulses in video signal. Sync pulses are added above the blanking level and occupy upper 25% of the composite video signal amplitude.

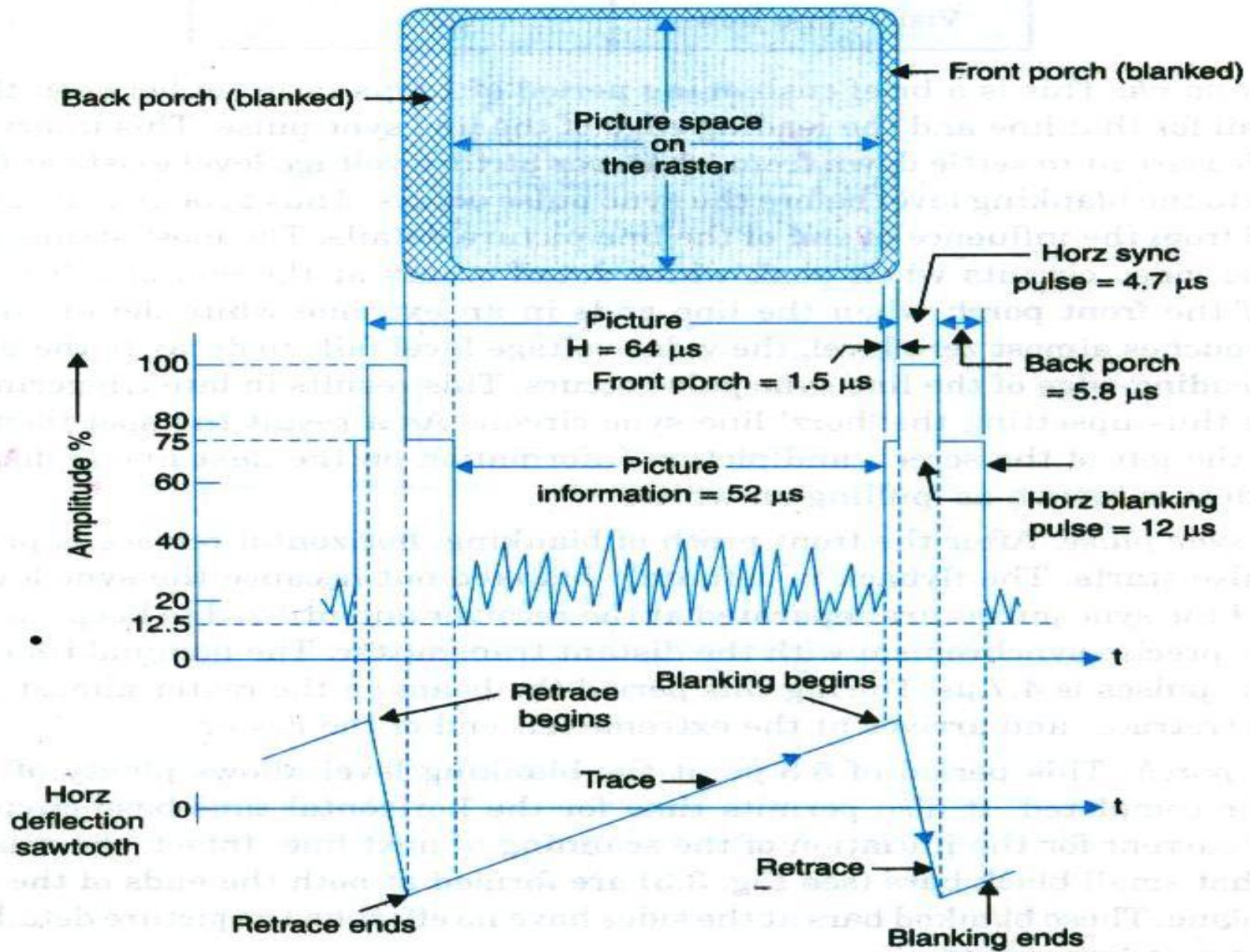


Fig. 3.3 Horz line and sync details compared to horizontal deflection sawtooth and picture space on the raster.

Details of Horizontal Scanning

<i>Period</i>	<i>Time (μs)</i>
Total line (H)	64
Horz blanking	$12 \pm .3$
Horz sync pulse	4.7 ± 0.2
Front porch	$1.5 \pm .3$
Back porch	$5.8 \pm .3$
Visible line time	52

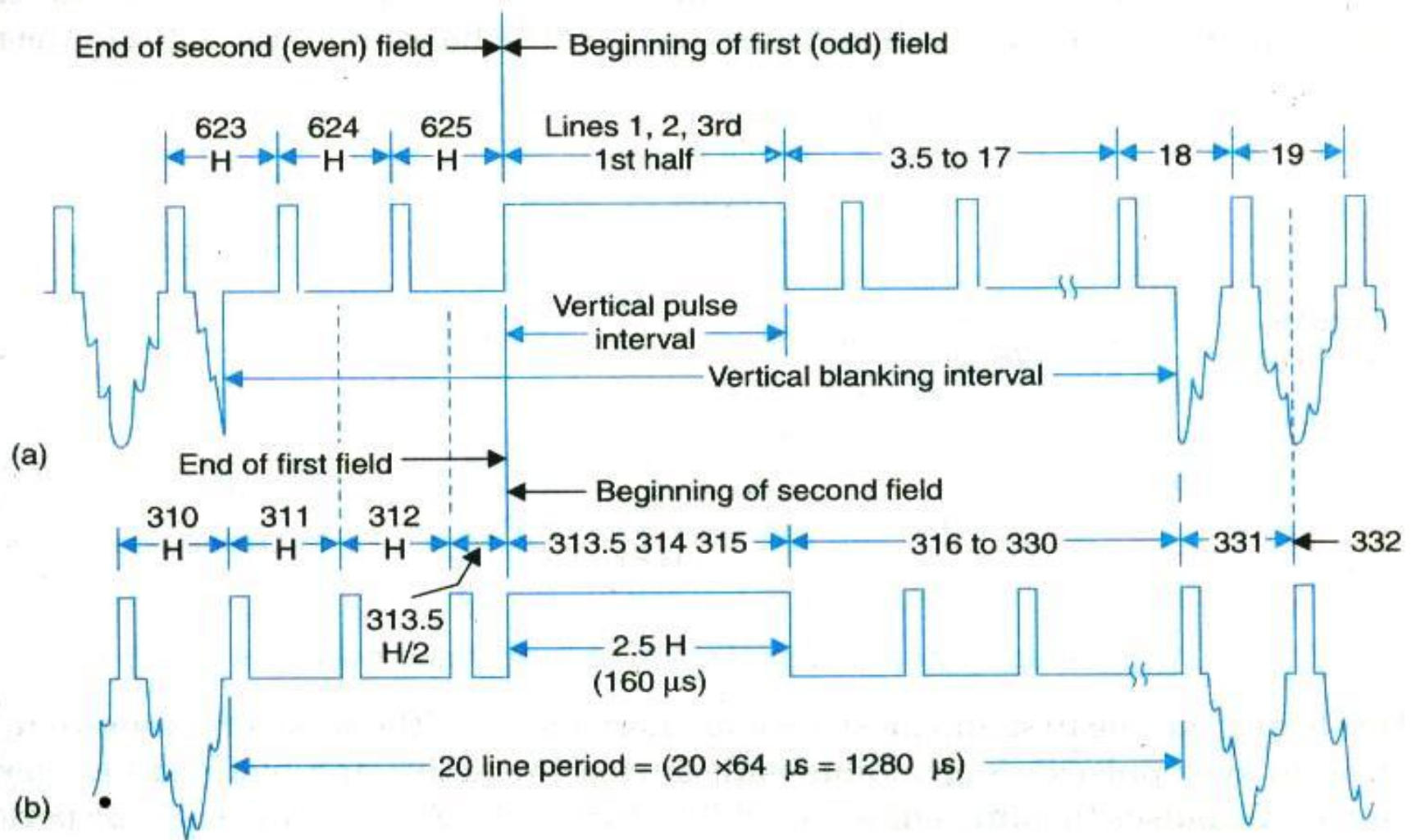


Fig. 3.4 Composite video waveforms showing horizontal and basic vertical sync pulses at the end of (a) second (even) field, (b) first (odd) field. Note, the widths of horizontal blanking intervals and sync pulses are exaggerated.

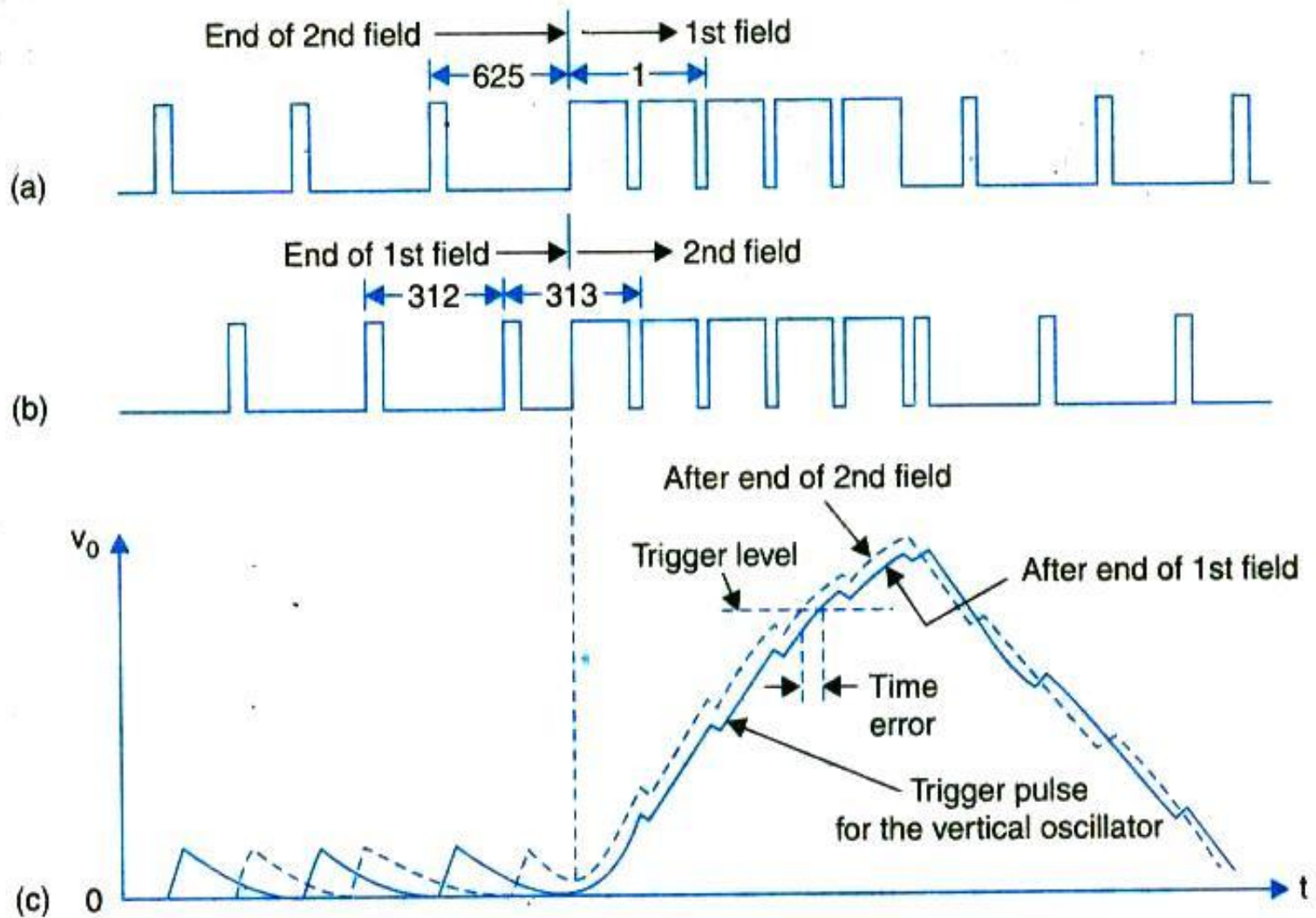


Fig. 3.7 Integrating waveforms (a) pulses at the end of 2nd (even) field (b) pulses at the end of 1st (odd) field (c) integrator output. Note the above sync pulses have purposely been drawn without equalizing pulses.

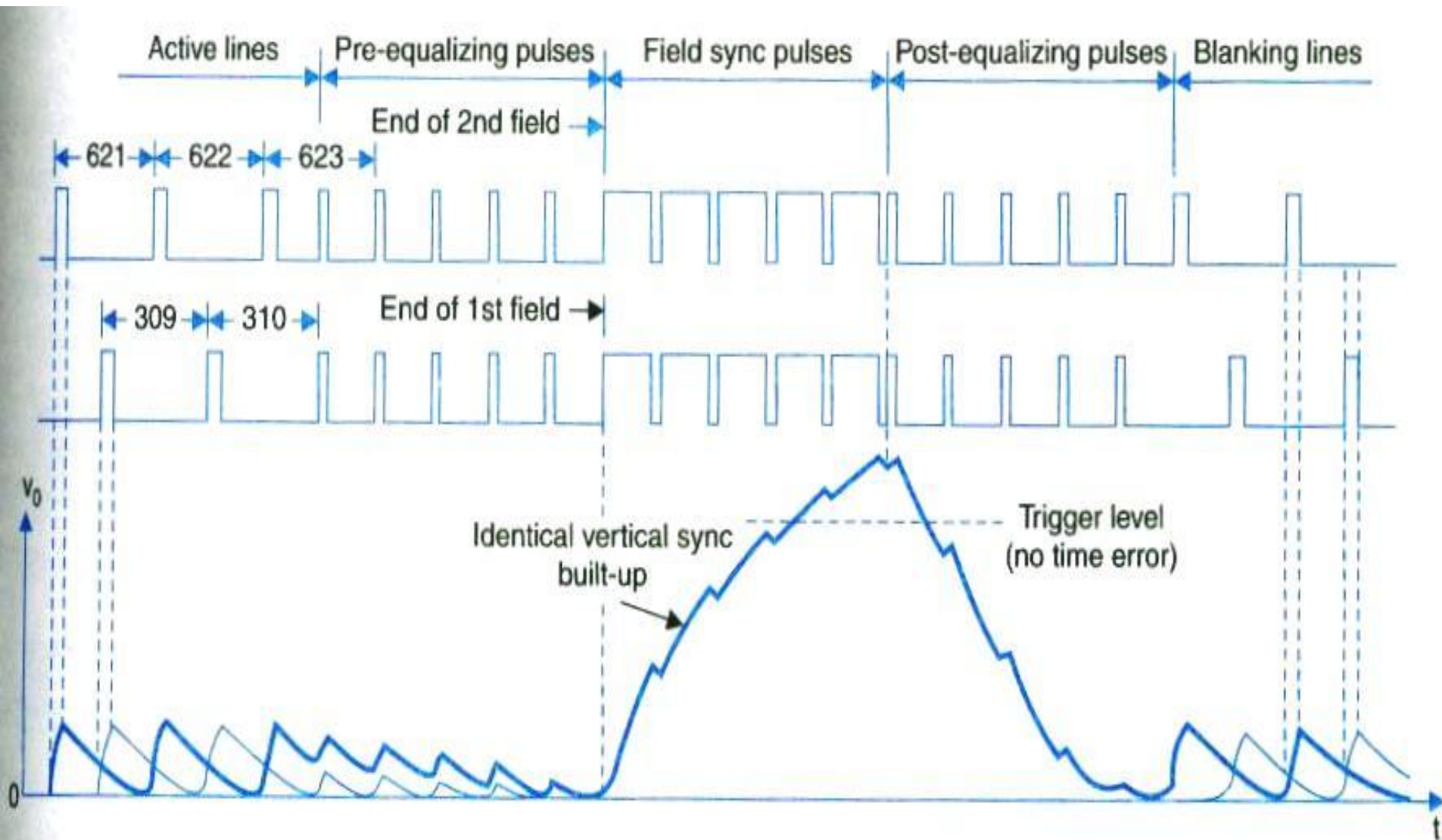


Fig. 3.9 Identical vertical sync voltage built-up across the integrating capacitor.

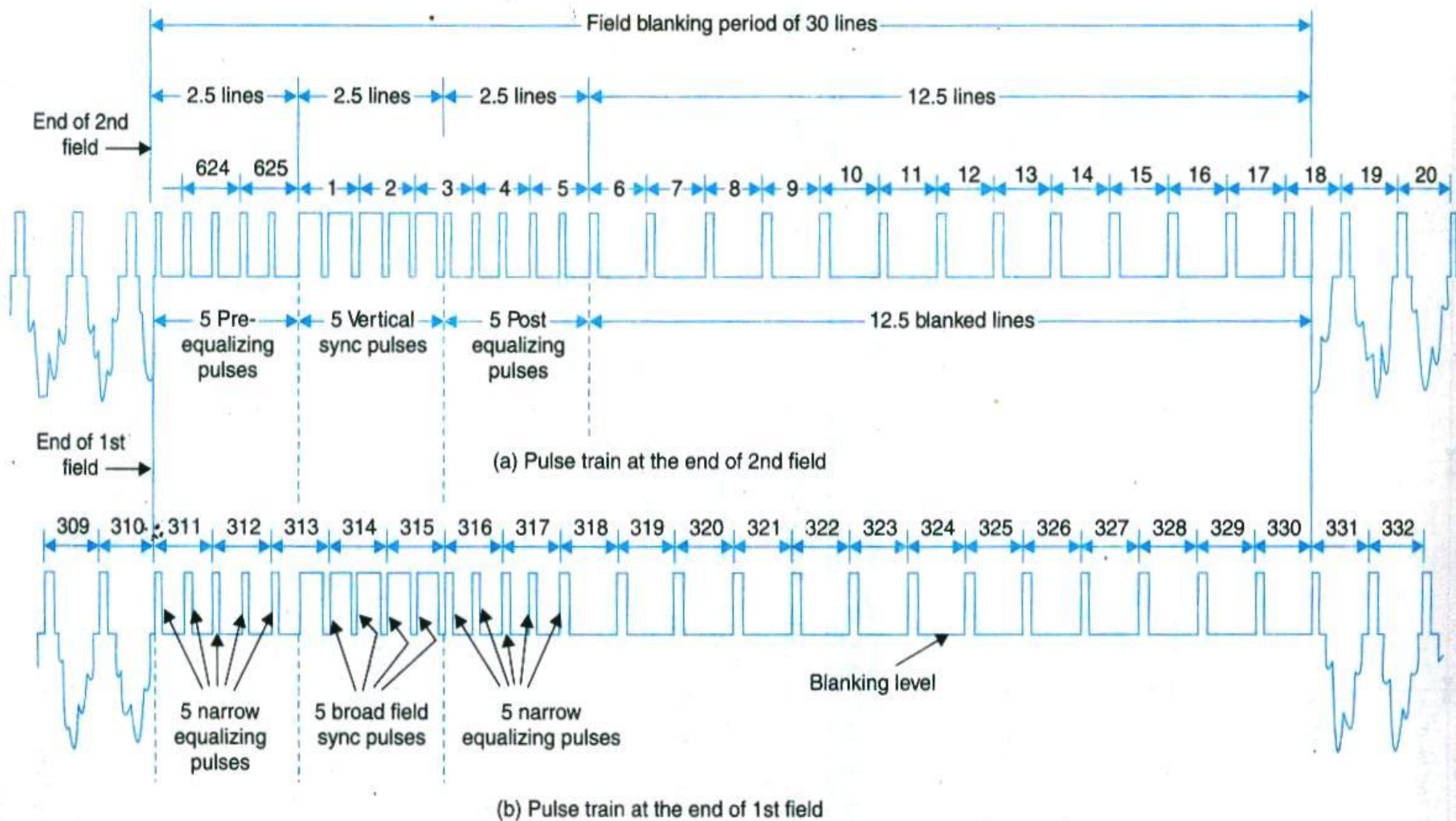


Fig. 3.10 Field synchronizing pulse trains of the 625 lines TV system.

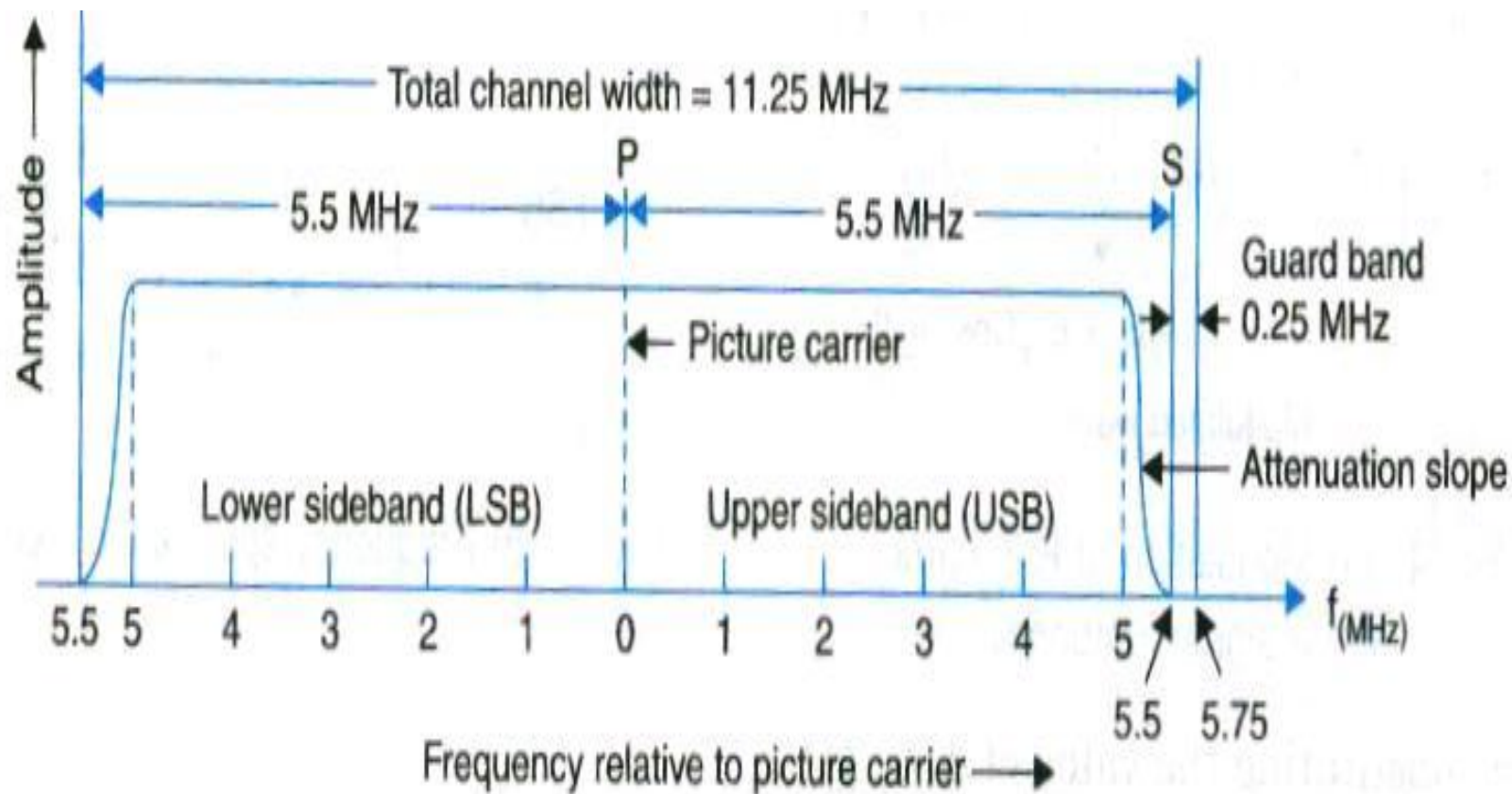


Fig. 4.2 Total channel bandwidth using double sideband picture signal.
P is picture carrier and S is sound carrier.

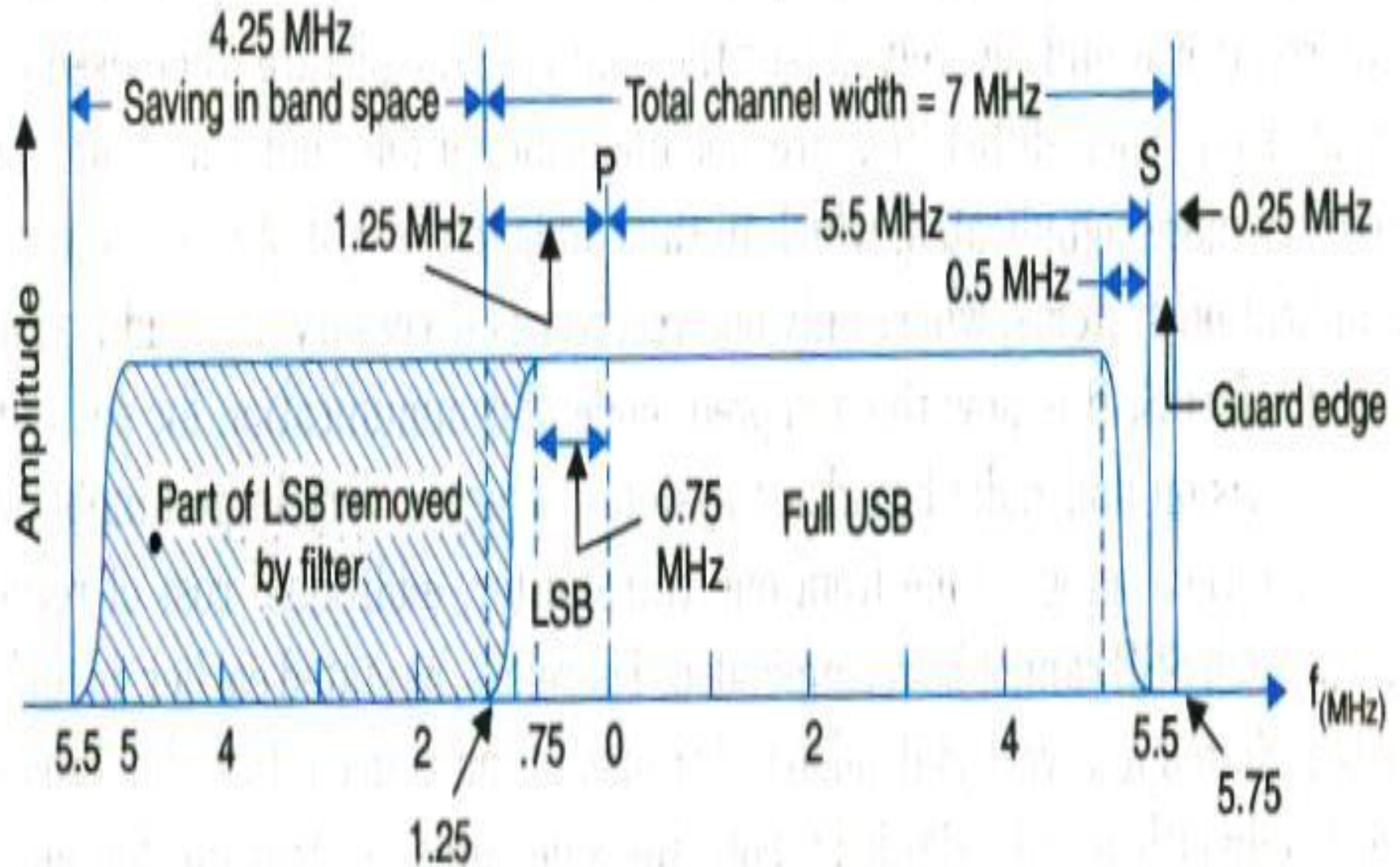


Fig. 4.3 Total channel bandwidth using vestigial lower sideband.

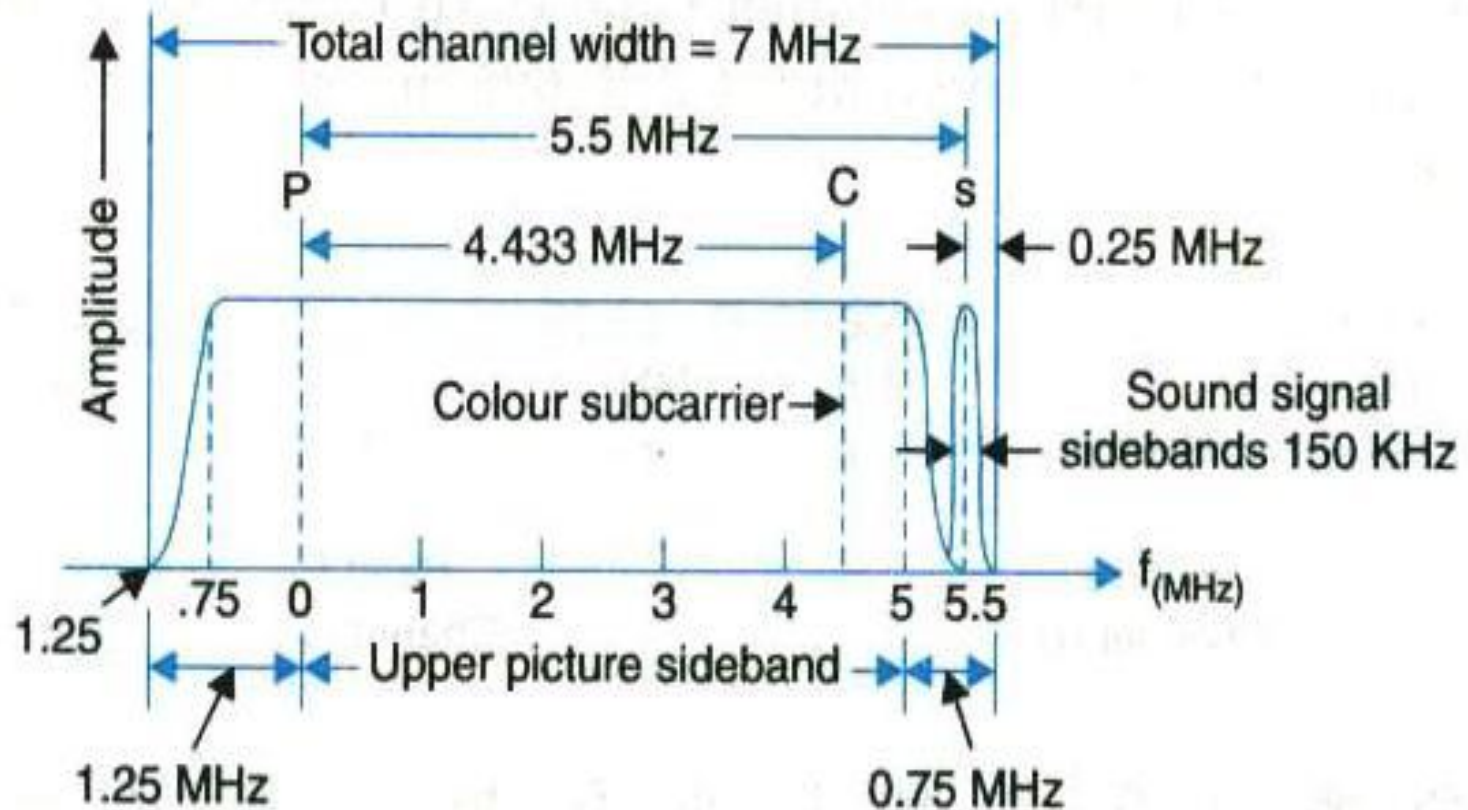


Fig. 4.4 C.C.I.R. (Indian and European) TV channel sideband spectrum.
C is colour subcarrier frequency.

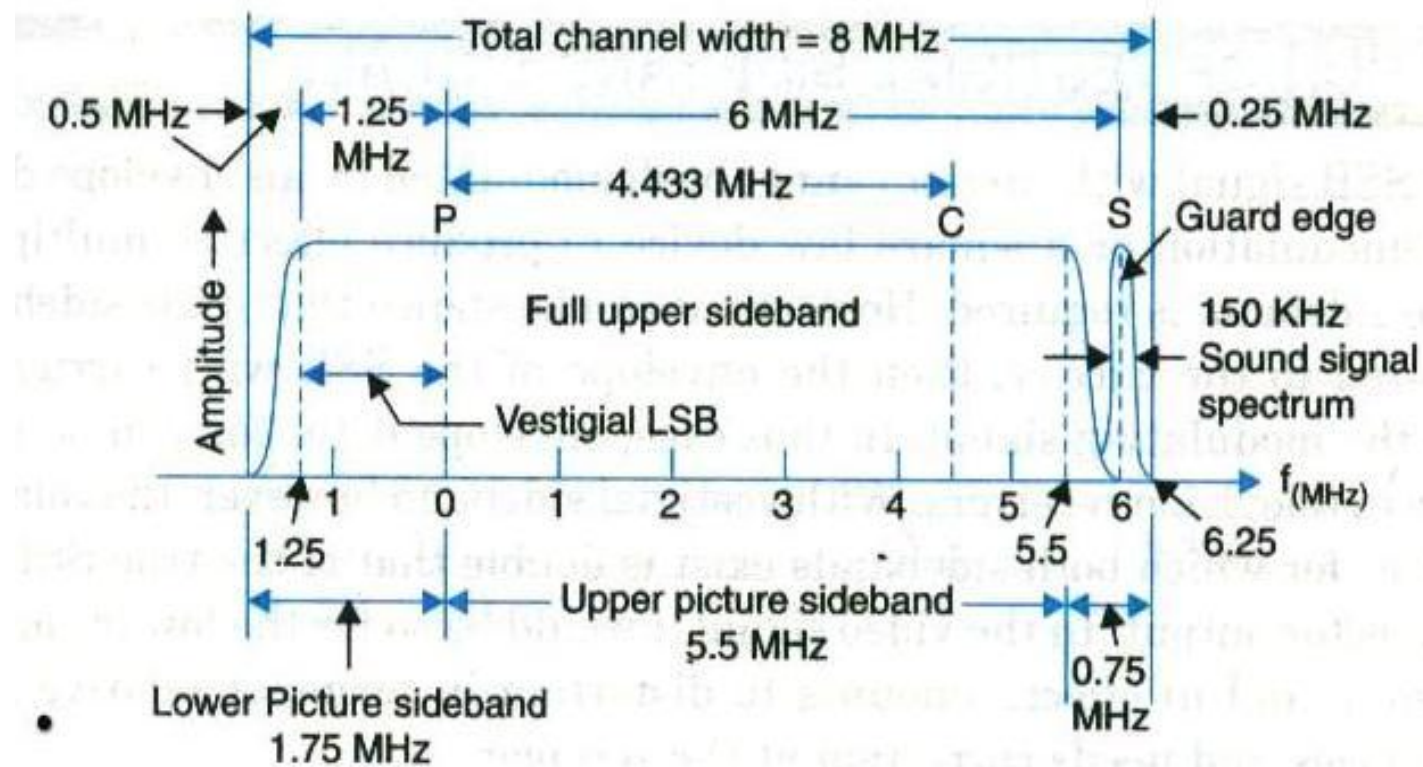


Fig. 4.5(a) U.K. TV channel standards.

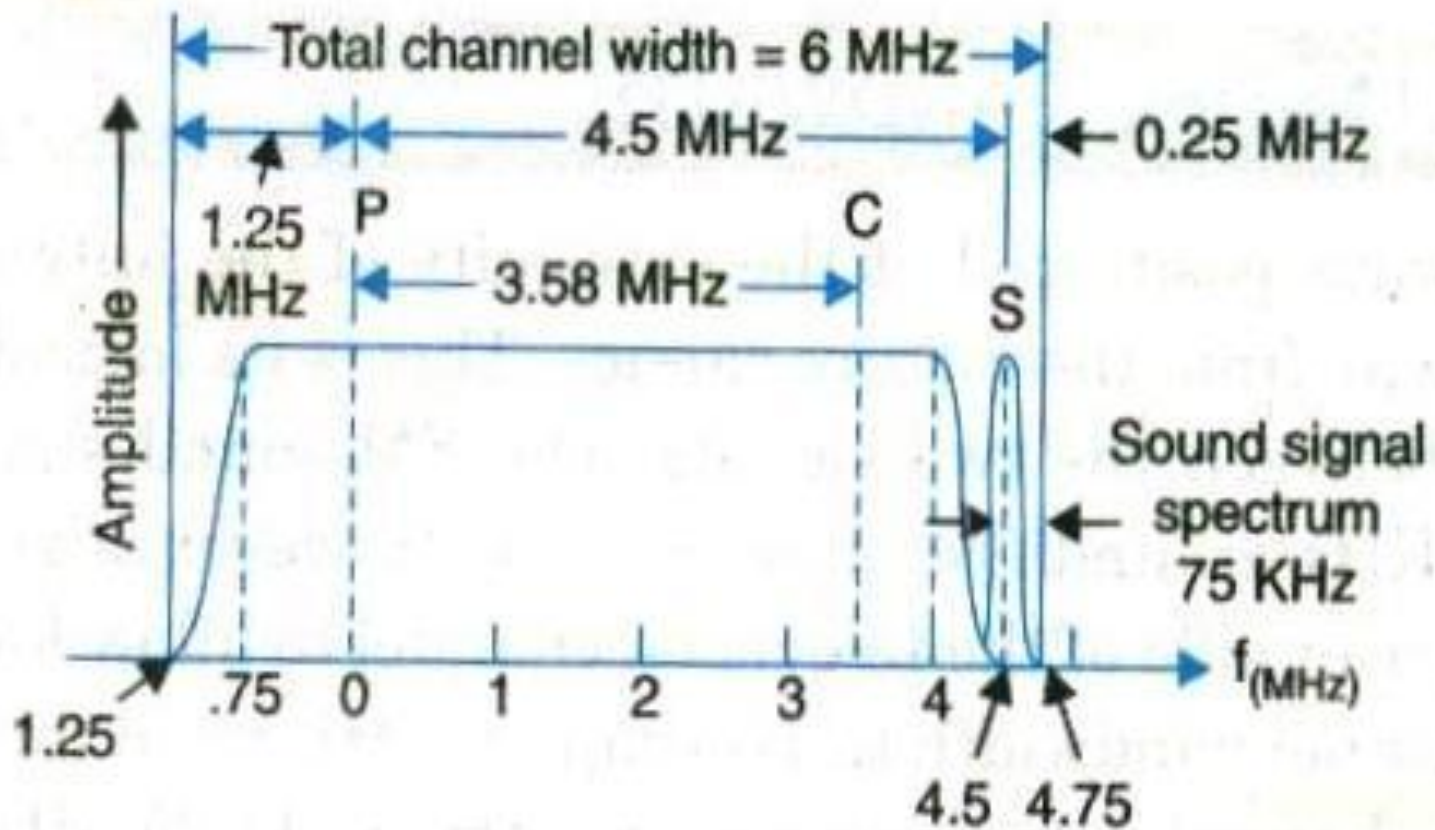


Fig. 4.5(b) American TV channel standards.

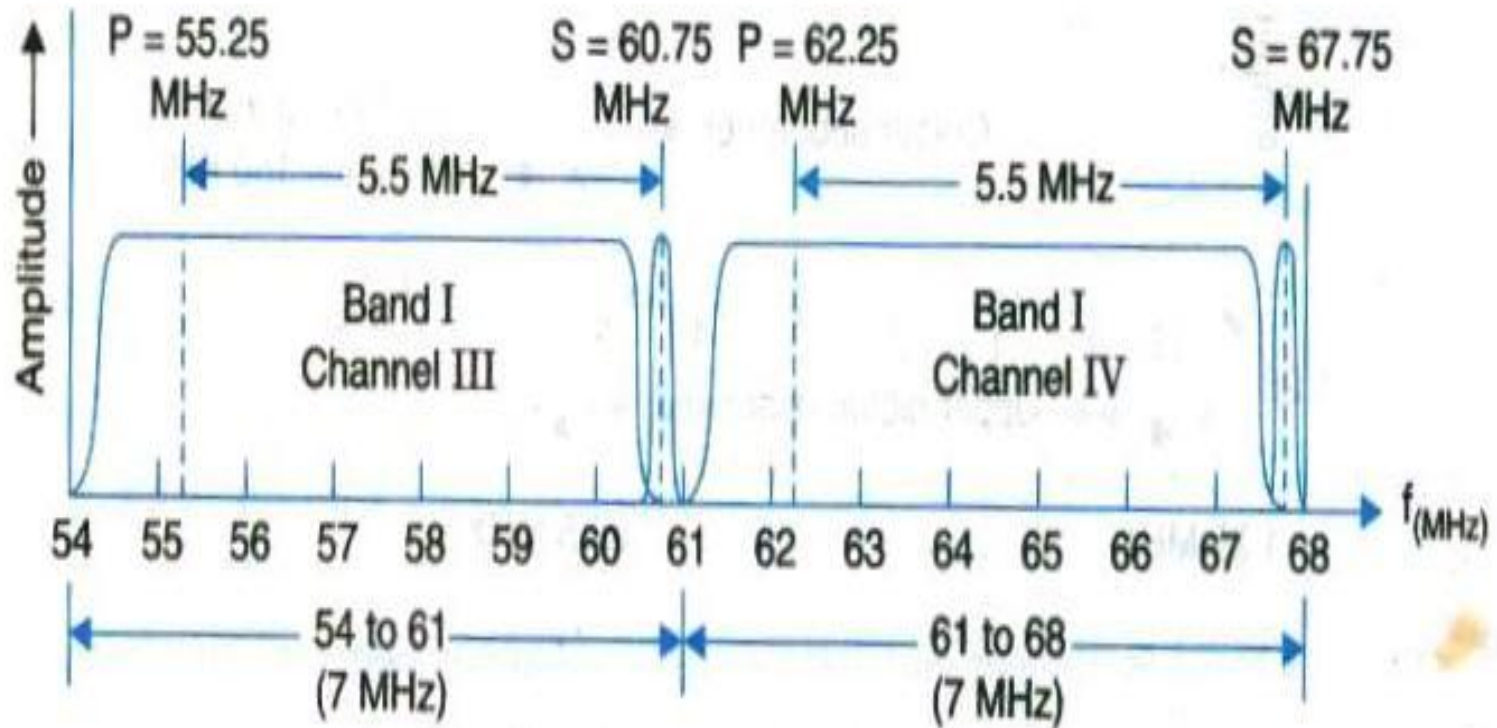


Fig. 4.6 Sideband spectrum of two adjacent channels of the lower VHF band of television station allocations.

Characteristics of the 625-B Monochrome TV System

No. of lines per picture (frame)	625
Field frequency (Fields/second)	50
Interlace ratio, <i>i.e.</i> , No. of fields/picture	2/1
Picture (frame) frequency, <i>i.e.</i> , Pictures/second	25
Line frequency and tolerance in lines/second, (when operated non-synchronously)	$15625 \pm 0.1\%$
Aspect Ratio (width/height)	4/3
Scanning sequence	(i) Line : Left to right (ii) Field : Top to bottom
System capable of operating independently of power supply frequency	YES
Approximate gamma of picture signal	0.5
Nominal video bandwidth, <i>i.e.</i> , highest video modulating frequency (MHz)	5
Nominal Radio frequency bandwidth, <i>i.e.</i> , channel bandwidth (MHz)	7
Sound carrier relative to vision carrier (MHz)	+ 5.5
Sound carrier relative to nearest edge of channel (MHz)	- 0.25
Nearest edge of channel relative to picture carrier (MHz)	- 1.25
Fully radiated sideband	Upper
Nominal width of main sideband (upper) (MHz)	5
Width of end-slope of full (Main) sideband (MHz)	0.5
Nominal width of vestigial sideband (MHz)	0.75
Vestigial (attenuated) sideband	Lower

TV

By KMN

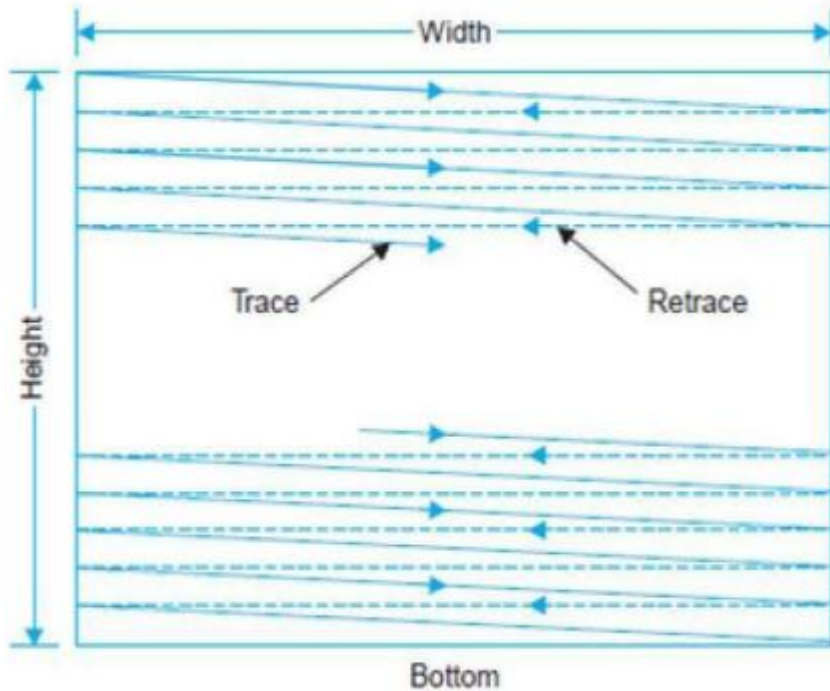
ASPECT RATIO

- The frame adopted in all television systems is rectangular with width/height ratio, i.e., aspect ratio = $4/3$.
- For human most of the motion occurs in the horizontal plane and so a larger width is desirable.
- The eyes can view with more ease and comfort when the width of a picture is more than its height.
- This enables direct television transmission of film programmes without wastage of any film area.
- This is achieved by setting the magnitudes of the current in the deflection coils to correct values, both at the TV camera and receiving picture tube.
- Synchronizing pulses are transmitted along with the picture information to achieve exact congruence between transmitter and receiver scanning systems

IMAGE CONTINUITY

- illusion of continuity and any motion in the scene appears on the picture tube screen as a smooth and continuous.
- the sensation produced when nerves of the eye's retina are stimulated by incident light does not cease immediately after the light is removed but persists for about 1/16th of a second.
- the scanning rate per second is made greater than sixteen, or the number of pictures shown per second is more than sixteen, the eye is able to integrate the changing levels of brightness in the scene.
- when the picture elements are scanned rapidly enough, they appear to the eye as a complete picture unit.
- In present day motion pictures twenty-four still pictures of the scene are taken per second and later projected on the screen at the same rate.
- Each picture or frame is projected individually as a still picture, but they are shown one after the other in rapid succession to produce the illusion of continuous motion of the scene

scanning

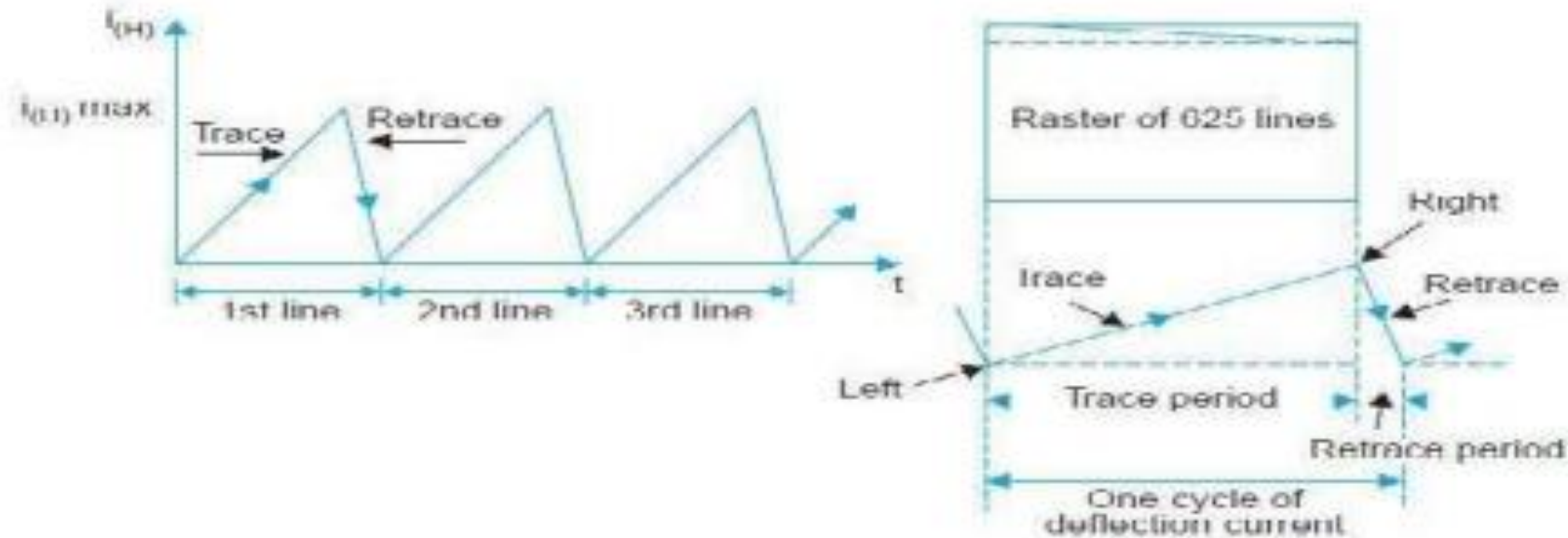


The scene is scanned rapidly both in the horizontal and vertical directions simultaneously to provide sufficient number of complete pictures or frames per second to give the illusion of continuous motion.

Instead of the 24 as in commercial motion picture practice, the frame repetition rate is 25 per second in most television systems

Figure. Path of scanning beam in covering picture area

Horizontal scanning

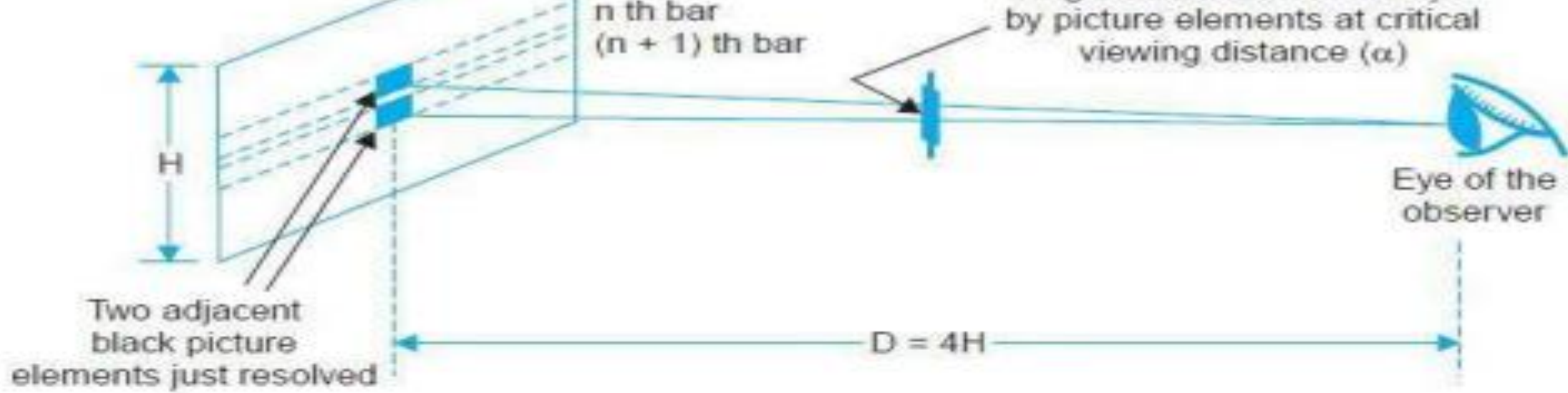


Waveform of current in the horizontal deflection coils producing linear (constant velocity) scanning in the horizontal direction.

- In Horizontal scanning. Fig. (a) shows the trace and retrace of several horizontal lines.
- The linear rise of current in the horizontal deflection coils Fig. (b) deflects the beam across the screen with a continuous, uniform motion for the trace from left to right.
- At the peak of the rise, the sawtooth wave reverses direction and decreases rapidly to its initial value.
- This fast reversal produces the retrace or flyback. The start of the horizontal trace is at the left edge of raster.
- In Vertical scanning The sawtooth current in the vertical deflection coils moves the electron beam from top to bottom of the raster at a uniform speed while the electron beam is being deflected horizontally.
- the rapid vertical retrace returns the beam to the top

continued

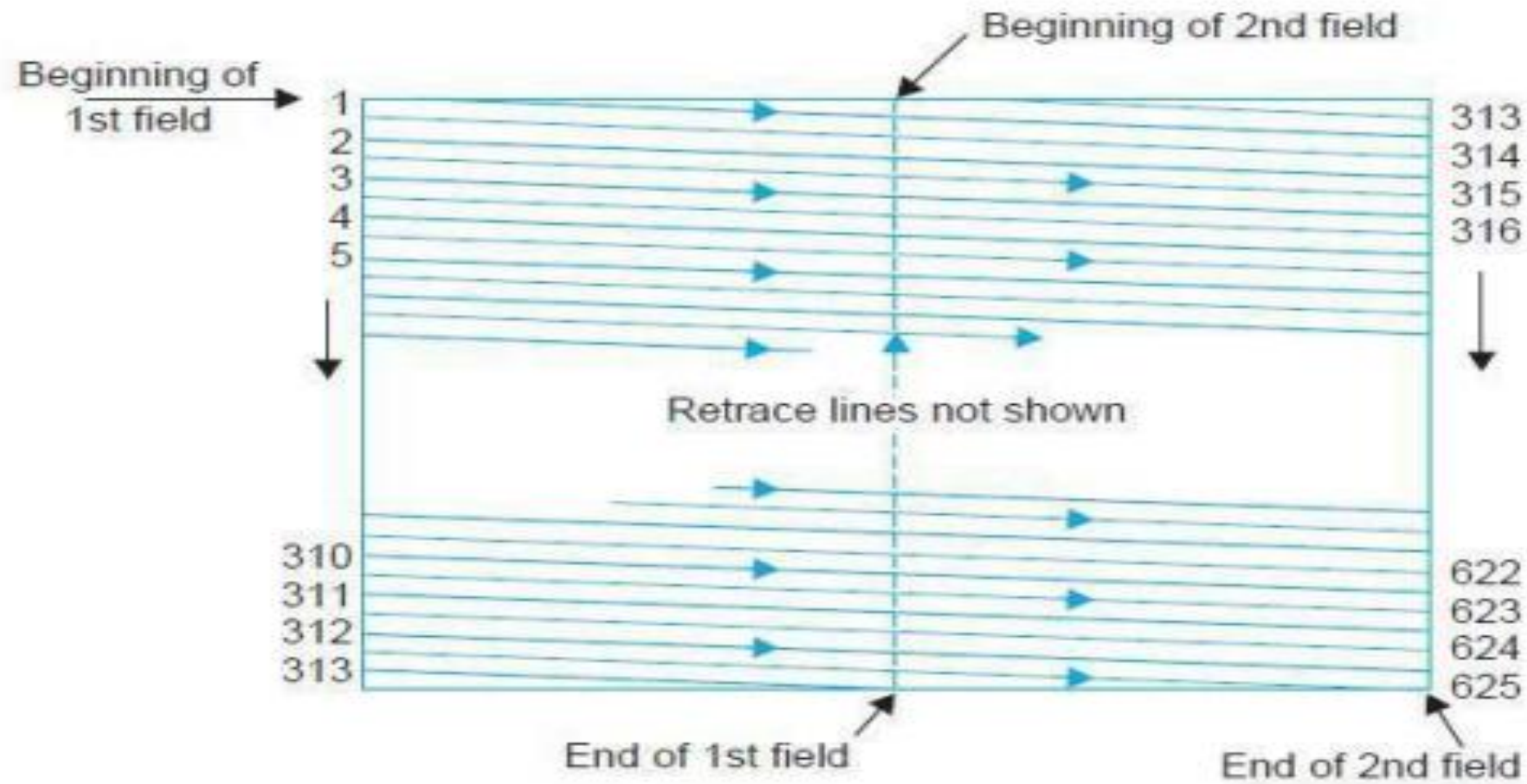
- The maximum number of alternate light and dark elements (lines) which can be resolved by the eye is given by
- $N_v = \alpha \rho$
- where N_v = total number of lines (elements) to be resolved in the vertical direction,
- α = minimum resolving angle of the eye expressed in radians,
- and $\rho = D/H$ = viewing-distance/picture height



Thus as a compromise between quality and cost, the total number of lines inclusive of those lost during vertical retrace has been chosen to be 625 in the 625-B monochrome TV system.

In the 525 line American system, the total number of lines has been fixed at 525 because of a somewhat higher scanning rate employed in this system.

Interlace scanning



Interlaced scanning.

- In television pictures an effective rate of 50 vertical scans per second is utilized to reduce flicker.
- This is accomplished by increasing the downward rate of travel of the scanning electron beam,
- so that every alternate line gets scanned instead of every successive line.
- when the beam reaches the bottom of the picture frame, it quickly returns to the top to scan those lines that were missed in the previous scanning.
- Thus the total number of lines are divided into two groups called 'fields'. Each field is scanned alternately. This method of scanning is known as interlaced scanning.
- the 625 lines of each frame or picture are divided into sets of 312.5 lines and each set is scanned alternately to cover the entire picture area.

- To achieve this the horizontal sweep oscillator is made to work at a frequency of 15625 Hz ($312.5 \times 50 = 15625$) to scan the same number of lines per frame ($15625/25 = 625$ lines).
- vertical sweep circuit is run at a frequency of 50 Hz.
- In the American TV system, a field frequency of 60 was adopted because the supply frequency is 60 Hz in USA.
- This brings the total number of lines scanned per second ($(525/2) \times 60 = 15750$) lines to practically the same as in the 625 line system.

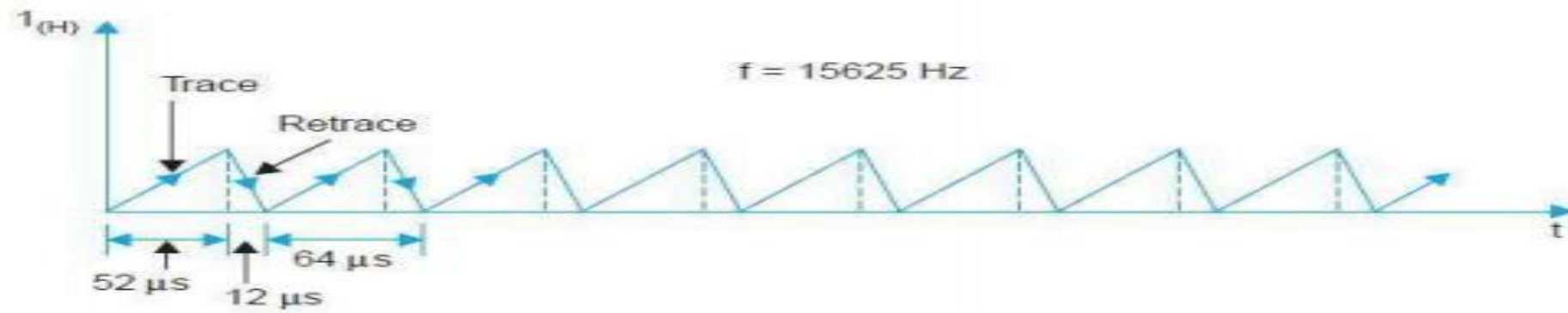


Figure. Horizontal deflection current

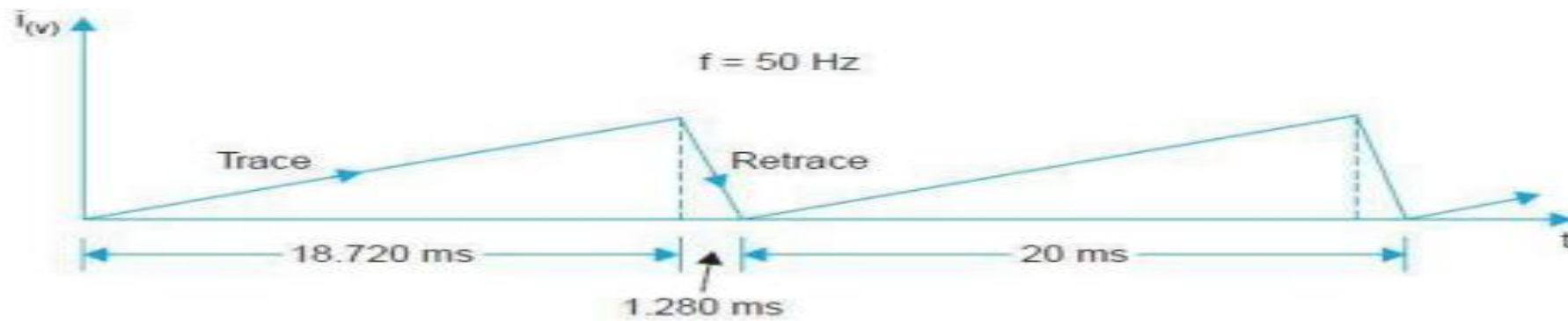


Figure. Vertical deflection current

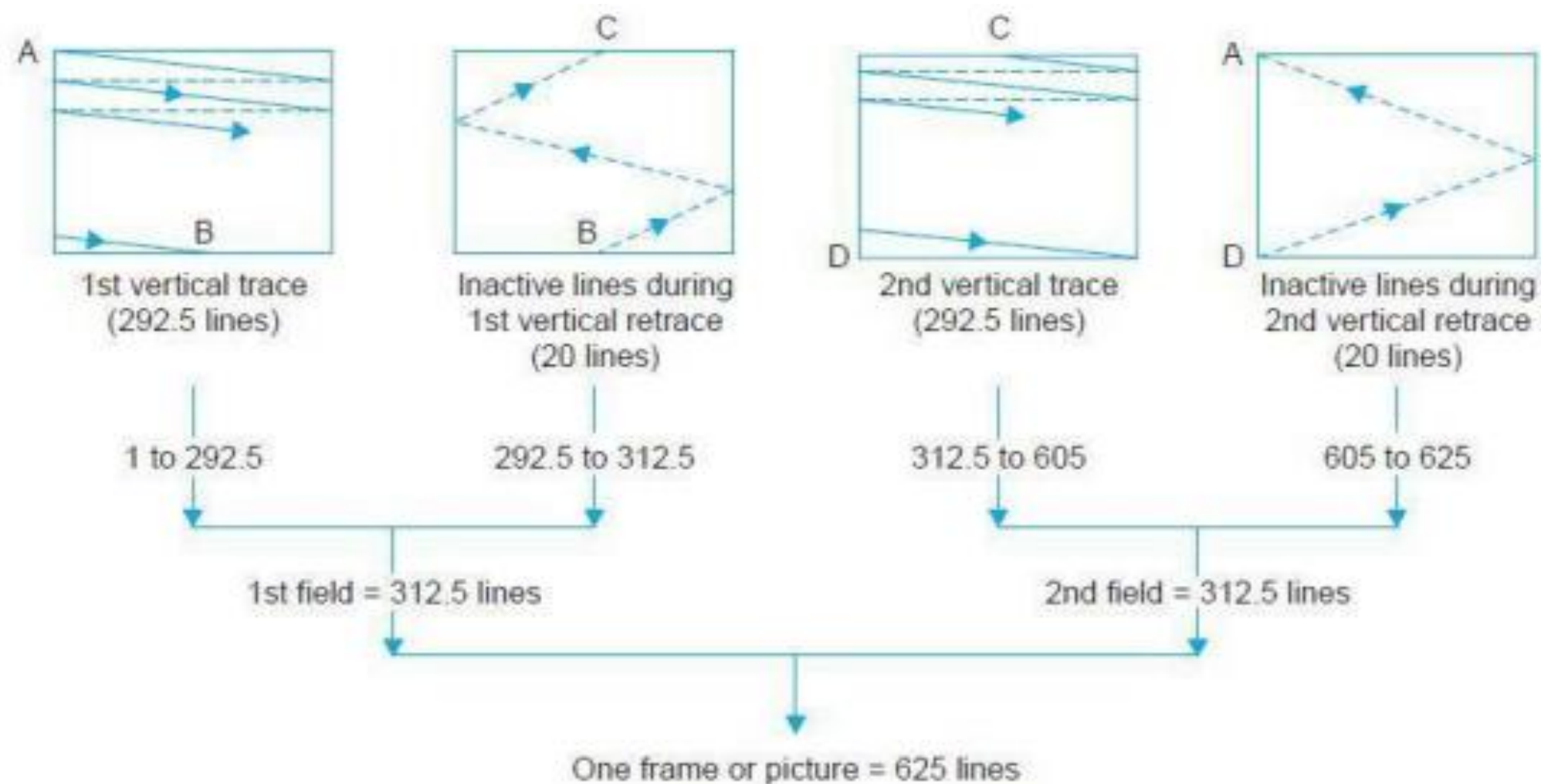


Figure. Odd line interlaced scanning procedure.

TELEVISION CAMERA TUBES

- Some of the more important functions,
 - (i) sensitivity to visible light,
 - (ii) wide dynamic range with respect to light intensity, and
 - (iii) ability to resolve details while viewing a multi-element scene.

Challenges:

- (i) poor sensitivity, (ii) poor resolution, (iii) high noise level, (iv) undesirable spectral response, (v) instability, (vi) poor contrast range and (vii) difficulties of processing.

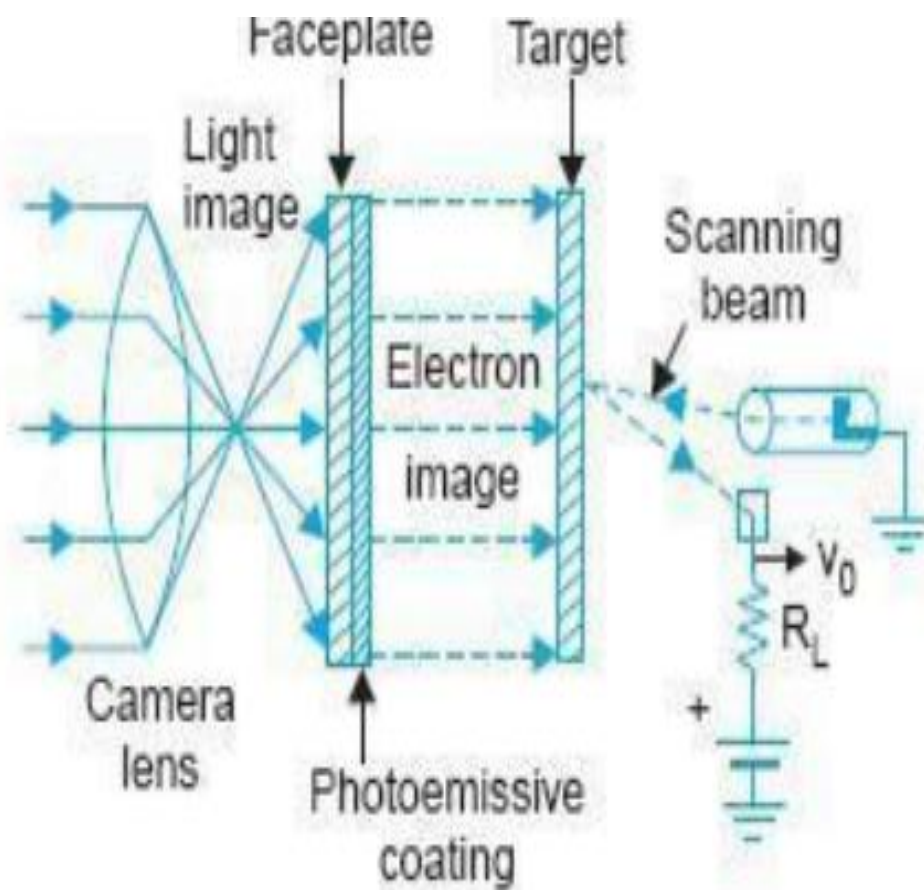


Figure. Production of video signal by photoemission

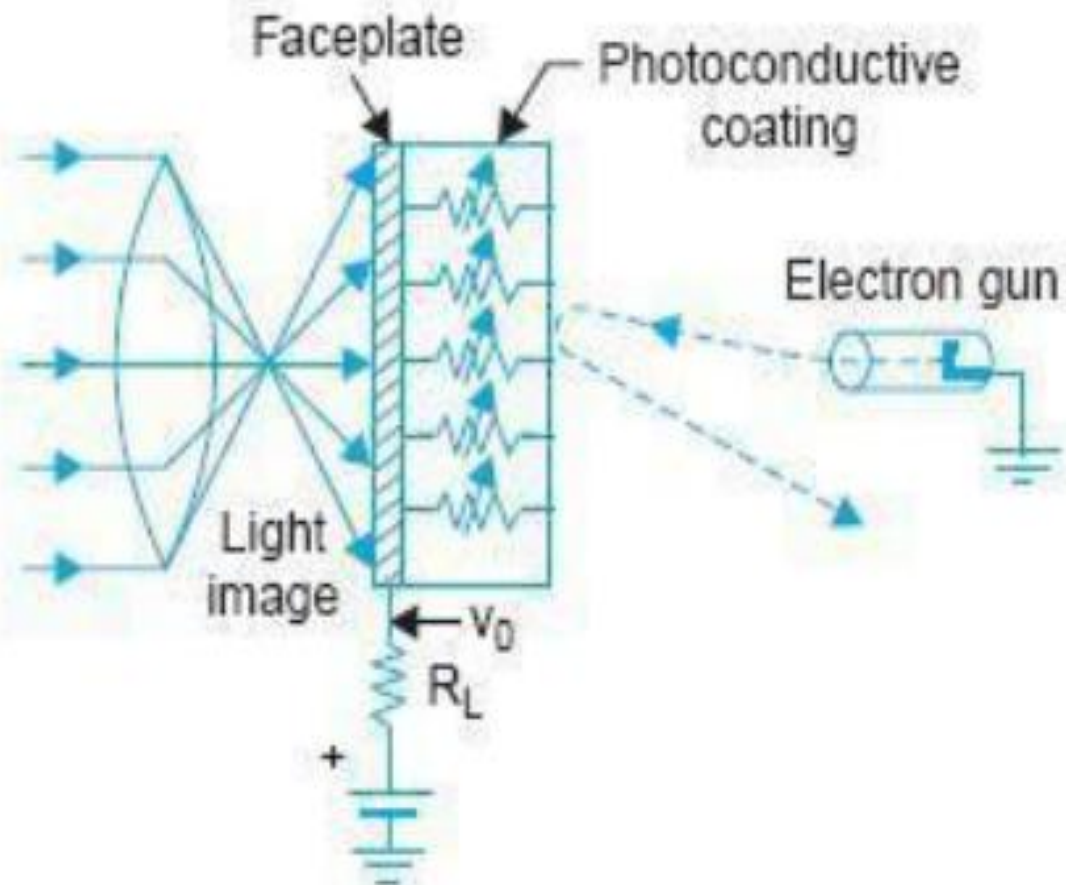


Figure. Production of video signal by photoconduction.

Electron multiplication

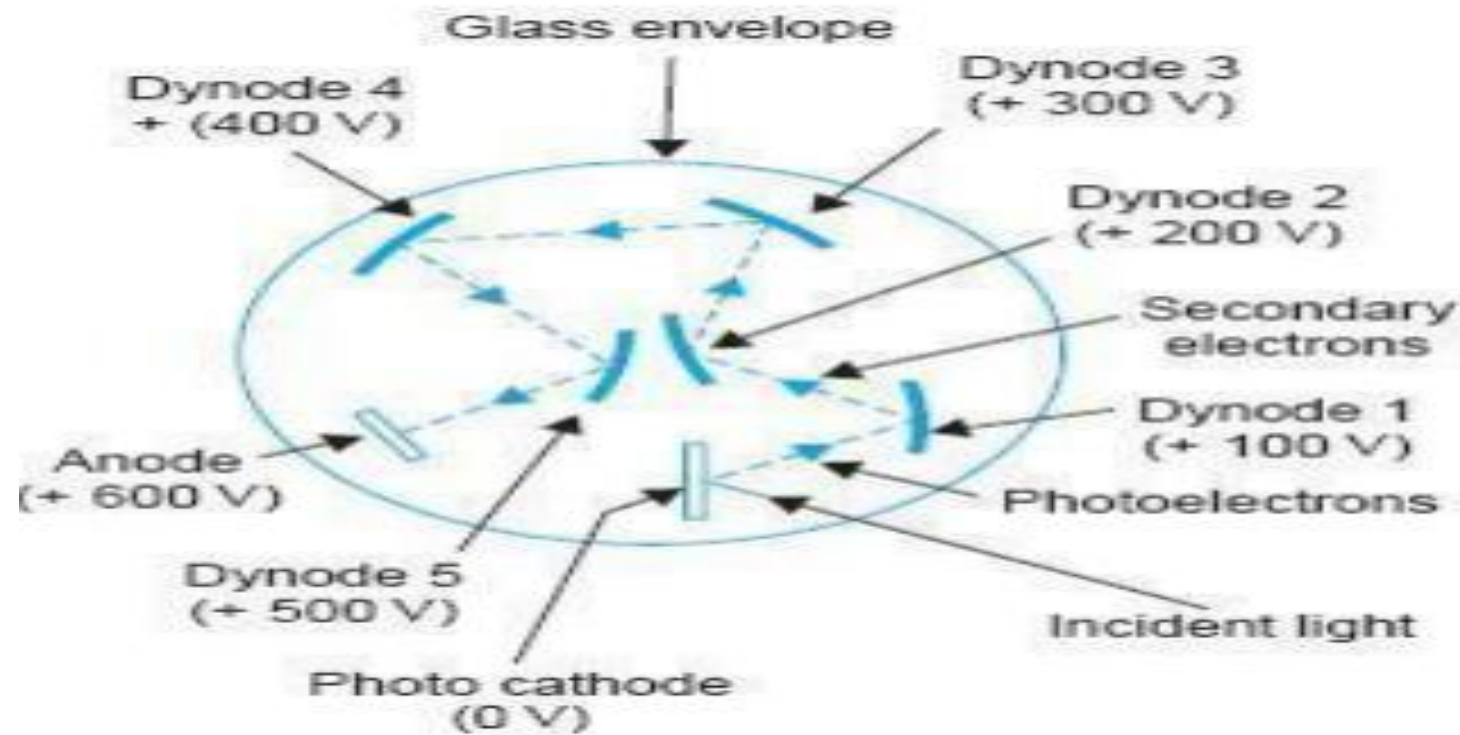


Figure. Illustration of an electron-multiplier structure.

Plumbicon camera tube

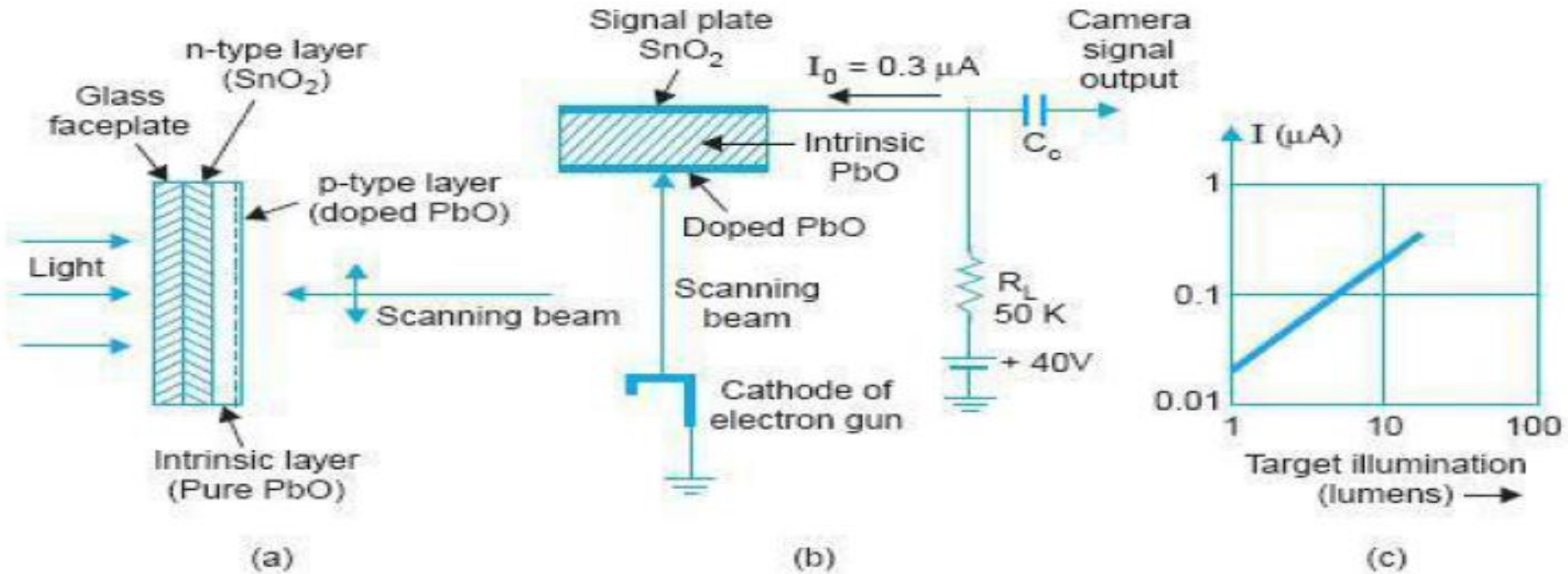
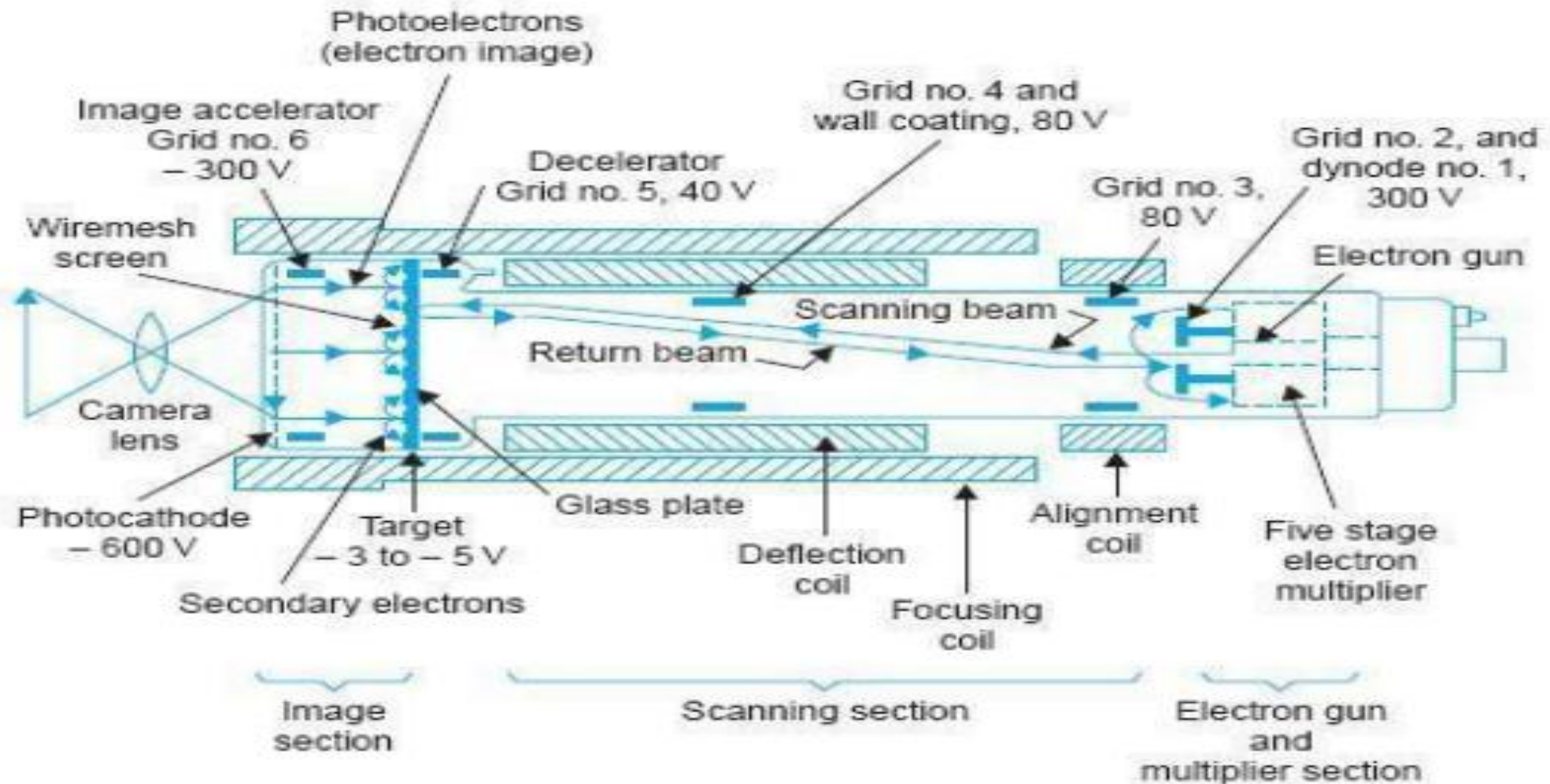


Figure. Plumbicon camera tube (a) target details (b) output signal current and (c) characteristics.

- plumbicon is very similar to the standard vidicon.
- the Focus and deflection are both obtained magnetically.
- it operates effectively as a P–I–N semi- conductor diode.
- The inner surface of the faceplate is coated with a thin transparent conductive layer of tin oxide (SnO_2).
- This forms a strong N layer and serves as the signal plate of the target.
- On the scanning side of this layer is deposited a photoconductive layer of pure lead monoxide (PbO) which is intrinsic or 'I' type.
- Finally the pure PbO is doped to form a P type semiconductor on which the scanning beam lands.

- The incidence of light on the target results in photo excitation of semiconductor junction between the pure PbO and doped layer.
- The resultant decrease in resistance causes signal current flow which is proportional to the incident light on each photo element.
- The overall thickness of the target is 10 to 20 μm .

IMAGE ORTHICON



Electron multiplication

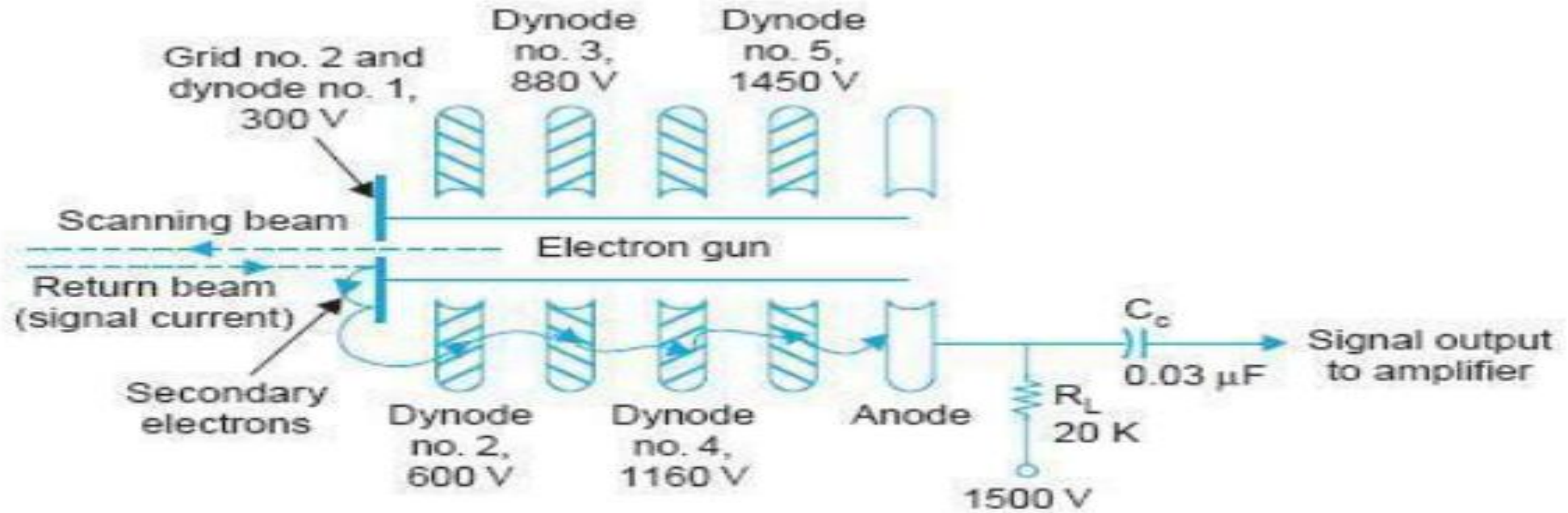


Figure. Electron-multiplier section of the Image Orthicon.

- This tube makes use of the high photo emissive sensitivity obtainable from photocathodes,
- image multiplication at the target caused by secondary emission and an electron multiplier
- It has three main sections: image section, scanning section and electron gun-cum-multiplier section.
- In Image Section The inside of the glass face plate at the front is coated with a silver, antimony coating sensitized with cesium, to serve as photocathode.

- the electron image produced at the photocathode is made to move towards the target plate located at a short distance from it. The target plate is made of a very thin sheet of glass and can store the charge received by it.
- Because of the high velocity attained by the electrons while in motion from photocathode to the target plate, secondary emission results, as the electrons bombard at the target surface.
- These secondary electrons are collected by a wire-mesh screen, which is located in front of the target on the image side and is maintained at a slightly higher potential with respect to the target

Scanning Section

- The electron gun structure produces a beam of electrons that is accelerated towards the target.
- Deflection of electron beams to scan the entire target plate is accomplished by magnetic fields of vertical and horizontal deflecting coils mounted on yoke external to the tube.
- These coils are fed from two oscillators, one working at 15625 Hz, for horizontal deflection, and the other operating at 50 Hz, for vertical deflection

Electron Multiplier

- When the returning electrons strike the disc which is at a positive potential of about 300 volts, with respect to the target, they produce secondary emission.
- The multiplication so obtained maintains a high signal to noise ratio.

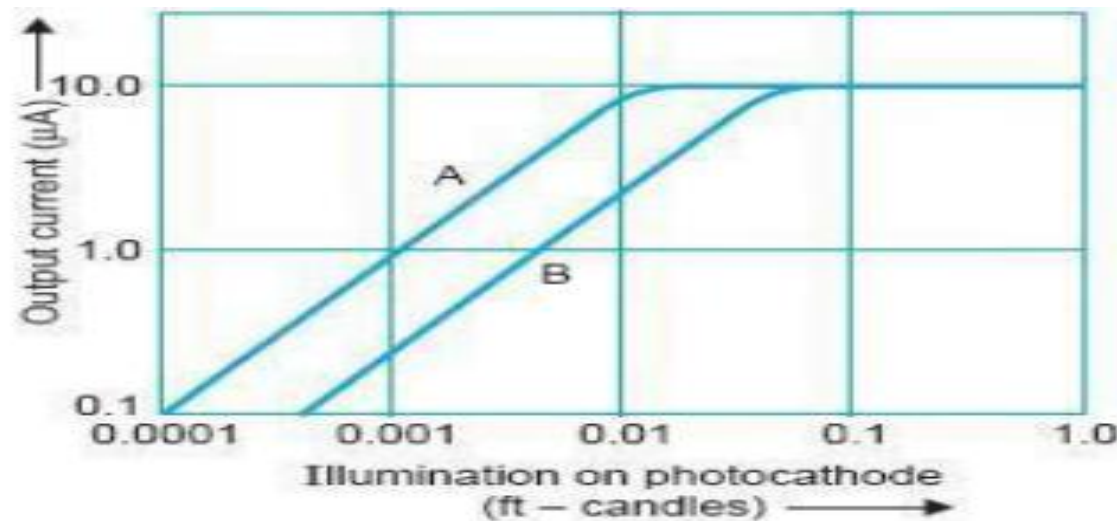


Figure. Light transfer characteristics of two different Image Orthicons.

Composite video signal

- A **composite video signal** is a single analog signal that combines **video (picture) information, synchronization pulses**, and sometimes **color information** into one waveform for transmission. Because all components Luminance (Y) and Chrominance (C) ie (Y + C + Sync) are **combined into one single signal** instead of being transmitted separately

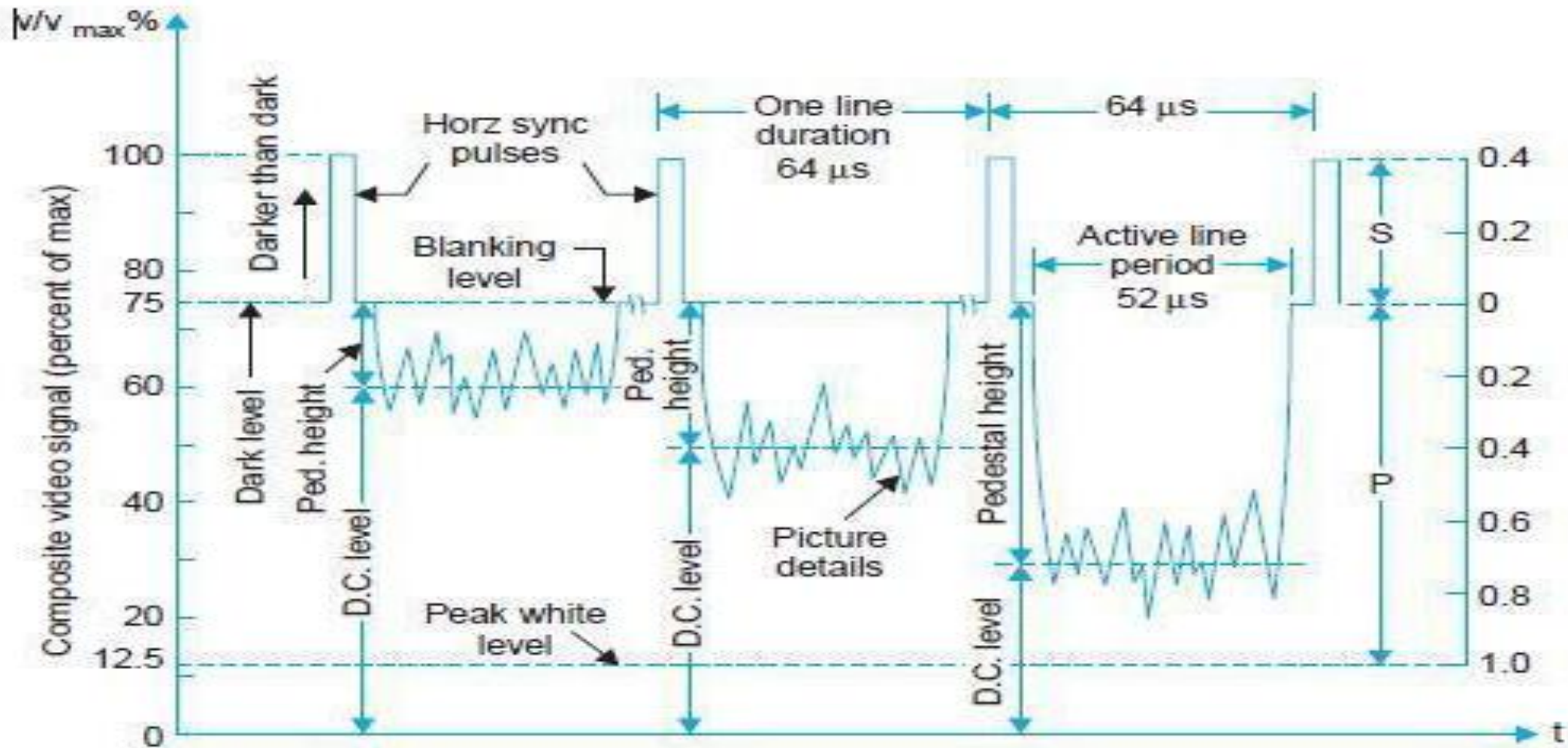
It was widely used in analog TV systems like:

NTSC

PAL

SECAM

Composite video signal



A composite video signal consists of three main parts:

Luminance (Y)

- Brightness information (black & white picture)
- Determines gray scale levels
- Contains picture details

Chrominance (C)

- Color information
- Superimposed on luminance using a color subcarrier
- Different frequency for NTSC and PAL systems

Synchronization Pulses

- Horizontal sync pulse → tells electron beam to move to next line
- Vertical sync pulse → tells beam to move to next frame
- Necessary for stable picture

Horizontal and Vertical Blanking pulses

